

WAYS OF WORKING WITH CELLS

Analyzing Cells, Molecules, and Systems

CHAPTER

8

Progress in science is often driven by advances in technology. The entire field of cell biology, for example, came into being when optical craftsmen learned to grind small lenses of sufficiently high quality to observe cells and their substructures. Innovations in lens grinding, rather than any conceptual or philosophical advance, allowed Hooke and van Leeuwenhoek to discover a previously unseen cellular world, where tiny creatures tumble and twirl in a droplet of water (**Figure 8-1**).

The twenty-first century is a particularly exciting time for biology. New methods for analyzing cells, proteins, DNA, and RNA are fueling an information explosion and allowing scientists to study cells and their macromolecules in previously unimagined ways. We now have access to the sequences of many billions of nucleotides, providing the complete molecular blueprints for hundreds of organisms—from microbes and mustard weeds to worms, flies, mice, dogs, chimpanzees, and humans. And powerful new techniques are helping us to decipher that information, allowing us not only to compile huge, detailed catalogs of genes and proteins but also to begin to unravel how these components work together to form functional cells and organisms. The long-range goal is nothing short of obtaining a complete understanding of what takes place inside a cell as it responds to its environment and interacts with its neighbors.

In this and the following chapter, we present some of the principal methods used to study cells and their molecular components. Chapter 9 describes the remarkable advances in microscopy that have helped fuel our understanding of the structure and function of cells. In this chapter, we first discuss the rapidly developing methods for analysis of the molecules and genes that drive cell behavior. We present the techniques used to determine protein structure, function, and interactions, and we discuss the breakthroughs in DNA technology that continue to revolutionize our understanding of cell function. We end the chapter with an overview of some of the mathematical approaches that are helping us understand the enormous complexity of cells. By considering cells as dynamic systems with many moving parts, mathematical approaches can reveal hidden insights into

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how the many components of cells work together to produce the special qualities of life.

ISOLATING CELLS AND GROWING THEM IN CULTURE

As described in Chapter 1, numerous model organisms are used by researchers to unravel the molecular basis of cell biology. In the first section of this chapter, we concern ourselves with the organisms and cells best suited for biochemical studies of proteins in the cell. Typically, we gain access to these proteins by obtaining large numbers of cells and physically breaking them open. Unicellular organisms, such as bacteria and yeast, are easy to produce in large amounts in the laboratory and are rich sources of the proteins involved in fundamental cell processes. But a deep understanding of human proteins in specific cell types requires human cells, or at the very least cells from a mammal. Specific animal tissues can be a useful solution, but these tend to be composed of a heterogeneous mixture of cell types. To obtain as much information as possible about specific cell types in a tissue, biologists have developed ways of dissociating cells from tissues and separating them according to type. These manipulations result in a relatively homogeneous population of cells that can then be analyzed—either directly or after their number has been greatly increased by allowing the cells to proliferate in culture.

Cells Can Be Isolated from Tissues and Grown in Culture

Intact tissues provide the most realistic source of material, as they represent the actual cells found within the body. For some biochemical preparations, the protein of interest can be obtained in sufficient quantity without having to separate the tissue or organ into cell types. Examples include the preparation of histones from calf thymus, actin from rabbit muscle, or tubulin from cow brain. For the majority of proteins, however, tissues are not an ideal source and it is preferable to use specific cell types grown in culture. Cultured cells provide a more homogeneous population of cells from which to extract material, and they are also much more convenient to work with in the laboratory. Given appropriate surroundings, most animal cells can live, multiply, and even express differentiated properties in a culture dish. The cells can be watched continually under the microscope or analyzed biochemically, and the effects of adding or removing specific molecules, such as hormones or growth factors, can be systematically explored.

Experiments performed on cultured cells are sometimes said to be carried out *in vitro* (literally, “in glass”) to contrast them with experiments using intact organisms, which are said to be carried out *in vivo* (literally, “in the living organism”). These terms can be confusing, however, because they are often used in a very different sense by biochemists. In the biochemistry lab, *in vitro* refers to reactions carried out in a test tube in the absence of living cells, whereas *in vivo* refers to any reaction taking place inside a living cell, even if that cell is growing in culture.

Tissue culture began in 1907 with an experiment designed to settle a controversy in neurobiology. The hypothesis under examination was known as the neuronal doctrine, which proposed that each nerve fiber is the outgrowth of a single nerve cell and not the product of the fusion of many cells. To test this contention, small pieces of spinal cord were placed in lymphatic fluid in a warm, moist chamber and observed at regular intervals under the microscope. After a day or so, individual nerve cells could be seen extending long, thin filaments (axons) into the clot. Thus, the neuronal doctrine received strong support, and the foundation was laid for the cell-culture revolution.

These original experiments on nerve fibers used cultures of small tissue fragments called explants. Today, cultures are more commonly made from suspensions of cells dissociated from tissues. The first step in isolating individual cells is to disrupt the extracellular matrix and cell–cell junctions that hold the cells together. For this purpose, a tissue sample is typically treated with proteolytic enzymes (such as trypsin and collagenase) to digest proteins in the extracellular

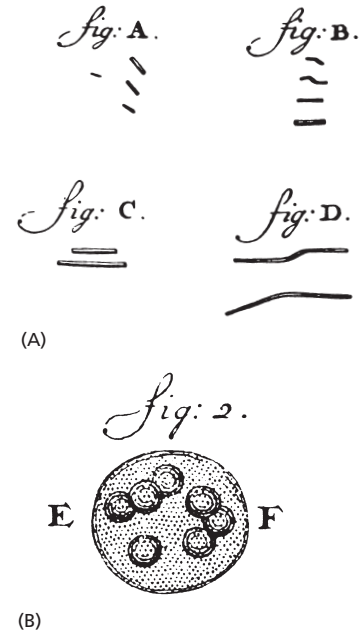


Figure 8–1 Microscopic life. A sample of “diverse animalcules” seen by van Leeuwenhoek using his simple microscope. (A) Bacteria seen in material he excavated from between his teeth. Those in fig. B he described as “swimming first forward and then backwards” (1692). (B) The eukaryotic green alga *Volvox* (1700). (Courtesy of the John Innes Foundation.)

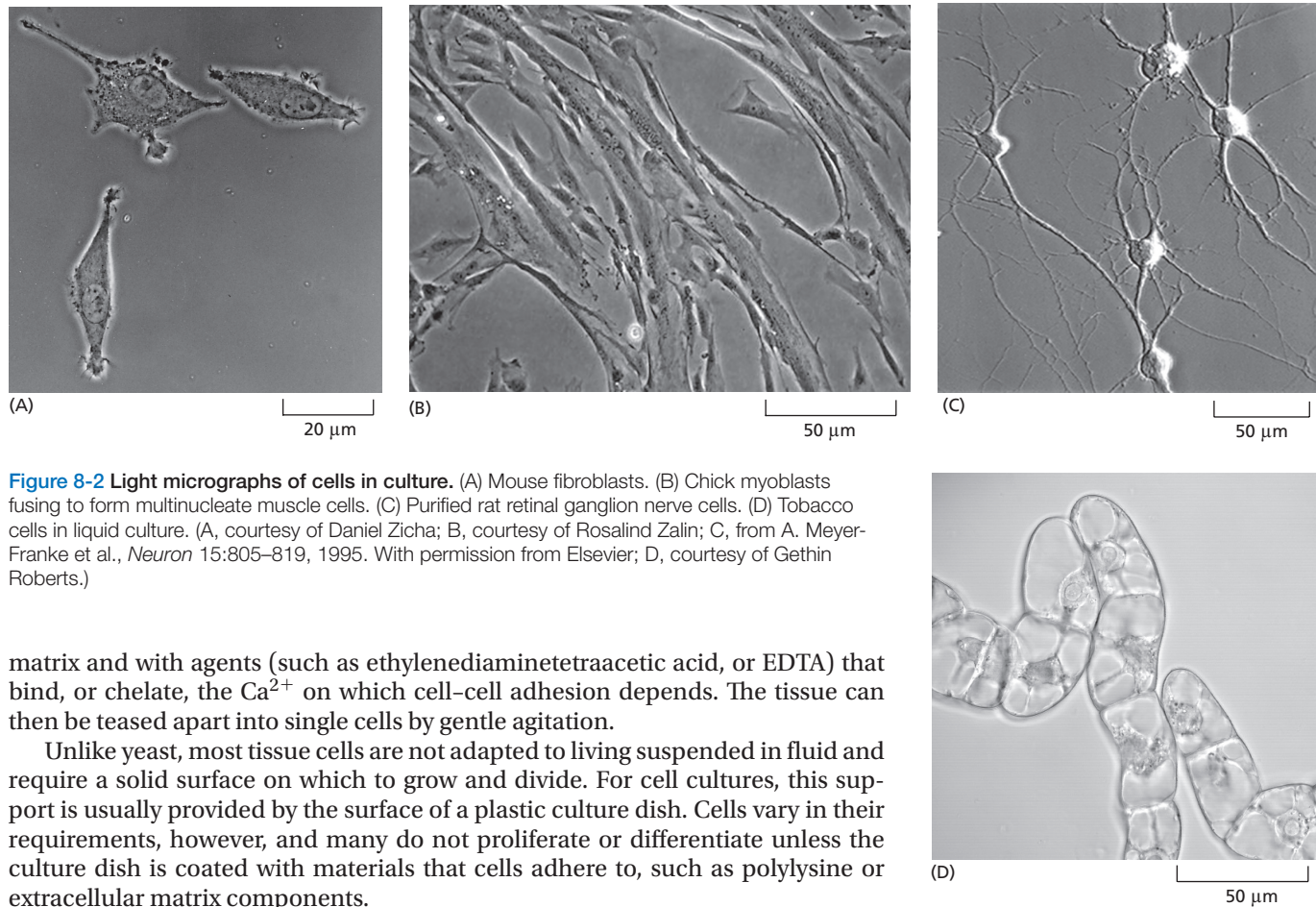


Figure 8-2 Light micrographs of cells in culture. (A) Mouse fibroblasts. (B) Chick myoblasts fusing to form multinucleate muscle cells. (C) Purified rat retinal ganglion nerve cells. (D) Tobacco cells in liquid culture. (A, courtesy of Daniel Zicha; B, courtesy of Rosalind Zalin; C, from A. Meyer-Franke et al., *Neuron* 15:805–819, 1995. With permission from Elsevier; D, courtesy of Gethin Roberts.)

matrix and with agents (such as ethylenediaminetetraacetic acid, or EDTA) that bind, or chelate, the Ca^{2+} on which cell–cell adhesion depends. The tissue can then be teased apart into single cells by gentle agitation.

Unlike yeast, most tissue cells are not adapted to living suspended in fluid and require a solid surface on which to grow and divide. For cell cultures, this support is usually provided by the surface of a plastic culture dish. Cells vary in their requirements, however, and many do not proliferate or differentiate unless the culture dish is coated with materials that cells adhere to, such as polylysine or extracellular matrix components.

Cultures prepared directly from the tissues of an organism are called *primary cultures*. These can be made with or without an initial fractionation step to separate different cell types. In most cases, cells in primary cultures can be removed from the culture dish and recultured repeatedly in so-called secondary cultures; in this way, they can be repeatedly subcultured (*passaged*) for weeks or months. Such cells often display many of the differentiated properties appropriate to their origin (**Figure 8–2**): fibroblasts continue to secrete collagen; cells derived from embryonic skeletal muscle fuse to form muscle fibers that contract spontaneously in the culture dish; nerve cells extend axons that are electrically excitable and make synapses with other nerve cells; and epithelial cells form extensive sheets with many of the properties of an intact epithelium. Because these properties are maintained in culture, they are accessible to study in ways that are often not possible in intact tissues.

Embryonic stem cells are an important cell type isolated from the early mammalian embryo. As described in Chapter 22, these cells are *pluripotent*; that is, they have the potential to differentiate into any cell type in the body. When cultured in the presence of the appropriate extracellular signaling factors and nutrients, stem cells can be directed to differentiate into a wide range of specific cell types. Under some conditions, it is even possible to stimulate these cells to assemble into three-dimensional multicellular structures that are miniature versions of certain organs, such as the gut. These *organoids* provide a powerful tool for the analysis of tissue function (see Chapter 22).

Cell culture is not limited to animal cells. When a piece of plant tissue is cultured in a sterile medium containing nutrients and appropriate growth regulators, many of the cells are stimulated to proliferate indefinitely in a disorganized manner, producing a mass of relatively undifferentiated cells called a *callus*. If the nutrients and growth regulators are carefully manipulated, one can induce the formation of a shoot and then root apical meristems within the callus, and,

in many species, regenerate a whole new plant. Similar to animal cells, callus cultures can be dissociated into single cells, which will grow and divide as a suspension culture (see Figure 8-2D).

Eukaryotic Cell Lines Are a Widely Used Source of Homogeneous Cells

The cell cultures obtained by disrupting tissues tend to suffer from a problem—eventually the cells die. Most vertebrate cells stop dividing after a finite number of cell divisions in culture, a process called *replicative cell senescence* (discussed in Chapter 17). Normal human fibroblasts, for example, typically divide only 25–40 times in culture before they stop. In these cells, the limited proliferation capacity reflects a progressive shortening and uncapping of the cell's telomeres, the repetitive DNA sequences and associated proteins that cap the ends of each chromosome (discussed in Chapter 5). Human somatic cells in the body have turned off production of the enzyme, called *telomerase*, that normally maintains the telomeres, which is why their telomeres shorten with each cell division. Human fibroblasts can often be coaxed to proliferate indefinitely by providing them with the gene that encodes the catalytic subunit of telomerase; in this case, they can be propagated as an *immortalized* cell line.

Some human cells, however, cannot be immortalized by this trick. Although their telomeres remain long, they still stop dividing after a limited number of divisions because culture conditions cause excessive stimulation of cell proliferation, which activates a poorly understood protective mechanism that stops cell division—a process sometimes called *culture shock*. To immortalize these cells, one has to do more than introduce telomerase. One must also inactivate the protective mechanisms, which can be done by introducing certain cancer-promoting oncogenes (discussed in Chapter 20).

Unlike human cells, most rodent cells do not turn off production of telomerase, and therefore their telomeres do not shorten with each cell division. Therefore, if culture shock can be avoided, some rodent cell types will divide indefinitely in culture. In addition, rodent cells often undergo spontaneous genetic changes in culture that inactivate their protective mechanisms, thereby producing immortalized cell lines.

Cell lines can often be most easily generated from cancer cells, but these cultures—referred to as *transformed cell lines*—differ from those prepared from normal cells in several ways. Transformed cell lines often grow without attaching to a surface, for example, and they can proliferate to a much higher density in a culture dish. Similar properties can be induced experimentally in normal cells by transforming them with a tumor-inducing virus or chemical. The resulting transformed cell lines can usually cause tumors if injected into a susceptible animal.

Transformed and nontransformed cell lines are extremely useful in cell research as sources of very large numbers of cells of a uniform type, especially because they can be stored in liquid nitrogen at -196°C for an indefinite period and retain their viability when thawed. It is important to keep in mind, however, that cell lines nearly always differ in important ways from their normal progenitors in the tissues from which they were derived.

Some widely used cell lines are listed in **Table 8-1**. Different lines have different advantages; for example, the PtK epithelial cell lines derived from the rat kangaroo remain flat during mitosis (unlike many other cell types), allowing the mitotic apparatus to be readily observed in action.

Hybridoma Cell Lines Are Factories That Produce Monoclonal Antibodies

As we see in this chapter and throughout this book, antibodies are particularly useful tools for cell biology. Their great specificity allows precise detection of selected proteins among the many thousands that each cell typically produces.

TABLE 8-1 Some Commonly Used Cell Lines

Cell line*	Cell type and origin
NIH-3T3	Fibroblast (mouse)
MDCK	Kidney epithelial cell (dog)
HeLa	Cervical epithelial cell (human)
PtK	Kidney epithelial cell (rat kangaroo)
L6	Myoblast (rat)
PC12	Chromaffin cell (rat)
COS	Kidney fibroblast (monkey)
HEK293	Kidney epithelial cell (human)
CHO	Ovary epithelial cell (Chinese hamster)
RPE	Retinal pigment epithelial cell (human)
Vero	Kidney epithelial cell (African green monkey)
Jurkat	White blood cell (human)

*Many of these cell lines were derived from tumors. All of them are capable of indefinite replication in culture and express at least some of the special characteristics of their cells of origin.

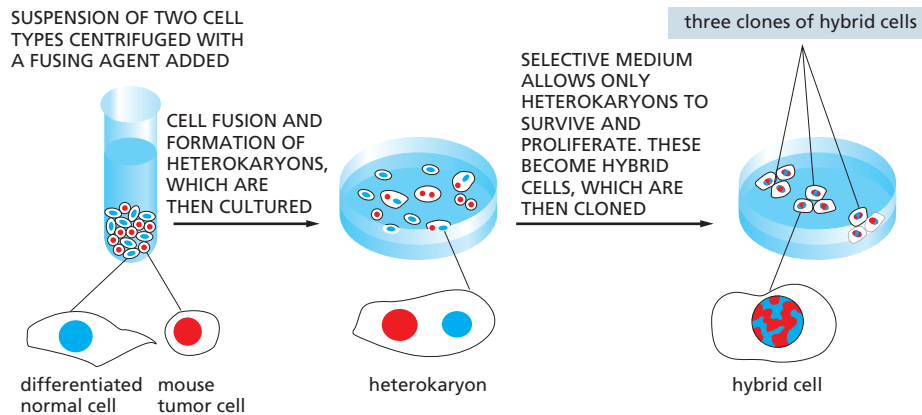


Figure 8-3 The production of hybrid cells. It is possible to fuse one cell with another to form a *heterokaryon*, a combined cell with two separate nuclei. Typically, a suspension of cells is treated with certain inactivated viruses or with polyethylene glycol, each of which alters the plasma membranes of cells in a way that induces them to fuse. Eventually, a heterokaryon proceeds to mitosis and produces a hybrid cell in which the two separate nuclear envelopes have been disassembled, allowing all the chromosomes to be brought together in a single large nucleus. Such hybrid cells can give rise to immortal hybrid cell lines. If one of the parent cells was from a tumor cell line, the hybrid cell is called a hybridoma.

Antibodies are often produced by inoculating animals with the protein of interest and subsequently isolating the antibodies specific to that protein from the serum of the animal. However, only limited quantities of antibodies can be obtained from a single inoculated animal, and the *polyclonal* antibodies produced will be a heterogeneous mixture of antibodies that recognize a variety of different antigenic sites on the protein. Moreover, antibodies specific for the antigen will constitute only a fraction of the antibodies found in the serum. An alternative technology, which allows the production of an unlimited quantity of identical antibodies and greatly increases the specificity and convenience of antibody-based methods, is the production of monoclonal antibodies by hybridoma cell lines.

This procedure involves propagating a clone of cells from a single antibody-secreting B lymphocyte to obtain a homogeneous preparation of antibodies in large quantities. B lymphocytes normally have a limited life span in culture, but individual antibody-producing B lymphocytes from an immunized mouse, when fused with cells derived from a transformed B lymphocyte cell line, can give rise to hybrids that have both the ability to make a particular antibody and the ability to multiply indefinitely in culture (**Figure 8-3**). These **hybridomas** are propagated as individual clones, each of which provides a permanent and stable source of a single type of **monoclonal antibody**. Each type of monoclonal antibody recognizes a single type of antigenic site; for example, a particular cluster of five or six amino acid side chains on the surface of a protein. Their uniform specificity makes monoclonal antibodies much more useful than conventional antisera for many purposes.

An important advantage of the hybridoma technique is that monoclonal antibodies can be made against molecules that constitute only a minor component of a complex mixture. In an ordinary antiserum made against such a mixture, the proportion of antibody molecules that recognize the minor component would be too small to be useful. But if the B lymphocytes that produce the various components of this antiserum are made into hybridomas, it becomes possible to screen individual hybridoma clones from the large mixture to select one that produces the desired type of monoclonal antibody and to propagate the selected hybridoma indefinitely so as to produce that antibody in unlimited quantities. In principle, therefore, a monoclonal antibody can be made against any protein in a biological sample. Once an antibody has been made, it can be used to localize the protein in cells and tissues, to follow its movement, and to purify the protein to study its structure and function.

Monoclonal antibodies are not just useful research tools but are valuable as treatments for a number of human diseases. Certain cancers, for example, can be treated by intravenous infusion of monoclonal antibodies that bind and inhibit signaling receptors on the cancer cell surface, thereby reducing proliferation of the tumor cells. In other cases, monoclonal antibodies that bind specific cell-surface immune regulators can promote immunological attack of cancer cells. Monoclonal antibodies that bind and inhibit specific immune-stimulatory molecules provide useful therapies in autoimmune diseases such

as rheumatoid arthritis. Monoclonal antibodies against specific viral proteins can reduce infection. Given their exquisite specificity, it is likely that monoclonal antibodies will continue to be developed as effective therapies for these and other diseases.

Summary

Tissues can be dissociated into their component cells, from which individual cell types can be purified and used for biochemical analysis or for the establishment of cell cultures. Many animal and plant cells survive and proliferate in a culture dish if they are provided with a suitable culture medium containing nutrients and appropriate signal molecules. Although many animal cells stop dividing after a finite number of cell divisions, cells that have been immortalized through spontaneous mutations or genetic manipulation can be maintained indefinitely as cell lines. Hybridoma cells are widely employed to produce unlimited quantities of uniform monoclonal antibodies, which are used to detect and purify cell proteins, as well as to diagnose and treat diseases.

PURIFYING PROTEINS

The challenge of isolating a single type of protein from the thousands of other proteins in a cell is formidable but must be overcome to study protein function *in vitro*. As we shall see later in this chapter, *recombinant DNA technology* can enormously simplify this task by engineering cells to produce large quantities of a given protein, thereby making its purification much easier. Whether the source of the protein is an engineered cell or a natural tissue, a purification procedure usually starts with subcellular fractionation to reduce the complexity of the material and is then followed by purification steps of increasing specificity.

Cells Can Be Separated into Their Component Fractions

To purify a protein, it must first be extracted from inside the cell. Cells can be broken up in various ways: they can be subjected to osmotic shock or ultrasonic vibration, forced through a small orifice, or ground up in a blender. These procedures break many of the membranes of the cell (including the plasma membrane and endoplasmic reticulum) into fragments that immediately reseal to form small closed vesicles. If carefully carried out, however, the disruption procedures leave organelles such as nuclei, mitochondria, the Golgi apparatus, lysosomes, and peroxisomes largely intact. The suspension of cells is thereby reduced to a thick slurry (called a *homogenate* or *extract*) that contains a variety of membrane-enclosed organelles, each with a distinctive size and density. Provided that the homogenization medium has been carefully chosen (by trial and error for each organelle), the various components—including the vesicles derived from the endoplasmic reticulum, called *microsomes*—retain most of their original biochemical properties.

The different components of the homogenate must then be separated. Such cell fractionations became possible only after the commercial development in the early 1940s of an instrument known as the *preparative ultracentrifuge*, which rotates extracts of broken cells at high speeds (**Figure 8–4**). This treatment separates cell components by size and density: in general, the largest objects experience the largest centrifugal force and move the most rapidly. At relatively low speed, large components such as nuclei sediment to form a pellet at the bottom of the centrifuge tube; at slightly higher speed, a pellet of mitochondria is deposited; and at even higher speeds and with longer periods of centrifugation, first vesicles and then ribosomes can be collected (**Figure 8–5**). All of these fractions are impure, but many of the contaminants can be removed by resuspending the pellet and repeating the centrifugation procedure several times.

Centrifugation is the first step in most fractionations, but it separates only components that differ greatly in size. A finer degree of separation can be achieved by

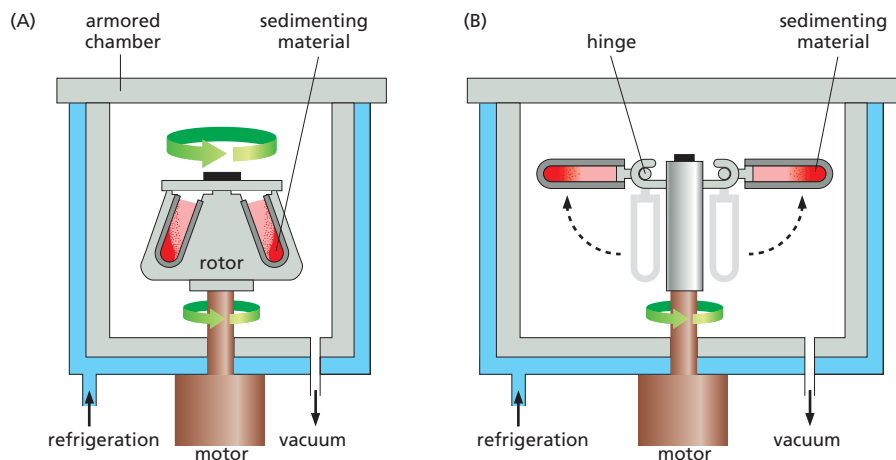


Figure 8-4 The preparative ultracentrifuge. (A) The sample is contained in tubes that are inserted into a ring of angled cylindrical holes in a metal *rotor*. Rapid rotation of the rotor generates enormous centrifugal forces, which cause particles in the sample to sediment against the bottom sides of the sample tubes, as shown here. The vacuum reduces friction, preventing heating of the rotor and allowing the refrigeration system to maintain the sample at 4°C. (B) Some fractionation methods require a different type of rotor called a *swinging-bucket rotor*. In this case, the sample tubes are placed in metal tubes on hinges that allow the tubes to swing outward when the rotor spins. Sample tubes are therefore horizontal during spinning, and samples are sedimented toward the bottom, not the sides, of the tube, providing better separation of differently sized components (see Figures 8-5 and 8-6).

layering the homogenate in a thin band on top of a salt solution that fills a centrifuge tube. When centrifuged, the various components in the mixture move as a series of distinct bands through the solution, each at a different rate, in a process called *velocity sedimentation* (Figure 8-6A). For the procedure to work effectively, the bands must be protected from convective mixing, which would normally occur whenever a denser solution (for example, one containing organelles) finds itself on top of a lighter one (the salt solution). This is achieved by augmenting the solution in the tube with a shallow gradient of sucrose prepared by a special mixing device. The resulting density gradient—with the dense end at the bottom of the tube—keeps each region of the solution denser than any solution above it, and it thereby prevents convective mixing from distorting the separation.

When sedimented through sucrose gradients, different cell components separate into distinct bands that can be collected individually. The relative rate at which each component sediments depends primarily on its size and shape—normally being described in terms of its sedimentation coefficient, or *S* value. Present-day ultracentrifuges rotate at speeds of up to 80,000 rpm and produce forces as high as 500,000 times gravity. These enormous forces drive even small macromolecules, such as tRNA molecules and simple enzymes, to sediment at an appreciable rate and allow them to be separated from one another by size.

The ultracentrifuge is also used to separate cell components on the basis of their buoyant density, independently of their size and shape. In this case, the sample is sedimented through a steep density gradient that contains a very high concentration of sucrose or cesium chloride. Each cell component begins to move down the gradient as in Figure 8-6A, but it eventually reaches a position where the density of the solution is equal to its own density. At this point, the component

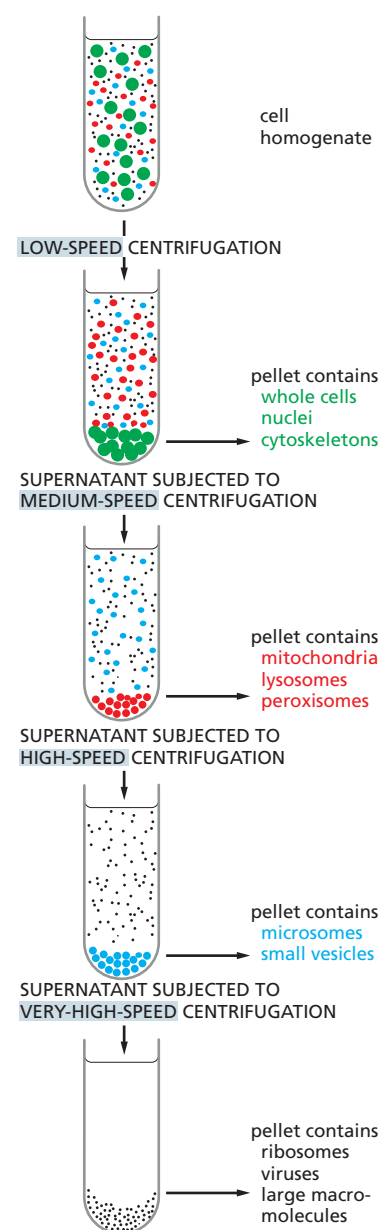


Figure 8-5 Cell fractionation by centrifugation. Repeated centrifugation at progressively higher speeds will fractionate homogenates of cells into their components. In general, the smaller the subcellular component, the greater the centrifugal force required to sediment it. Typical values for the various centrifugation steps referred to in the figure are:

low speed: 1000 times gravity for 10 minutes
medium speed: 20,000 times gravity for 20 minutes
high speed: 80,000 times gravity for 1 hour
very high speed: 150,000 times gravity for 3 hours

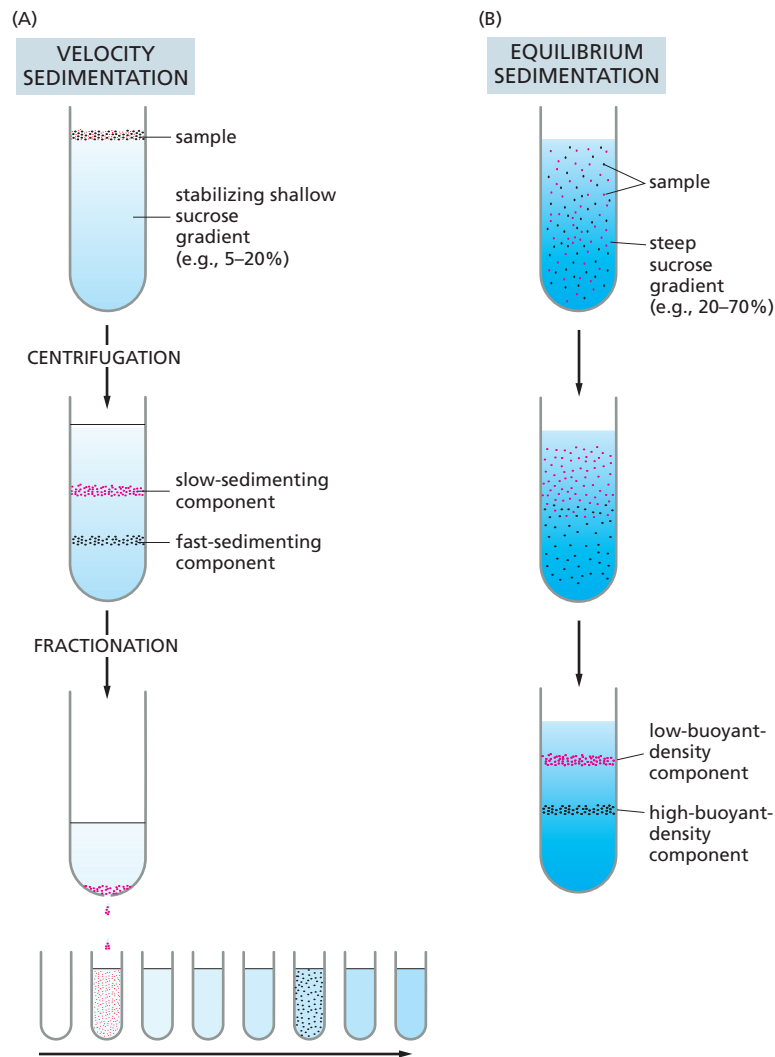


Figure 8–6 Comparison of velocity sedimentation and equilibrium sedimentation. (A) In velocity sedimentation, subcellular components sediment at different speeds according to their size and shape when layered over a solution containing sucrose. To stabilize the sedimenting bands against convective mixing caused by small differences in temperature or solute concentration, the tube contains a continuous shallow gradient of sucrose, which increases in concentration toward the bottom of the tube (typically from 5 to 20% sucrose). After centrifugation, the different components can be collected individually, most simply by puncturing the plastic centrifuge tube with a needle and collecting drops from the bottom, as illustrated here. (B) In equilibrium sedimentation, subcellular components move up or down when centrifuged in a gradient until they reach a position where their density matches that of their surroundings. Although a sucrose gradient is shown here, denser gradients, which are especially useful for protein and nucleic acid separation, can be formed from cesium chloride. The final bands, at equilibrium, can be collected as in A.

floats and can move no farther. A series of distinct bands is thereby produced in the centrifuge tube, with the bands closest to the bottom of the tube containing the components of highest buoyant density (**Figure 8–6B**). This method, called *equilibrium sedimentation*, is so sensitive that it can separate macromolecules that have incorporated heavy isotopes, such as ^{13}C or ^{15}N , from the same macromolecules that contain the lighter, common isotopes (^{12}C or ^{14}N). In fact, the cesium-chloride method was developed in 1957 to separate the labeled from the unlabeled DNA produced after exposure of a growing population of bacteria to nucleotide precursors containing ^{15}N ; this classic experiment provided direct evidence for the semiconservative replication of DNA (see **Figure 5–5**).

Cell Extracts Provide Accessible Systems to Study Cell Functions

Studies of organelles and other large subcellular components isolated in the ultracentrifuge have contributed enormously to our understanding of the functions of different cell components. Experiments on mitochondria and chloroplasts purified by centrifugation, for example, demonstrated the central function of these organelles in converting energy into forms that the cell can use. Similarly, resealed vesicles formed from fragments of rough and smooth endoplasmic reticulum (microsomes) have been separated from each other and analyzed as functional models of these compartments of the intact cell.

Similarly, highly concentrated cell extracts, especially undiluted cytoplasm from *Xenopus laevis* (African clawed frog) oocytes and fertilized eggs, have played

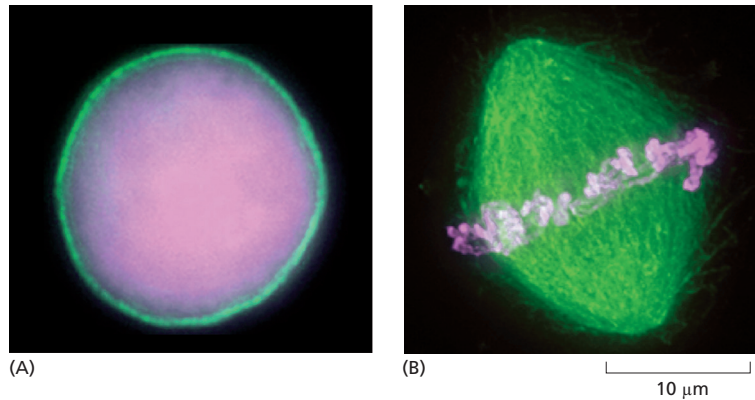


Figure 8-7 The formation of subcellular structures in a cytoplasmic extract. Concentrated cytoplasm was prepared from unfertilized eggs of the frog *Xenopus laevis*. After addition of *Xenopus* sperm chromosomes, these extracts were stimulated to progress through the cell cycle. (A) In an interphase extract, a nuclear envelope (green) forms around the sperm chromosomes (pink). (B) In a metaphase extract, the nuclear envelope breaks down and the microtubules (green) form a mitotic spindle that is attached to the condensed chromosomes (pink). [B, T. Maresca et al., *J Cell Biol.* 169(6):859–69, 2005, doi 10.1083/jcb.200503031. With permission from Rockefeller University Press.]

a critical role in the study of such complex and highly organized processes as the cell-division cycle, the assembly and function of the mitotic spindle (Figure 8-7), and the vesicular-transport steps involved in the movement of proteins through the secretory pathway.

Cell extracts also provide, in principle, the starting material for the complete separation of all of the individual macromolecular components of the cell. We now consider how this separation is achieved, focusing on proteins.

Proteins Can Be Separated by Chromatography

Proteins are most often fractionated by **column chromatography**, in which a mixture of proteins in solution is passed through a column containing a porous gel matrix. Different proteins are retarded to different extents by their interaction with the matrix, and they can be collected separately as they flow out of the bottom of the column (Figure 8-8). Depending on the choice of matrix, proteins can be separated according to their charge (*ion-exchange chromatography*), their hydrophobicity (*hydrophobic chromatography*), their size (*gel-filtration*

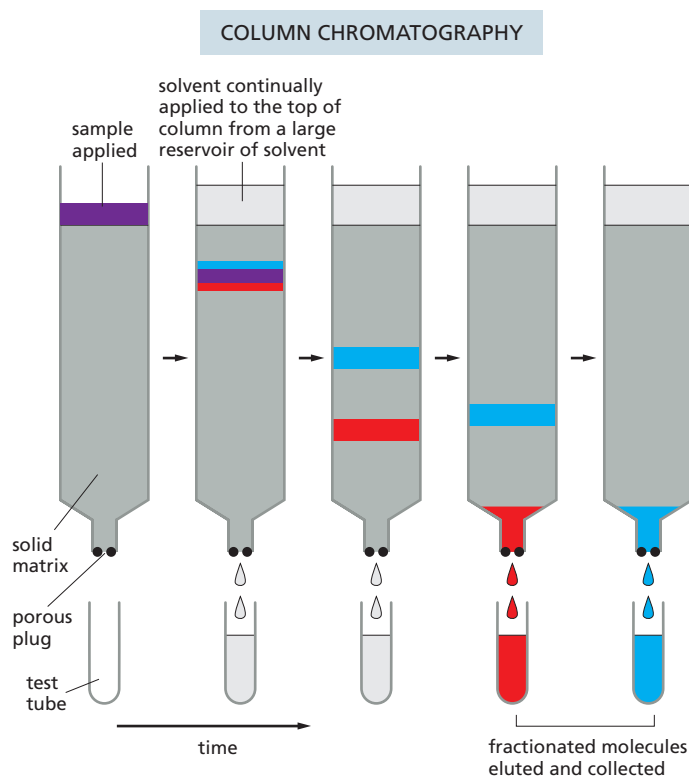


Figure 8-8 The separation of molecules by column chromatography. The sample, a solution containing a mixture of different molecules, is applied to the top of a cylindrical glass or plastic column filled with a permeable gel matrix, such as cellulose. A large amount of solvent is then passed slowly through the column and collected in separate tubes as it emerges from the bottom. Because various components of the sample travel at different rates through the column, they are fractionated into different tubes.

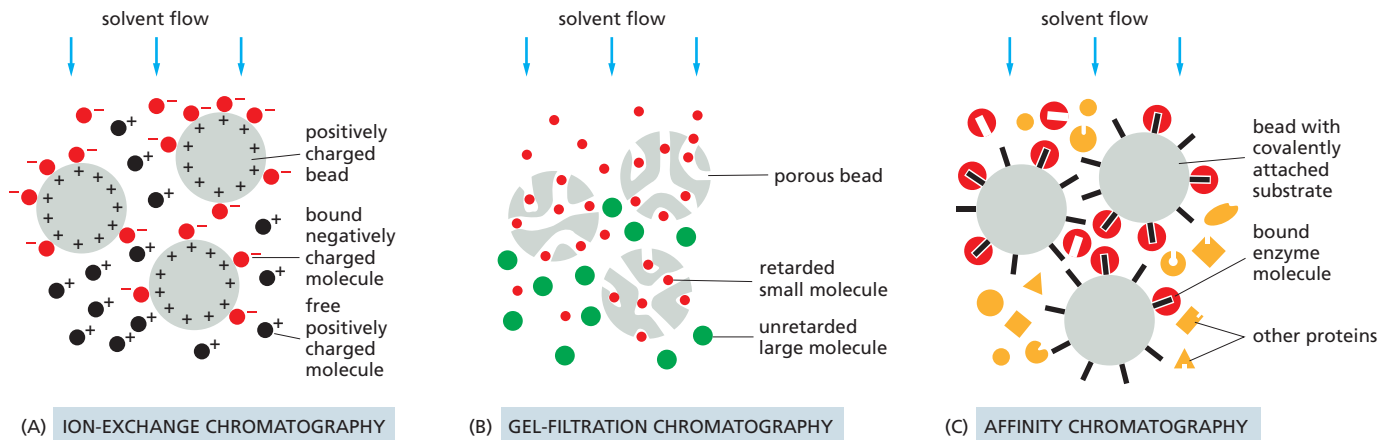


Figure 8-9 Three types of matrices used for chromatography. (A) In ion-exchange chromatography, the insoluble matrix carries ionic charges that retard the movement of molecules of opposite charge. Matrices used for separating proteins include diethylaminoethylcellulose (DEAE-cellulose), which is positively charged, and carboxymethylcellulose (CM-cellulose) and phosphocellulose, which are negatively charged. Analogous matrices based on agarose or other polymers are also frequently used. The strength of the association between the dissolved molecules and the ion-exchange matrix depends on both the ionic strength and the pH of the solution that is passing down the column, which may therefore be varied systematically (as in Figure 8-10) to achieve an effective separation. (B) In gel-filtration chromatography, the small beads that form the matrix are inert but porous. Molecules that are small enough to penetrate into the matrix beads are thereby delayed and travel more slowly through the column than do larger molecules that cannot penetrate. Beads of cross-linked polysaccharide (dextran, agarose, or acrylamide) are available commercially in a wide range of pore sizes, making them suitable for the fractionation of molecules of various masses, from less than 500 daltons to more than 5×10^6 daltons. (C) Affinity chromatography uses an insoluble matrix that is covalently linked to a specific ligand, such as an antibody molecule or an enzyme substrate, that will bind a specific protein. Enzyme molecules that bind to immobilized substrates on such columns can be eluted with a concentrated solution of the free form of the substrate molecule, while molecules that bind to immobilized antibodies can be eluted by dissociating the antibody-antigen complex with concentrated salt solutions or solutions of high or low pH. High degrees of purification can be achieved in a single pass through an affinity column.

chromatography), or their ability to bind to particular small molecules or to other macromolecules (*affinity chromatography*).

Many types of matrices are available. Ion-exchange columns (**Figure 8-9A**) are packed with small beads that carry either a positive or a negative charge, so that proteins are fractionated according to the arrangement of charges on their surface. Hydrophobic columns are packed with beads from which hydrophobic side chains protrude, selectively retarding proteins with exposed hydrophobic regions. Gel-filtration columns (**Figure 8-9B**), which separate proteins according to their size, are packed with tiny porous beads: molecules that are small enough to enter the pores linger inside successive beads as they pass down the column, while larger molecules remain in the solution flowing between the beads and therefore move more rapidly, emerging from the column first. Besides providing a means of separating molecules, gel-filtration chromatography is a convenient way to estimate their size.

Affinity chromatography (**Figure 8-9C**) takes advantage of the biologically important binding interactions that occur on protein surfaces. If a substrate molecule is covalently coupled to an inert matrix such as a polysaccharide bead, the enzyme that operates on that substrate will often be specifically retained by the matrix and can then be eluted (washed out) in nearly pure form. Likewise, short DNA oligonucleotides of a specifically designed sequence can be immobilized in this way and used to purify DNA-binding proteins that normally recognize this sequence of nucleotides in chromosomes. Alternatively, specific antibodies can be coupled to a matrix to purify protein molecules recognized by the antibodies (called *immunoaffinity chromatography*). Because of the great specificity of all such affinity columns, 1000- to 10,000-fold purifications can sometimes be achieved in a single pass.

If one starts with a complex mixture of proteins, a single passage through an ion-exchange or a gel-filtration column does not produce very highly purified

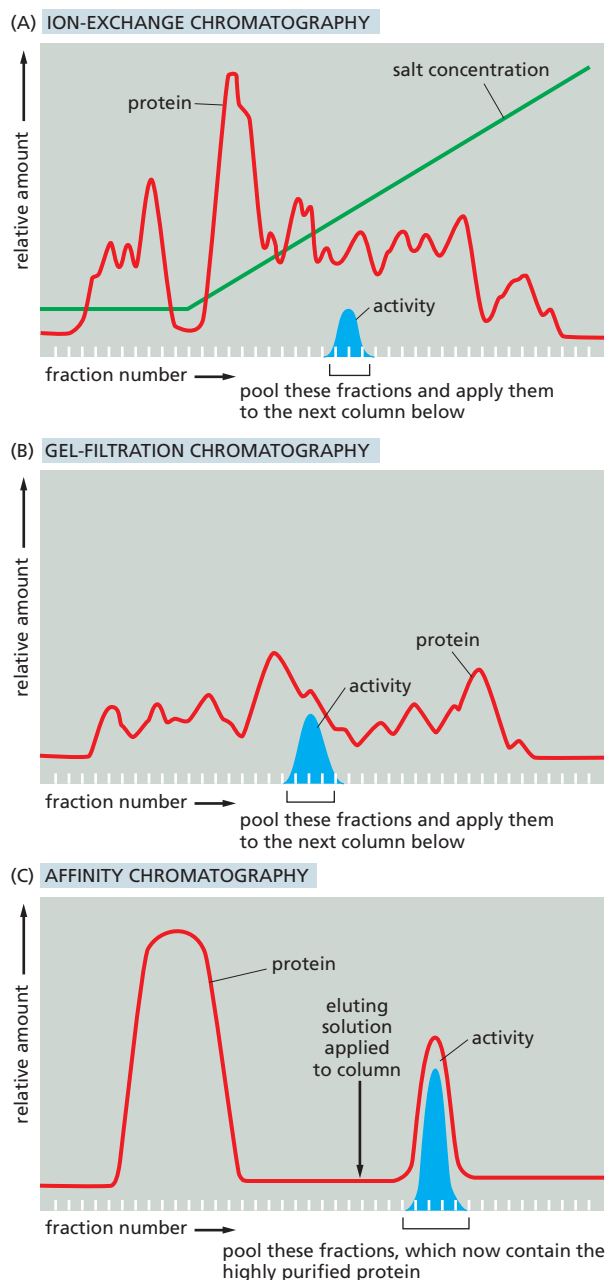


Figure 8-10 Protein purification by chromatography. Typical results obtained when three different chromatographic steps are used in succession to purify a protein. In this example, a homogenate of cells was first fractionated by allowing it to percolate through an ion-exchange resin packed into a column (A). The column was washed to remove all unbound contaminants, and the bound proteins were then eluted by pouring a solution containing a gradually increasing concentration of salt onto the top of the column. Proteins with the lowest affinity for the ion-exchange resin passed directly through the column and were collected in the earliest fractions eluted from the bottom of the column. The remaining proteins were eluted in sequence according to their affinity for the resin—those proteins binding most tightly to the resin requiring the highest concentration of salt to remove them. The protein of interest was eluted in several fractions and was detected by its enzymatic activity. The fractions with activity were pooled and then applied to a gel-filtration column (B). The elution position of the still-impure protein was again determined by its enzymatic activity, and the active fractions were pooled and purified to homogeneity on an affinity column (C) that contained an immobilized substrate of the enzyme.

fractions, because these methods individually increase the proportion of a given protein in the mixture no more than twentyfold. Because most individual proteins represent less than 1/1000 of the total cell protein, it is usually necessary to use several different types of columns in succession to attain sufficient purity, with affinity chromatography being the most efficient (Figure 8-10).

Imperfections in the matrices (such as cellulose), which cause an uneven flow of solvent through the column, limit the resolution of conventional column chromatography. Special chromatography resins (usually silica-based) composed of tiny spheres (3–10 μm in diameter) can be packed with a special apparatus to form a uniform column bed. Such **high-performance liquid chromatography (HPLC)** columns attain a high degree of resolution. In HPLC, the solutes are passed through the column at high pressure and equilibrate very rapidly with the interior of the tiny spheres, and so solutes with different affinities for the matrix are efficiently separated from one another even at very fast flow rates. HPLC is therefore the method of choice for separating many proteins and small molecules.

Immunoprecipitation Is a Rapid Affinity Purification Method

Immunoprecipitation is a useful variation on the theme of affinity chromatography. Specific antibodies that recognize the protein to be purified are attached to small agarose beads. Rather than being packed into a column, as in affinity chromatography, a small quantity of the antibody-coated beads is simply added to a protein extract in a test tube and mixed in suspension for a short period of time—thereby allowing the antibodies to bind the desired protein. The beads are then collected by low-speed centrifugation, and the unbound proteins in the supernatant are discarded. This method is commonly used to purify small amounts of enzymes from cell extracts for analysis of enzymatic activity or for identification of associated proteins. As we describe later in this chapter, immunoprecipitation also provides a method to identify DNA or RNA sequences recognized by specific proteins.

Genetically Engineered Tags Provide an Easy Way to Purify Proteins

Using the recombinant DNA methods discussed later in this chapter, any gene can be modified to produce its protein with an extra amino acid sequence that provides a specific recognition tag, so as to make subsequent purification of the protein simple and rapid. Often the recognition tag is an antigenic determinant, or *epitope*, which can be recognized by a highly specific monoclonal antibody. The antibody can then be used to purify the protein by affinity chromatography or immunoprecipitation (Figure 8–11). Other types of tags are specifically designed for protein purification. For example, a repeated sequence of the amino acid histidine binds to certain metal ions, including nickel and copper. If genetic-engineering techniques are used to attach a short string of histidines to one end of a protein, the slightly modified protein can be retained selectively on an affinity column containing immobilized nickel ions. Metal affinity chromatography can thereby be used to purify the modified protein from a complex molecular mixture.

In other cases, an entire protein is used as the recognition tag. When cells are engineered to synthesize the small enzyme glutathione *S*-transferase (GST) attached to a protein of interest, the resulting **fusion protein** can be purified from the other contents of the cell with an affinity column containing glutathione, a substrate molecule that binds specifically and tightly to GST.

As a further refinement of purification methods using recognition tags, an amino acid sequence that forms a cleavage site for a highly specific proteolytic enzyme can be engineered between the protein of choice and the recognition tag. Because the amino acid sequences at the cleavage site are very rarely found by chance in proteins, the tag can later be cleaved off without destroying the purified protein.

This type of specific cleavage is used in an especially powerful purification methodology known as *tandem affinity purification tagging* (*TAP-tagging*). Here, one end of a protein is engineered to contain two recognition tags that are separated by a protease cleavage site. The tag on the very end of the construct is chosen to bind irreversibly to an affinity column, allowing the column to be washed extensively to remove all contaminating proteins. Protease cleavage then releases the protein, which is then further purified using the second tag. Because this two-step strategy provides an especially high degree of protein purification with relatively little effort, it is used extensively in cell biology.

Purified Cell-free Systems Are Required for the Precise Dissection of Molecular Functions

Purified cell-free systems provide a means of studying biological processes free from all of the complex side reactions that occur in a living cell. To make this possible, cell homogenates are fractionated with the aim of purifying each of the

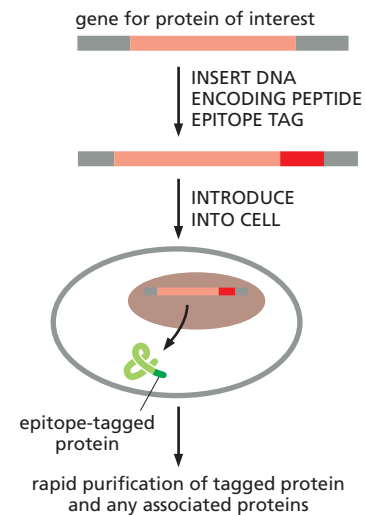


Figure 8–11 Epitope tagging for the purification of proteins. Using standard genetic-engineering techniques, a short amino acid sequence can be added to a protein of interest. If the tag is an antigenic determinant, or *epitope*, it can be targeted by an appropriate antibody, which can be used to purify the protein by immunoprecipitation or affinity chromatography.

individual macromolecules that are needed to catalyze a biological process of interest. For example, the experiments to decipher the mechanisms of protein synthesis began with a cell homogenate that could translate RNA molecules to produce proteins. Fractionation of this homogenate, step by step, produced in turn the ribosomes, tRNAs, and various enzymes that together constitute the protein-synthetic machinery. Once individual pure components were available, each could be added or withheld separately to define its exact role in the overall process.

A major goal for cell biologists is the reconstitution of every biological process in a purified cell-free system. Only in this way can we define all of the components needed for the process and control their concentrations, which is required to work out their precise mechanism of action. Although much remains to be done, a great deal of what we know today about the molecular biology of the cell has been discovered by studies in such cell-free systems. They have been used, for example, to decipher the molecular details of DNA replication and DNA transcription, RNA splicing, protein translation, muscle contraction, particle transport along microtubules, and many other processes that occur in cells.

Summary

Populations of cells can be analyzed biochemically by disrupting them and fractionating their contents, allowing functional cell-free systems to be developed. Highly purified cell-free systems are needed for determining the molecular details of complex cell processes, and the development of such systems requires extensive purification of all the proteins and other components involved. The proteins in soluble cell extracts can be purified by column chromatography; depending on the type of column matrix, biologically active proteins can be separated on the basis of their molecular weight, hydrophobicity, charge characteristics, or affinity for other molecules. In a typical purification, the sample is passed through several different columns in turn, with the enriched fractions obtained from one column being applied to the next. Recombinant DNA techniques (described later) allow special recognition tags to be attached to proteins, thereby greatly simplifying their purification.

ANALYZING PROTEINS

Proteins perform most cellular processes: they catalyze metabolic reactions, use nucleotide hydrolysis to do mechanical work, and serve as the major structural elements of the cell. The great variety of protein structures and functions has stimulated the development of a multitude of techniques to study them.

Proteins Can Be Separated by SDS Polyacrylamide-Gel Electrophoresis

Proteins usually possess a net positive or negative charge, depending on the mixture of charged amino acids they contain. An electric field applied to a solution containing a protein molecule causes the protein to migrate at a rate that depends on its net charge and on its size and shape. The most useful application of this property is **sodium dodecyl sulfate polyacrylamide-gel electrophoresis (SDS-PAGE)**. It uses a highly cross-linked gel of polyacrylamide as the inert matrix through which the proteins migrate. The gel is prepared by polymerization of monomers; the pore size of the gel can be adjusted so that it is small enough to retard the migration of the protein molecules of interest. The proteins are dissolved in a solution that includes a powerful negatively charged detergent, sodium dodecyl sulfate, or SDS (**Figure 8-12**). Because this detergent binds to hydrophobic regions of the protein molecules, causing them to unfold into extended polypeptide chains, the individual protein molecules are released from their associations with other proteins or lipid molecules and rendered freely soluble in the detergent solution. In addition, a reducing agent such as β -mercaptoethanol

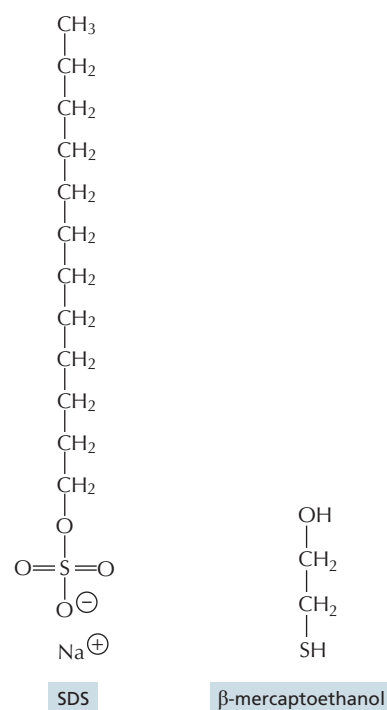


Figure 8-12 The detergent sodium dodecyl sulfate (SDS) and the reducing agent β -mercaptoethanol. These two chemicals are used to solubilize proteins for SDS polyacrylamide-gel electrophoresis. The SDS is shown here in its ionized form.

(see Figure 8-12) is usually added to break any disulfide linkages in the proteins, so that all of the constituent polypeptides in multisubunit proteins can be analyzed separately.

What happens when a mixture of SDS-solubilized proteins is run through a slab of polyacrylamide gel? Each protein molecule binds large numbers of the negatively charged detergent molecules, which mask the protein's intrinsic charge and cause it to migrate toward the positive electrode when a voltage is applied. Proteins of the same size tend to move through the gel with similar speeds because (1) their native structure is completely unfolded by the SDS, so that their shapes are the same, and (2) they bind the same amount of SDS and therefore have the same amount of negative charge. Larger proteins, with more charge, are subjected to larger electrical forces but also to a larger drag. In free solution, the two effects would cancel out, but, in the mesh of the polyacrylamide gel, which acts as a molecular sieve, large proteins are retarded much more than small ones. As a result, a complex mixture of proteins is fractionated into a series of discrete protein bands mostly according to their mass (Figure 8-13). The major proteins are readily detected by staining the proteins in the gel with a dye such as Coomassie blue. When small amounts of protein are present and more sensitive methods are required, gels can be treated with a silver stain, which will detect as little as 10 ng of protein in a band. For some purposes, specific proteins can also be labeled with a radioactive isotope tag; exposure of the gel to film or a radiation detector results in an *autoradiograph* on which the labeled proteins are visible (see Figure 8-16).

SDS-PAGE is widely used because it can separate all types of proteins, including those that are normally insoluble in water—such as the many proteins in

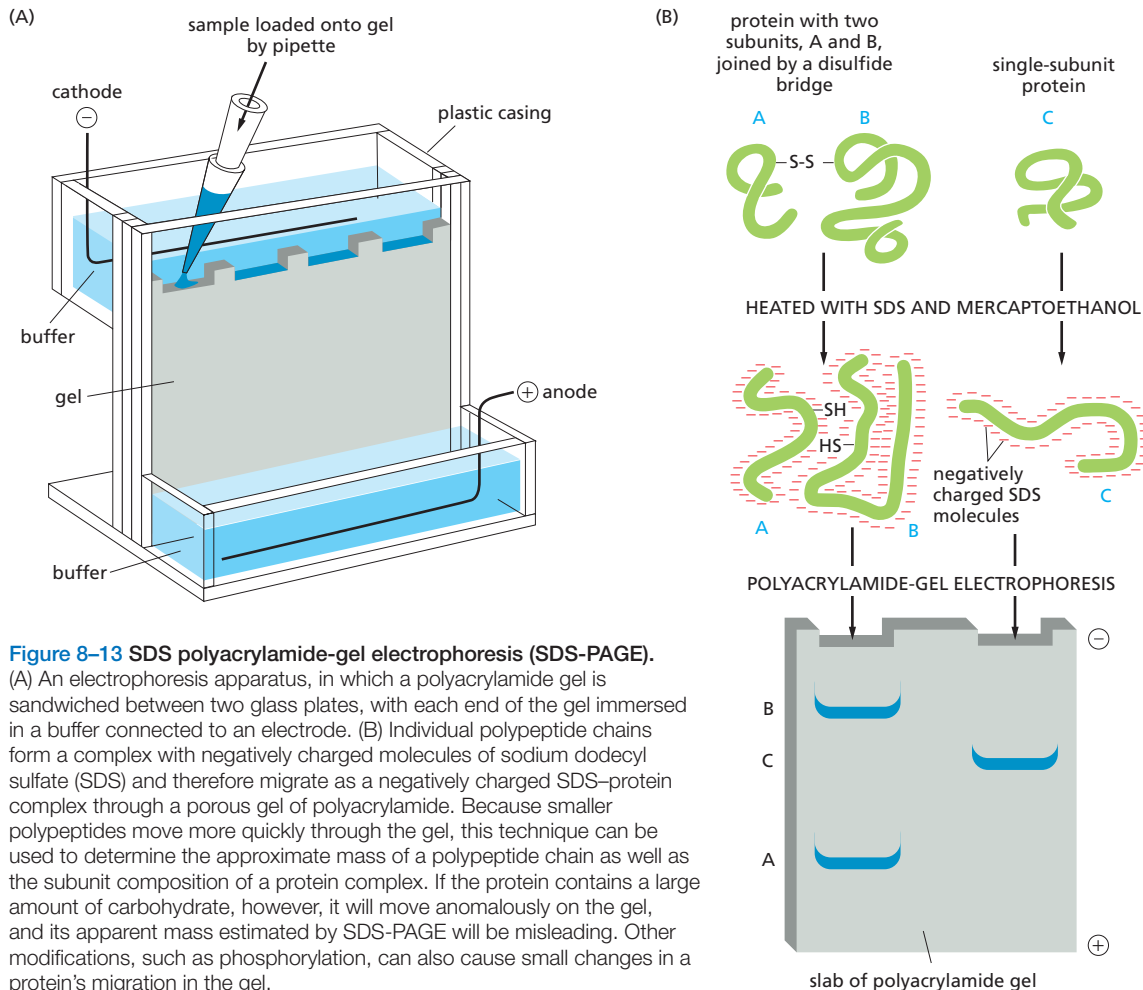
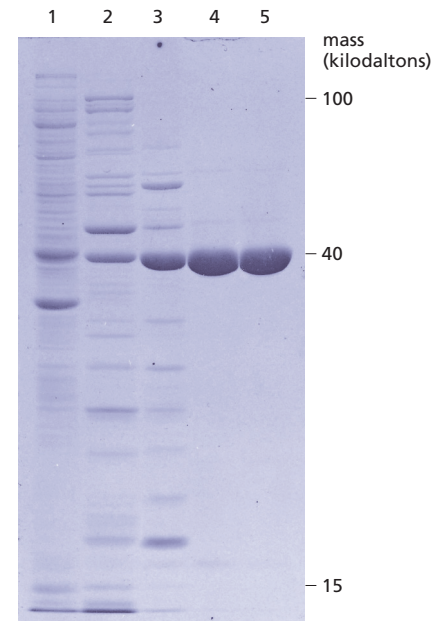


Figure 8-13 SDS polyacrylamide-gel electrophoresis (SDS-PAGE).

(A) An electrophoresis apparatus, in which a polyacrylamide gel is sandwiched between two glass plates, with each end of the gel immersed in a buffer connected to an electrode. (B) Individual polypeptide chains form a complex with negatively charged molecules of sodium dodecyl sulfate (SDS) and therefore migrate as a negatively charged SDS-protein complex through a porous gel of polyacrylamide. Because smaller polypeptides move more quickly through the gel, this technique can be used to determine the approximate mass of a polypeptide chain as well as the subunit composition of a protein complex. If the protein contains a large amount of carbohydrate, however, it will move anomalously on the gel, and its apparent mass estimated by SDS-PAGE will be misleading. Other modifications, such as phosphorylation, can also cause small changes in a protein's migration in the gel.

Figure 8–14 Analysis of protein samples by SDS polyacrylamide-gel electrophoresis. The photograph shows a Coomassie blue–stained gel that has been used to detect the proteins present at successive stages in the purification of an enzyme. The leftmost lane (lane 1) contains the complex mixture of proteins in the starting cell extract, and each succeeding lane analyzes the proteins obtained after a chromatographic fractionation of the protein sample analyzed in the previous lane (see Figure 8–10). The same total amount of protein (10 μ g) was loaded onto the gel at the top of each lane. Individual proteins normally appear as sharp, dye-stained bands; a band broadens, however, when it contains a large amount of protein. (From T. Formosa and B.M. Alberts, *J. Biol. Chem.* 261:6107–6118, 1986.)



membranes. And because the method separates polypeptides by size, it provides information about the mass and subunit composition of proteins. **Figure 8–14** presents a photograph of a gel that has been used to analyze each of the successive stages in the purification of a protein.

Two-dimensional Gel Electrophoresis Provides Greater Protein Separation

Because different proteins can have similar sizes, shapes, masses, and overall charges, most separation techniques such as SDS polyacrylamide-gel electrophoresis or ion-exchange chromatography cannot typically separate all the proteins in a cell or even in an organelle. In contrast, **two-dimensional gel electrophoresis**, which combines two different separation procedures, can resolve up to 2000 proteins in the form of a two-dimensional protein map.

In the first step, the proteins are separated by their intrinsic charges. The sample is dissolved in a small volume of a solution containing a nonionic (uncharged) detergent, together with β -mercaptoethanol and the denaturing reagent urea. This solution solubilizes, denatures, and dissociates all the polypeptide chains but leaves their intrinsic charge unchanged. The polypeptide chains are then separated in a pH gradient by a procedure called *isoelectric focusing*, which takes advantage of the variation in the net charge on a protein molecule with the pH of its surrounding solution. Every protein has a characteristic isoelectric point, the pH at which the protein has no net charge and therefore does not migrate in an electric field. In isoelectric focusing, proteins are separated electrophoretically in a narrow tube of polyacrylamide gel in which a gradient of pH is established by a mixture of special buffers. Each protein moves to a position in the gradient that corresponds to its isoelectric point and remains there (**Figure 8–15**). This is the first dimension of two-dimensional polyacrylamide-gel electrophoresis.

In the second step, the narrow tube of gel containing the separated proteins is soaked in SDS and placed along the top edge of an SDS polyacrylamide-gel slab. Electrophoresis is then carried out as in one-dimensional SDS-PAGE, and each polypeptide chain migrates into the gel to form a discrete spot. The only proteins left unresolved are those that have both identical sizes and identical isoelectric points, a relatively rare situation. Even trace amounts of each polypeptide chain

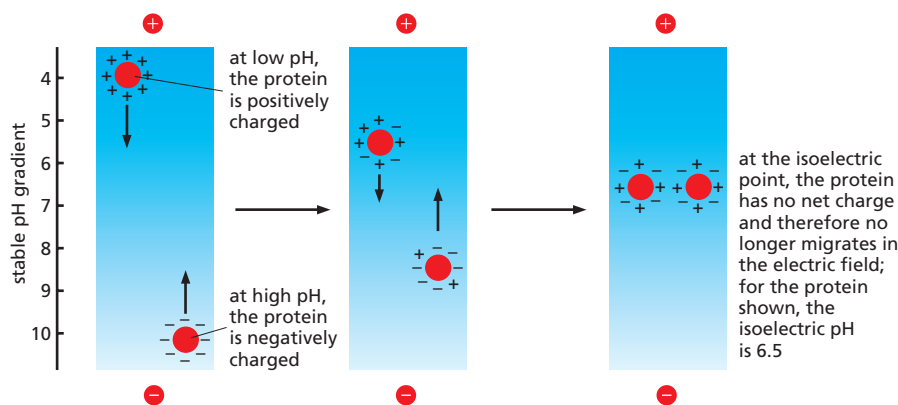


Figure 8–15 Separation of protein molecules by isoelectric focusing.

At low pH (high H^+ concentration), the carboxylic acid groups of proteins tend to be uncharged ($-COOH$) and their nitrogen-containing basic groups fully charged (for example, $-NH_3^+$), giving most proteins a net positive charge. At high pH, the carboxylic acid groups are negatively charged ($-COO^-$) and the basic groups tend to be uncharged (for example, $-NH_2$), giving most proteins a net negative charge. At its *isoelectric point*, a protein has no net charge because the positive and negative charges balance. Thus, when a tube containing a fixed pH gradient is subjected to a strong electric field in the appropriate direction, each protein species migrates until it forms a sharp band at its isoelectric point, as shown.

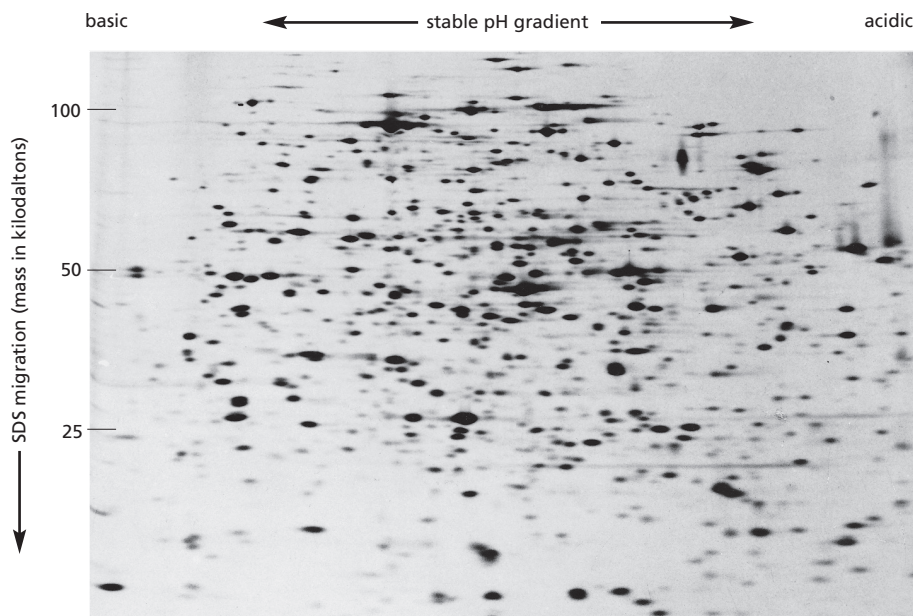


Figure 8-16 Two-dimensional polyacrylamide-gel electrophoresis. All the proteins in an *Escherichia coli* bacterial cell are separated in this gel, in which each spot corresponds to a different polypeptide chain. The proteins were first separated on the basis of their isoelectric points by isoelectric focusing in the horizontal dimension. They were then further fractionated according to their mass by electrophoresis from top to bottom in the presence of SDS. Note that different proteins are present in very different amounts. The bacteria were fed with a mixture of radioisotope-labeled amino acids so that all of their proteins were radioactive and could be detected by autoradiography. (Courtesy of Patrick O’Farrell.)

can be detected on the gel by various staining procedures—or by autoradiography if the protein sample was initially labeled with a radioisotope (Figure 8-16). The technique has such great resolving power that it can distinguish between two proteins that differ in only a single charged amino acid or a single negatively charged phosphorylation site.

Specific Proteins Can Be Detected by Blotting with Antibodies

A specific protein can be identified after its fractionation on a polyacrylamide gel by exposing all the proteins on the gel to a specific antibody that has been labeled with a radioactive isotope or a fluorescent dye. This procedure is normally carried out after transferring all of the separated proteins in the gel onto a sheet of nitrocellulose paper or nylon membrane. Placing the membrane against the gel and driving the proteins out of the gel with a strong electric current transfers the protein onto the membrane. The membrane is then soaked in a solution of labeled antibody to reveal the protein of interest. This method of detecting proteins is called **Western blotting**, or **immunoblotting** (Figure 8-17). Sensitive Western blotting methods can detect very small amounts of a specific protein (1 nanogram or less) in a total cell extract or some other heterogeneous protein mixture. The method can be very useful when assessing the amounts of a specific protein in the cell or when measuring changes in those amounts under various conditions.

Hydrodynamic Measurements Reveal the Size and Shape of a Protein Complex

Most proteins in a cell are subunits of larger complexes, and knowledge of the size and shape of these complexes often leads to insights regarding their function. This information can be obtained in several ways. Sometimes, a complex can be directly visualized using electron microscopy, as described in Chapter 9. A complementary approach relies on the hydrodynamic properties of a complex; that is, its behavior as it moves through a liquid medium. Usually, two separate measurements are made. One measure is the velocity of a complex as it moves under the influence of a centrifugal field produced by an ultracentrifuge (see Figure 8-6A). The sedimentation coefficient (or S value) obtained depends on both the size and the shape of the complex and does not, by itself, convey especially useful information. However, once a second hydrodynamic measurement is performed—by charting the migration of a complex through a gel-filtration chromatography

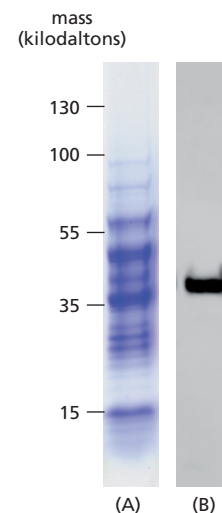


Figure 8-17 Western blotting. In this experiment, proteins of the budding yeast *Saccharomyces cerevisiae* were separated on a polyacrylamide gel. In (A), the gel was stained with Coomassie blue to reveal the most abundant proteins. In (B), the proteins in the gel were transferred to a membrane and exposed to antibodies directed against a specific protein. Unbound antibodies were washed away, and antibodies bound to the protein were detected with a fluorescent label. By this sensitive Western blotting method, small amounts of a single rare protein can be detected in a complex mixture of other proteins. (Courtesy of Jonathan Asfaha.)

column (see Figure 8-9B)—both the approximate shape of a complex and its mass can be calculated.

The mass of a protein complex can also be determined more directly by using an analytical ultracentrifuge, a complex device that allows protein absorbance measurements to be made on a sample while it is subjected to centrifugal forces. In this approach, the sample is centrifuged until it reaches equilibrium, where the centrifugal force on a protein complex exactly balances its tendency to diffuse away. Because this balancing point is dependent on a complex's mass but not on its particular shape, the mass can be directly calculated.

Mass Spectrometry Provides a Highly Sensitive Method for Identifying Unknown Proteins

A frequent problem in cell biology and biochemistry is the identification of a protein or collection of proteins that has been obtained by one of the purification procedures discussed in the preceding pages. Because the genome sequences of most experimental organisms are known, catalogs of all the proteins produced in those organisms are available. The task of identifying an unknown protein (or collection of unknown proteins) thus reduces to matching some of the amino acid sequences present in the unknown sample with known cataloged genes. This task is now performed almost exclusively by mass spectrometry in conjunction with computer searches of databases.

Charged particles have very precise dynamics when subjected to electric and magnetic fields in a vacuum. *Mass spectrometry* exploits this principle to separate ions according to their mass-to-charge (m/z) ratio. It is an enormously sensitive technique. It requires very little material and is capable of determining the precise mass of intact proteins and of peptides derived from them by enzymatic or chemical cleavage. Masses can be obtained with great accuracy, often with an error of less than one part in a million.

Mass spectrometry is performed using complex instruments with three major components (Figure 8-18A). The first is the *ion source*, which transforms tiny amounts of a peptide sample into a gas containing individual charged peptide molecules. These ions are accelerated by an electric field into the second component, the *mass analyzer*, where electric or magnetic fields are used to separate the ions on the basis of their mass-to-charge ratios. Finally, the separated ions collide with a *detector*, which generates a mass spectrum containing a series of peaks representing the masses of the molecules in the sample.

There are many different types of mass spectrometer, varying mainly in the nature of their ion sources and mass analyzers. One of the most common ion sources depends on a technique called *matrix-assisted laser desorption ionization* (MALDI). In this approach, the proteins in the sample are first cleaved into short peptides by a protease such as trypsin. These peptides are mixed with an organic acid and then dried onto a metal or ceramic slide. A brief laser burst is directed toward the sample, producing a gaseous puff of ionized peptides, each carrying one or more positive charges. In many cases, the MALDI ion source is coupled to a mass analyzer called a *time-of-flight* (TOF) analyzer, which is a long chamber through which the ionized peptides are accelerated by an electric field toward a detector. Their mass and charge determine the time it takes them to reach the detector: large peptides move more slowly, and more highly charged molecules move more quickly. By analyzing those ionized peptides that bear a single charge, the precise masses of peptides present in the original sample can be determined. This information is then used to search genomic databases, in which the masses of all proteins and of all their predicted peptide fragments have been tabulated from the genomic sequences of the organism. An unambiguous match to a particular open reading frame can often be made by knowing the mass of only a few peptides derived from a given protein.

By employing two mass analyzers in tandem (an arrangement known as tandem mass spectrometry, or MS/MS; Figure 8-18B), it is possible to directly determine the amino acid sequences of individual peptides in a complex mixture.

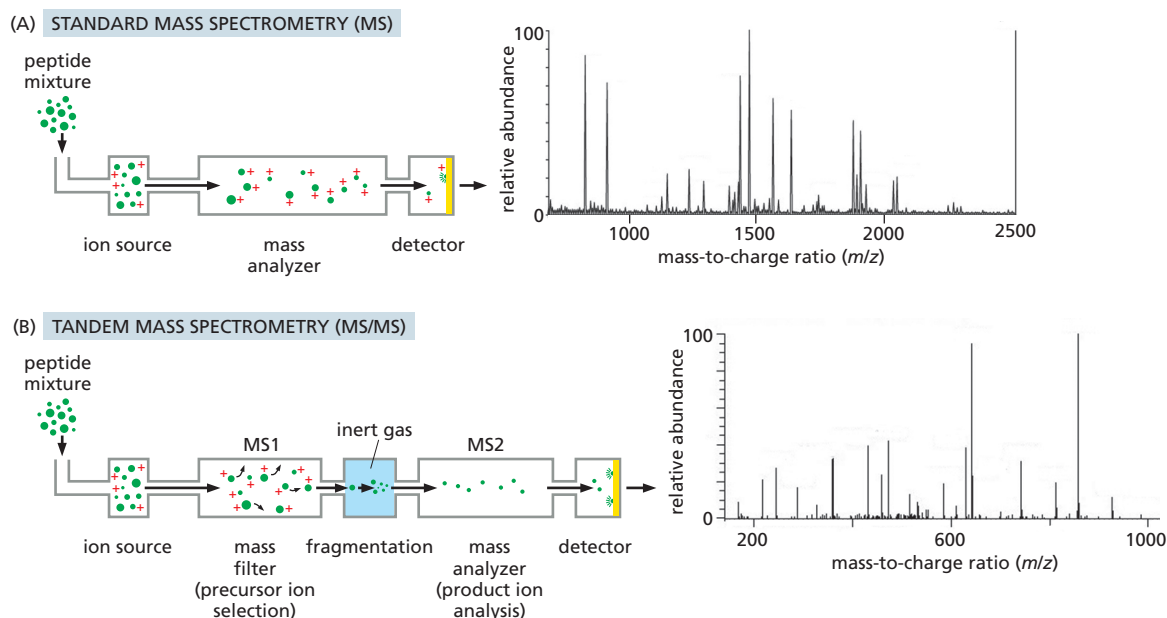


Figure 8-18 The mass spectrometer. (A) Mass spectrometers used in biology contain an ion source that generates gaseous peptides or other molecules under conditions that render most molecules positively charged. The two major types of ion source are MALDI and electrospray, as described in the text. Ions are accelerated into a mass analyzer, which separates the ions on the basis of their mass and charge by one of three major methods: (1) *Time-of-flight (TOF)* analyzers determine the mass-to-charge ratio of each ion in the mixture from the rate at which it travels from the ion source to the detector. (2) *Quadrupole* mass filters contain a long chamber lined by four electrodes that produce oscillating electric fields that govern the trajectory of ions; by varying the properties of the electric field over a wide range, a spectrum of ions with specific mass-to-charge ratios is allowed to pass through the chamber to the detector, while other ions are discarded. (3) *Ion traps* contain doughnut-shaped electrodes producing a three-dimensional electric field that traps all ions in a circular chamber; the properties of the electric field can be varied over a wide range to eject a spectrum of specific ions to a detector. (B) Tandem mass spectrometry typically involves two mass analyzers separated by a collision chamber containing an inert, high-energy gas. The electric field of the first mass analyzer is adjusted to select a specific peptide ion, called a *precursor ion*, which is then directed to the collision chamber. Collision of the peptide with gas molecules causes random peptide fragmentation, primarily at peptide bonds, resulting in a highly complex mixture of fragments containing one or more amino acids from throughout the original peptide. The second mass analyzer is then used to measure the masses of the fragments (called *product* or *daughter ions*). With computer assistance, the pattern of fragments can be used to deduce the amino acid sequence of the original peptide.

The MALDI-TOF instrument described above is not ideal for this method. Instead, MS/MS typically involves an *electrospray* ion source, which produces a continuous thin stream of peptides that are ionized and accelerated into the first mass analyzer. The mass analyzer is typically either a *quadrupole* or *ion trap*, which employs large electrodes to produce oscillating electric fields inside the chamber containing the ions. These instruments act as *mass filters*: the electric field is adjusted over a broad range to select a single peptide ion and discard all the others in the peptide mixture. In tandem mass spectrometry, this single ion is then exposed to an inert, high-energy gas, which collides with the peptide, resulting in fragmentation, primarily at peptide bonds. The second mass analyzer then determines the masses of the peptide fragments, which can be used by computational methods to determine the amino acid sequence of the original peptide and thereby identify the protein from which it came.

Tandem mass spectrometry is also useful for detecting and precisely mapping post-translational modifications of proteins, such as phosphorylation or acetylation. Because these modifications impart a characteristic mass increase to an amino acid, they are easily detected during the analysis of peptide fragments in the second mass analyzer, and the precise site of the modification can often be deduced from the spectrum of peptide fragments.

A powerful, “two-dimensional” mass spectrometry technique can be used to determine all of the proteins present in an organelle or another complex mixture of proteins. First, the mixture of proteins present is digested with trypsin to

produce short peptides. Next, these peptides are separated by an automated high-performance form of liquid chromatography (LC). Every peptide fraction from the chromatographic column is injected directly into an electrospray ion source on a tandem mass spectrometer, providing the amino acid sequence and post-translational modifications for every peptide in the mixture. This method, often called LC-MS/MS, is used to identify hundreds or thousands of proteins in complex protein mixtures from specific organelles or from whole cells. It can also be used to map all of the phosphorylation sites in the cell, or all of the proteins targeted by other post-translational modifications such as acetylation or ubiquitylation.

Sets of Interacting Proteins Can Be Identified by Biochemical Methods

Because most proteins in the cell function as part of complexes with other proteins, an important way to begin to characterize the biological role of an unknown protein is to identify all of the other proteins to which it specifically binds.

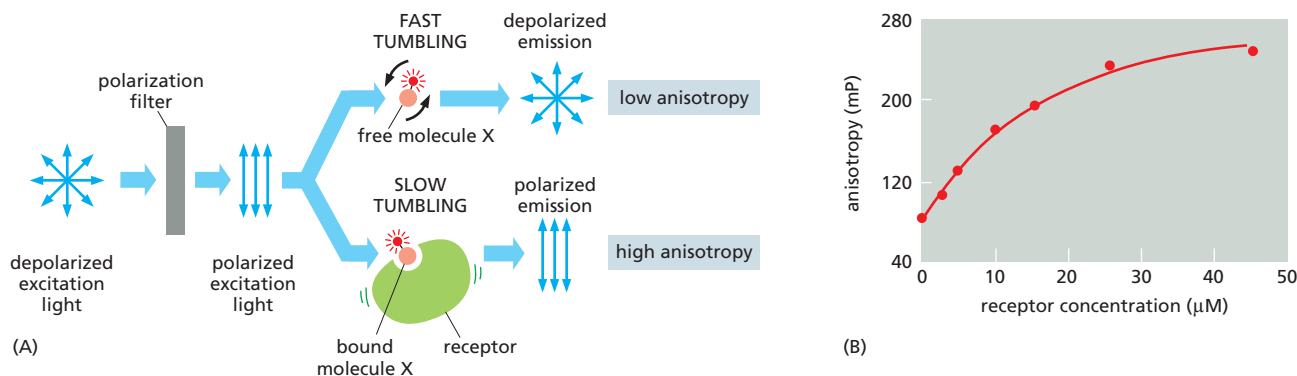
A key method for identifying proteins that bind to one another tightly is *co-immunoprecipitation*. A target protein is immunoprecipitated from a cell lysate using specific antibodies coupled to beads, as described earlier. If the target protein is associated tightly enough with another protein when it is captured by the antibody, the partner precipitates as well and can be identified by mass spectrometry. This method is useful for identifying proteins that are part of a complex inside cells, including those that interact only transiently; for example, when extracellular signal molecules stimulate cells (discussed in Chapter 15).

Optical Methods Can Monitor Protein Interactions

Once two proteins—or a protein and a small molecule—are known to associate, it becomes important to characterize their interaction in more detail. Proteins can associate with each other more or less permanently (like the subunits of RNA polymerase or the proteasome), or engage in transient encounters that may last only a few milliseconds (like a protein kinase and its substrate). To understand how a protein functions inside a cell, we need to determine how tightly it binds to other proteins and how covalent modifications, small molecules, or other proteins influence these interactions.

As we discussed in Chapter 3 (see Figure 3-42), the extent to which two proteins interact is determined by the rates at which they associate and dissociate. These rates depend, respectively, on the association rate constant (k_{on}) and dissociation rate constant (k_{off}). The kinetic rate constant k_{off} is a particularly useful number because it provides valuable information about how long two proteins remain bound to one another. The ratio of the two kinetic constants ($k_{\text{on}}/k_{\text{off}}$) yields another very useful number called the equilibrium constant (K , also known as K_{eq} or K_{a}), the inverse of which is the more commonly used dissociation constant, K_{d} . The equilibrium constant is useful as a general indicator of the affinity of the interaction, and it can be used to estimate the amount of bound complex at different concentrations of the two protein partners—thereby providing insights into the importance of the interaction at the protein concentrations found inside the cell.

A wide range of methods can be used to determine binding constants for a two-protein complex. In a simple *equilibrium binding experiment*, two proteins are mixed at a range of concentrations, allowed to reach equilibrium, and the amount of bound complex is measured; half of the protein complex will be bound at a concentration that is equal to K_{d} . Equilibrium experiments often involve the use of radioactive or fluorescent tags on one of the protein partners, coupled with biochemical or optical methods for measuring the amount of bound protein. In a more complex *kinetic binding experiment*, the kinetic rate constants are determined using rapid methods that allow real-time measurement of the formation of a bound complex over time (to determine k_{on}) or the dissociation of a bound complex over time (to determine k_{off}).



Optical techniques provide particularly rapid, convenient, and accurate binding measurements, and in some cases the proteins do not even need to be labeled. Certain amino acids (tryptophan, for example) exhibit weak fluorescence that can be detected with sensitive fluorimeters. In many cases, the fluorescence intensity, or the emission spectrum of fluorescent amino acids located in a protein-protein interface, will change when two proteins associate. When this change can be detected by fluorimetry, it provides a simple and sensitive measure of protein binding that is useful in both equilibrium and kinetic binding experiments. A related but more widely useful optical binding technique is based on *fluorescence anisotropy*, a change in the polarized light that is emitted by a fluorescently tagged protein in the bound and free states (Figure 8–19).

Another optical method for probing protein interactions uses *green fluorescent protein* (discussed in detail later) and its derivatives of different colors. In this application, two proteins of interest are each labeled with a different fluorescent protein, such that the emission spectrum of one fluorescent protein overlaps the absorption spectrum of the second. If the two proteins come very close to each other (within about 1–5 nm), the energy of the absorbed light is transferred from one fluorescent protein to the other. The energy transfer, called *fluorescence resonance energy transfer (FRET)*, is determined by illuminating the first fluorescent protein and measuring emission from the second (see Figure 9–19). When combined with fluorescence microscopy, this method can be used to characterize protein-protein interactions at specific locations inside living cells (discussed in Chapter 9).

Protein Structure Can Be Determined Using X-ray Diffraction

A deep understanding of protein function in the cell requires knowledge of its three-dimensional structure at atomic resolution, revealing the precise position of every amino acid in the protein. Structural analysis provides powerful insights into the mechanisms underlying a protein's enzymatic activity and its interactions with other proteins or small molecules. Numerous methods are used to unravel protein structure. In this chapter, we discuss well-established methods that depend on x-ray crystallography and nuclear magnetic resonance spectroscopy. Chapter 9 describes recently developed methods by which electron microscopy is being used to determine the high-resolution structures of larger proteins and protein complexes.

For many decades, the primary technique for protein structural analysis has been **x-ray crystallography**. X-rays, like light, are a form of electromagnetic radiation, but they have a much shorter wavelength, typically around 0.1 nm (the diameter of a hydrogen atom). If a narrow beam of parallel x-rays is directed at a sample of a pure protein, most of the x-rays pass straight through it. A small fraction, however, is scattered by the atoms in the sample. If the sample is a well-ordered crystal, the scattered waves reinforce one another at certain points and appear as diffraction spots when recorded by a suitable detector (Figure 8–20).

The slowest step in this technique is likely to be the generation of suitable protein crystals. This step requires large amounts of very pure protein and often involves

Figure 8–19 Measurement of binding with fluorescence anisotropy. This method depends on a fluorescently tagged protein that is illuminated with polarized light at the appropriate wavelength for excitation; a fluorimeter is used to measure the intensity and polarization of the emitted light.

If the fluorescent protein is fixed in position and therefore does not rotate during the brief period between excitation and emission, then the emitted light will be polarized at the same angle as the excitation light. This directional effect is called *fluorescence anisotropy*. Protein molecules in solution rotate or tumble rapidly, however, so that there is a decrease in the amount of anisotropic fluorescence. Larger molecules tumble at a slower rate and therefore have higher fluorescence anisotropy. (A) To measure the binding between a small molecule and a large receptor protein, the smaller molecule is labeled with a fluorophore. In the absence of its binding partner, the molecule tumbles rapidly, resulting in low fluorescence anisotropy (*top*). When the small molecule binds to its larger partner, however, it tumbles less rapidly, resulting in an increase in fluorescence anisotropy (*bottom*). (B) In the equilibrium binding experiment shown here, a small, fluorescent peptide ligand was present at a low concentration, and the amount of fluorescence anisotropy (in millipolarization units; mP) was measured after incubation with various concentrations of a larger protein receptor for the ligand. From the hyperbolic curve that fits the data, it can be seen that 50% binding occurred at about 10 μM , which is equal to the dissociation constant K_d for the binding interaction.

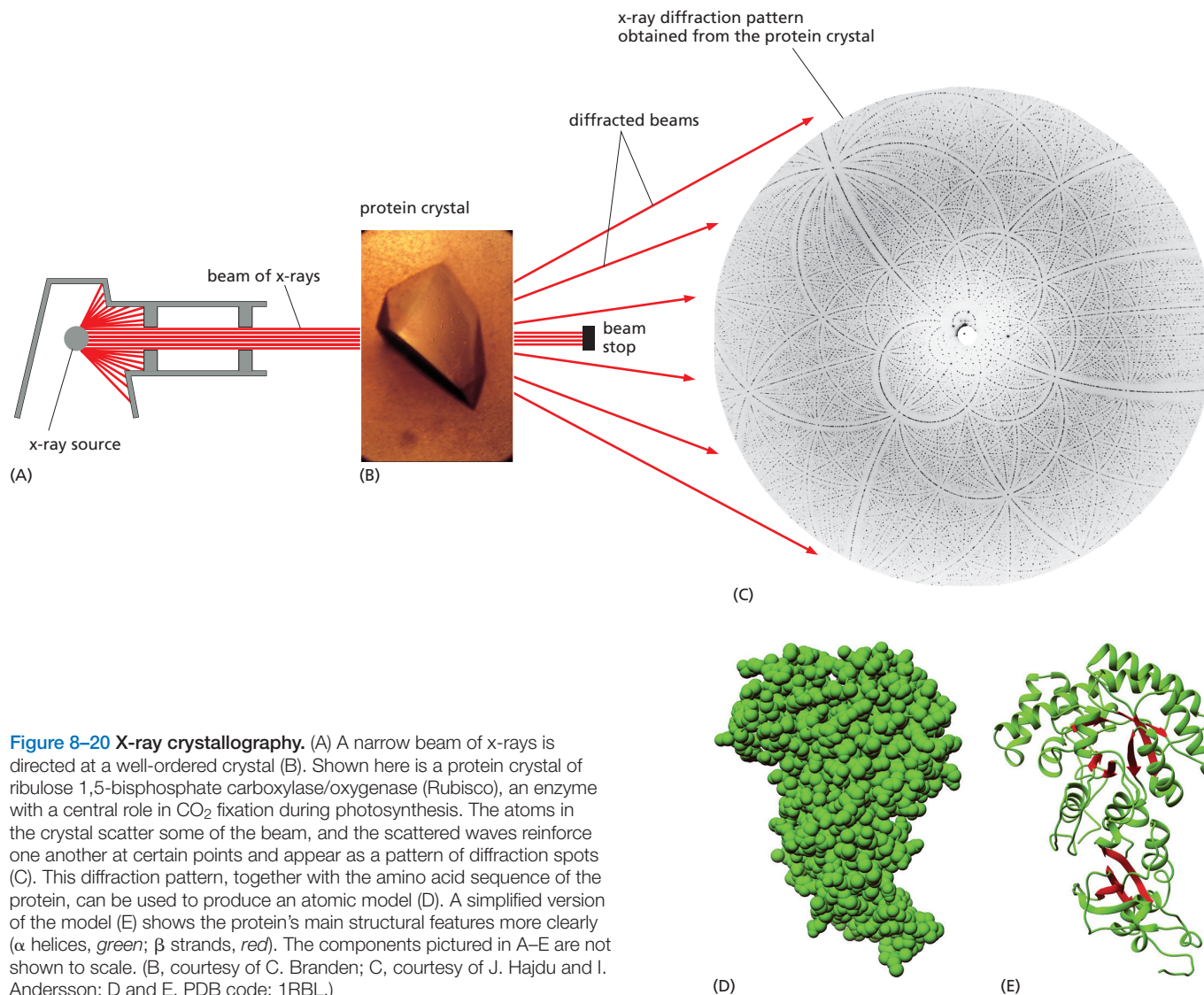


Figure 8-20 X-ray crystallography. (A) A narrow beam of x-rays is directed at a well-ordered crystal (B). Shown here is a protein crystal of ribulose 1,5-bisphosphate carboxylase/oxygenase (Rubisco), an enzyme with a central role in CO_2 fixation during photosynthesis. The atoms in the crystal scatter some of the beam, and the scattered waves reinforce one another at certain points and appear as a pattern of diffraction spots (C). This diffraction pattern, together with the amino acid sequence of the protein, can be used to produce an atomic model (D). A simplified version of the model (E) shows the protein's main structural features more clearly (α helices, green; β strands, red). The components pictured in A–E are not shown to scale. (B, courtesy of C. Branden; C, courtesy of J. Hajdu and I. Andersson; D and E, PDB code: 1RBL.)

years of trial and error to discover the proper crystallization conditions; the pace has greatly accelerated with the use of recombinant DNA techniques to produce pure proteins and robotic techniques to test large numbers of crystallization conditions.

The position and intensity of each spot in the x-ray diffraction pattern contain information about the locations of the atoms in the crystal that gave rise to it. Computer-assisted computational methods process the diffraction pattern to generate a three-dimensional electron-density map. Interpreting this map—translating its contours into a three-dimensional structure—can be a laborious procedure. Largely by trial and error, the sequence and the electron-density map are correlated by computer to give the best possible fit. The reliability of the final atomic model depends on the resolution of the original crystallographic data: 0.5-nm resolution might produce a low-resolution map of the polypeptide backbone, whereas a resolution of 0.15 nm allows all of the non-hydrogen atoms in the molecule to be reliably positioned.

A complete atomic model is often too complex to appreciate directly, but simplified versions that show a protein's essential structural features can be readily derived from it. The three-dimensional structures of tens of thousands of different proteins have been determined—enough to allow the grouping of common structures into families (**Movie 8.1**). These structures or protein folds often seem to be more conserved in evolution than are the amino acid sequences that form them (see Figure 3-13).

NMR Can Be Used to Determine Protein Structure in Solution

Nuclear magnetic resonance (NMR) spectroscopy has been widely used for many years to analyze the structure of small molecules, small proteins, or protein domains. Unlike x-ray crystallography, NMR does not depend on having a crystalline sample. It simply requires a small volume of concentrated protein solution that is placed in a strong magnetic field; indeed, it is the main technique that yields detailed evidence about the three-dimensional structure of molecules in solution.

Certain atomic nuclei, particularly hydrogen nuclei, have a magnetic moment or spin; that is, they have an intrinsic magnetization, like a bar magnet. When exposed to a strong magnetic field in an NMR experiment, the spin of these nuclei aligns with the magnetic field, but it can be forced into a misaligned, excited state by radiofrequency (RF) pulses of electromagnetic radiation. As the excited hydrogen nuclei return to their aligned state, they emit RF radiation, which can be measured and displayed as a spectrum. The signal of the emitted radiation, called the *chemical shift*, depends on the environment of each hydrogen nucleus, such that every nucleus in a protein displays a slightly different chemical shift. Furthermore, if one nucleus is excited, it influences the absorption and emission of radiation by other nuclei that lie close to it. It is consequently possible, by an ingenious elaboration of the basic NMR technique known as two-dimensional (2D) NMR, to distinguish the signals from hydrogen nuclei in different amino acid residues and to identify and measure the small changes in these signals that occur when these hydrogen nuclei lie close enough together to interact. Because the size of such a change reveals the distance between the interacting pair of hydrogen atoms, NMR can provide information about the distances between the parts of the protein molecule. Additional information can be gained by *heteronuclear* 2D NMR, which analyzes hydrogen nuclei in parallel with the nuclei of a nitrogen isotope. By combining this information with a knowledge of the amino acid sequence, it is possible in principle to compute the three-dimensional structure of the protein (**Figure 8–21**). For technical reasons, NMR spectroscopy is used primarily to determine the structure of small proteins of about 30,000 daltons or less.

Because NMR studies are performed in solution, this method also offers a convenient means of monitoring changes in protein structure; for example, during protein folding or when the protein binds to another molecule. NMR is also used widely to investigate molecules other than proteins and is valuable, for example, as a method to determine the three-dimensional structures of RNA molecules and the complex carbohydrate side chains of glycoproteins.

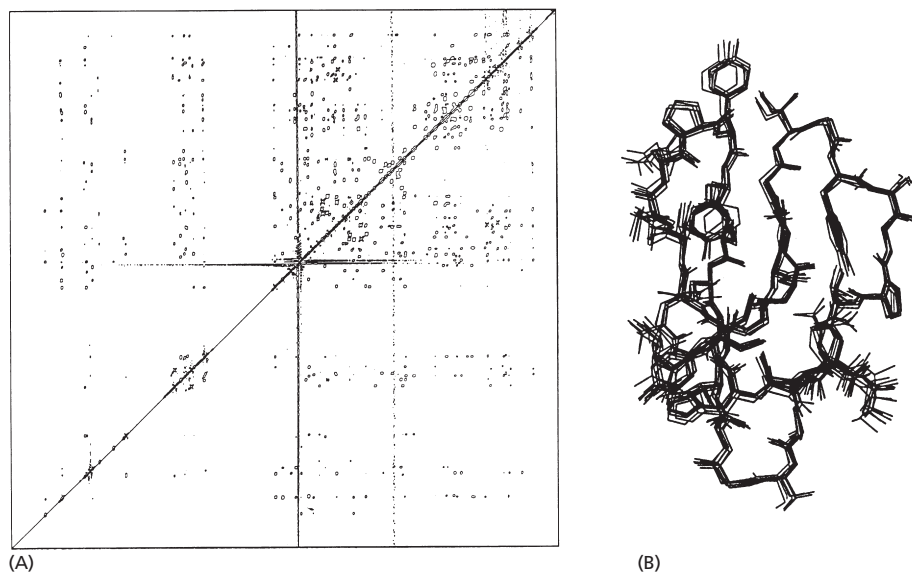


Figure 8–21 NMR spectroscopy. (A) An example of the data from an NMR machine. This NMR spectrum, displaying chemical shifts of hydrogen nuclei in two dimensions, is derived from the C-terminal domain of the enzyme cellulase. The spots away from the diagonal represent interactions between hydrogen atoms that are near neighbors in the protein; their positions reflect the distance that separates them. Complex computing methods, in conjunction with the known amino acid sequence, enable possible compatible structures to be derived. (B) Ten structures of the enzyme, all of which satisfy the distance constraints equally well, are shown superimposed on one another, giving a good indication of the probable three-dimensional structure. (Courtesy of P. Kraulis.)

A major problem in structural biology is the analysis of large protein complexes, which are difficult to crystallize and too large for NMR analysis. Single-particle analysis by cryo-electron microscopy provides a relatively straightforward approach to the analysis of large macromolecular assemblies, as we describe in Chapter 9.

Protein Sequence and Structure Provide Clues About Protein Function

Having discussed methods for purifying and analyzing proteins, we now turn to a common situation in cell and molecular biology: an investigator has identified a gene important for a biological process but has no direct knowledge of the biochemical properties of its protein product.

Thanks to the proliferation of protein and nucleic acid sequences that are cataloged in genome databases, the function of a gene—and its encoded protein—can often be predicted by simply comparing its sequence with those of previously characterized genes. Because amino acid sequence determines protein structure, and structure dictates biochemical function, proteins that share a similar amino acid sequence usually have the same structure and usually perform similar biochemical functions, even when they are found in distantly related organisms. Thus, the study of a newly discovered protein usually begins with a search for previously characterized proteins that are similar in their amino acid sequences.

Searching a collection of known sequences for similar genes or proteins simply involves selecting a database and entering the desired sequence. A sequence-alignment program—the most popular is *BLAST* (*Basic Local Alignment Search Tool*)—scans the database for similar sequences by sliding the submitted sequence along the archived sequences until a cluster of residues falls into full or partial alignment (Figure 8–22).

As was explained in Chapter 3, many proteins that adopt the same conformation and have related functions are too distantly related to be identified from a comparison of their amino acid sequences alone (see Figure 3–13). Thus, an ability to reliably predict the three-dimensional structure of a protein from its amino acid sequence would improve our ability to infer protein function from the sequence information in genomic databases. In recent years, major progress has been made in predicting the precise structure of a protein. These predictions are based, in part, on our knowledge of the thousands of protein structures that have already been determined by x-ray crystallography and NMR spectroscopy and, in part, on computations using our knowledge of the physical forces acting on the atoms. However, it remains a substantial and important challenge to predict

Figure 8–22 Results of a BLAST search.

Sequence databases can be searched to find similar amino acid or nucleic acid sequences. Here, a search for proteins similar to the human cell-cycle regulatory protein Cdk1 (*Query*) locates maize Cdk1 (*Sbjct*), which is 68% identical to human Cdk1 in its amino acid sequence. The alignment begins at residue 57 of the *Query* protein, suggesting that the human protein has an N-terminal region that is absent from the maize protein. The *green blocks* indicate differences in sequence, and the *yellow bars* summarize the similarities: when the two amino acid sequences are identical, the residue is shown; similar amino acid substitutions are indicated by a plus sign (+). Only one small gap has been introduced—indicated by the *red arrowhead* at position 194 in the *Query* sequence—to align the two sequences maximally. The alignment score (*Score*), which is expressed in two different types of units, takes into account penalties for substitutions and gaps; the higher the alignment score, the better the match. The significance of the alignment is reflected in the *Expectation* (*E*) value, which specifies how often a match this good would be expected to occur by chance. The lower the *E* value, the more significant the match; the extremely low value here (e^{-111}) indicates certain significance. *E* values much higher than 0.1 are unlikely to reflect true relatedness. For example, an *E* value of 0.1 means there is a 1 in 10 likelihood that such a match would arise solely by chance.

```

Score = 399 bits (1025), Expect = e-111
Identities = 198/290 (68%), Positives = 241/290 (82%), Gaps = 1/290

Query: 57  MENFQVEKIGEGTYGVVYKARNKLTGEVVALKKIRLDTEEGVPSTAIRESISLLKELNH 116
          ME ++KVEKIGEGTYGVVYKA +K T E +ALKKIRL+ E EGVPSTAIRESISLLKE+NH
Sbjct: 1   MEQYERVEKIGEGTYGVVYKALDKATNETIALKKIRLEQDEGVPSTAIRESISLLKEMNH 60

Query: 117 PNIVKLLDVIHTE NKLYLVFEFLHODLKKFMDASALTGIPLELIKSYLQQLGLAFCHS 176
          NIV+L DV+H+E ++YLVEE+L DLKKFMD+ LIKSYL+Q+L G+A+CHS
Sbjct: 61  GNIVRLHDVVHSEKRYFLVFEYLDL DLKKFMDSCPEFAKNPTLIKSYLQQLHGVAYCHS 120

Query: 177 HRVLHRDLKPQNLLINTE- GAIKLADFGLARAFGVPVRTYTHEVVTWYRAPEILLGCKY 235
          HRVLHRDLKPQNLLI+ A+KLADFGLARAFG+PVRT+THEVVTWYRAPEILLG +
Sbjct: 121 HRVLHRDLKPQNLLIDRRTNALKLADFGLARAFGIPVRTFTHEVVTWYRAPEILLGARQ 180

Query: 236 YSTAVDIWSLGCIFAEMVTRRLFPDSEIDQLFRIFRILGTPDEVVWPGVTSMPDYKPS 295
          YST VD+WS+GCIFAEMV ++ LFPDSEID+LF+IFR LGTP+E WPGV+ +PD+K +
Sbjct: 181 YSTPVDVWSVGCIFAEMVNQKELFPDSEIDELFKIFRILGTPNEQSWPGVSCLPDFKTA 240

Query: 296 FPKWARQDFSKVVPFLDEDEGRSLLSQMLHYDPNKRISAKAALAHFFQDV 345
          FP+W QD + VVP LD G LLS+ML Y+P+KRI+A+ AL H +F+D+
Sbjct: 241 FPRWQAQDLATVVPNLDPAGLDLLSKMLRYEPSKRITARQALEHYEFKDL 290

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the structures of proteins that are large or have multiple domains or to predict structures at the very high levels of resolution needed to assist in computer-based drug discovery.

While finding related sequences and structures for a new protein will provide many clues about its function, it is usually necessary to test these insights through direct experimentation. However, the clues generated from sequence comparisons typically point the investigator in the correct experimental direction, and their use has therefore become one of the most important strategies in modern cell biology.

Summary

Many methods exist for identifying proteins and analyzing their biochemical properties, structures, and interactions with other proteins. The most powerful and commonly used methods include protein separation by polyacrylamide gel electrophoresis, protein analysis by mass spectrometry, and high-resolution structural determination. Because proteins with similar structures often have similar functions, the biochemical activity of a protein can often be predicted by searching databases for previously characterized proteins that are similar in their amino acid sequences.

ANALYZING AND MANIPULATING DNA

Until the early 1970s, DNA was the most difficult biological molecule for the biochemist to analyze. Enormously long and chemically monotonous, the string of nucleotides that forms the genetic material of an organism could be determined only indirectly, from protein sequence or by genetic analysis. Today, the situation has changed entirely. From being the most difficult macromolecule of the cell to analyze, DNA has become the easiest. It is now possible to determine the entire nucleotide sequence of a bacterial or fungal genome in a matter of hours and the sequence of an individual human genome in less than a day. Once the nucleotide sequence of a genome is known, any individual gene can be easily isolated, and large quantities of the gene product (be it RNA or protein) can be made either by introducing the gene into bacteria or animal cells and coaxing these cells to over-express the foreign gene or by synthesizing the gene product *in vitro*. In this way, proteins and RNA molecules that might be present in only tiny amounts in living cells can be produced in large quantities for biochemical and structural analyses. And this approach can also be used to produce large quantities of human proteins (such as insulin, or interferon, or blood-clotting proteins) for use as human pharmaceuticals. As we will see later in this chapter, it is also possible for scientists to alter an isolated gene and transfer it back into the germ line of an animal or plant, so as to become a functional and heritable part of the organism's genome. In this way, the biological roles of any gene can be assessed by observing—in the whole organism—the results of modifying it.

The ability to manipulate DNA with precision in a test tube or an organism, known as **recombinant DNA technology**, has had a dramatic impact on all aspects of cell and molecular biology. We now describe the key features of these techniques.

Restriction Nucleases Cut Large DNA Molecules into Specific Fragments

Unlike a protein, a gene does not exist as a discrete entity in cells, but rather as a small region of a much longer DNA molecule. Although the DNA molecules in a cell can be randomly broken into small pieces by mechanical force, a fragment containing a single gene in a mammalian genome would still be only one among a hundred thousand or more DNA fragments, indistinguishable in their average size. How could such a gene be separated from all the others? Because all DNA molecules consist of an approximately equal mixture of the same four

nucleotides, they cannot be readily separated, as proteins can, on the basis of their different charges and biochemical properties. The solution to this problem began to emerge with the discovery of **restriction nucleases**. These enzymes, which are purified from bacteria, cut the DNA double helix at specific sites defined by the local nucleotide sequence, thereby cleaving a long, double-strand DNA molecule into fragments of strictly defined sizes.

Like many of the tools of recombinant DNA technology, restriction nucleases were discovered by researchers trying to understand an intriguing biological phenomenon. It had been observed that certain bacteria always degraded “foreign” DNA that was introduced into them experimentally. A search for the mechanism responsible revealed a then unanticipated class of bacterial nucleases that cleave DNA at specific nucleotide sequences. The bacterium’s own DNA is protected from cleavage by methylation of these same sequences, thereby protecting the bacterium from being overrun by foreign DNA. Because these enzymes restrict the transfer of DNA into bacteria, they were called restriction nucleases. The pursuit of this seemingly arcane biological puzzle set off the development of technologies that have forever changed the way cell and molecular biologists study living things.

Different bacterial species produce different restriction nucleases, each cutting at a different, specific nucleotide sequence (Figure 8-23). Because these target sequences are short—generally four to eight nucleotide pairs—many sites of cleavage will occur, purely by chance, in any long DNA molecule. The reason restriction nucleases are so useful in the laboratory is that each enzyme will always cut a particular DNA molecule at the same sites. Thus for a given sample of DNA (which contains many identical molecules), a particular restriction nuclease will reliably generate the same set of DNA fragments.

The size of the resulting fragments depends on the length of the target sequences of the restriction nucleases. As shown in Figure 8-23, the enzyme *HaeIII* cuts at a sequence of four nucleotide pairs; a sequence this long would be expected to occur purely by chance approximately once every 256 nucleotide pairs (1 in 4^4). In comparison, a restriction nuclease with a target sequence that is eight nucleotides long would be expected to cleave DNA on average once every 65,536 nucleotide pairs (1 in 4^8). This difference in sequence selectivity makes it possible to cleave a long DNA molecule into the fragment sizes that are most suitable for a given application.

Gel Electrophoresis Separates DNA Molecules of Different Sizes

The same types of gel-electrophoresis methods that have proved so useful in the analysis of proteins (see Figure 8-13) can be applied to DNA molecules. The procedure is actually simpler than for proteins: because each nucleotide in a nucleic acid molecule carries a single negative charge (on the phosphate group), there is no need to add the negatively charged detergent SDS that is required to make protein molecules move uniformly toward the positive electrode. Larger DNA

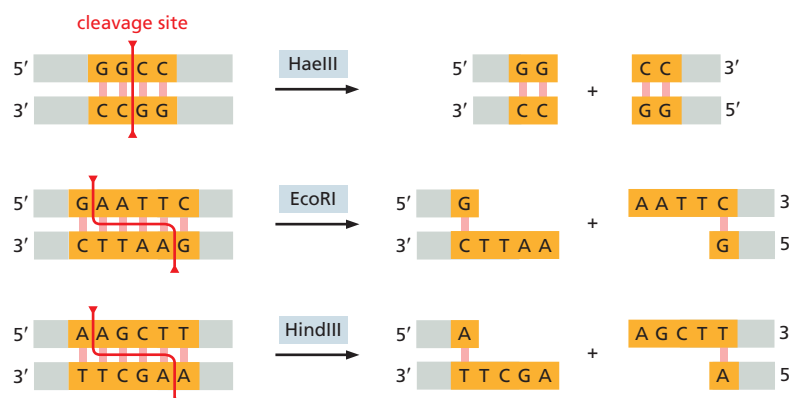


Figure 8-23 Restriction nucleases cleave DNA at specific nucleotide sequences.

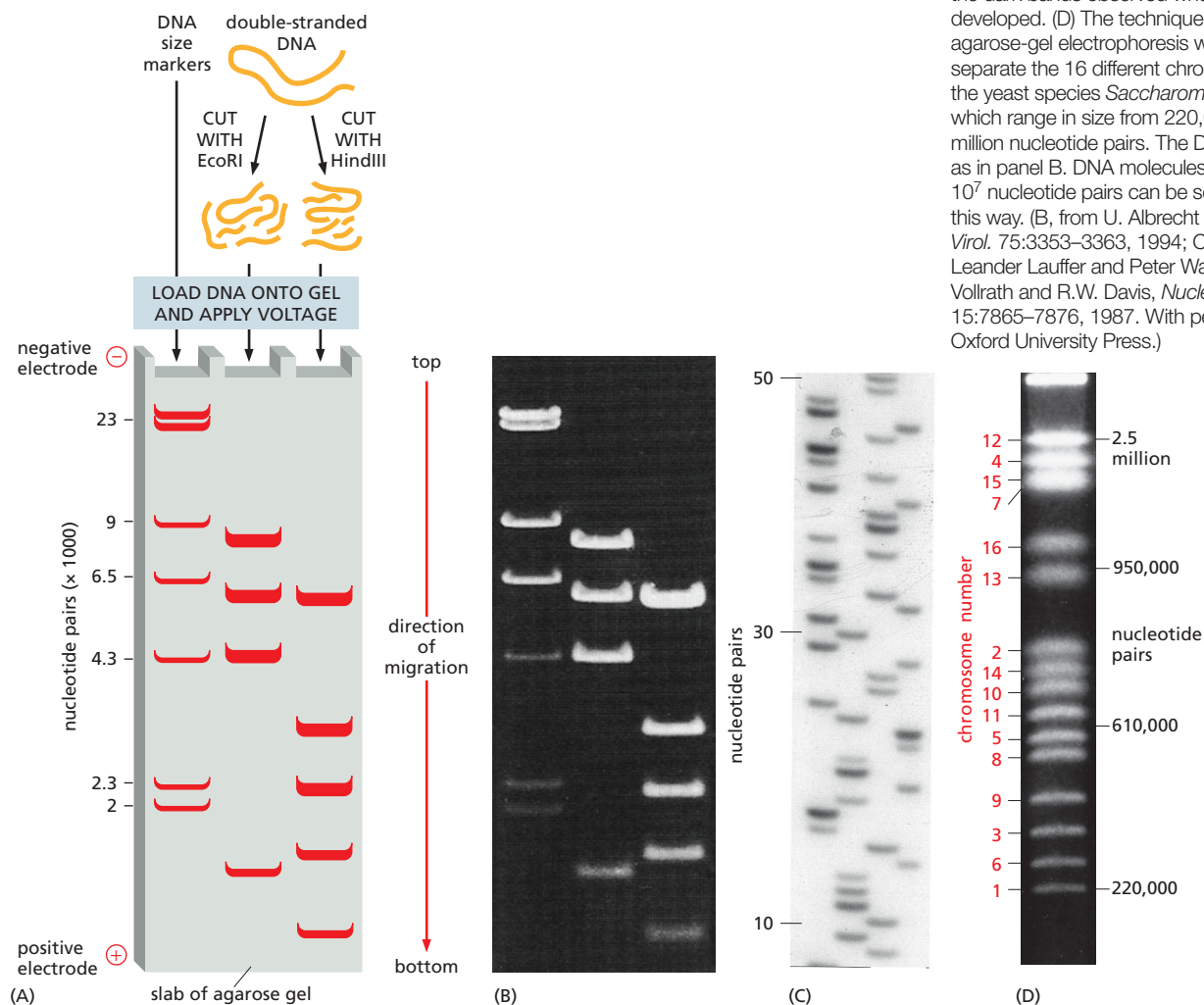
Like the sequence-specific DNA-binding proteins we encountered in Chapter 7 (see Figure 7-9), restriction enzymes often work as dimers, and the DNA sequence that each restriction enzyme recognizes and cleaves is often symmetrical around a central point. Here, both strands of the DNA double helix are cut at specific points within the target sequence (orange). Some enzymes, such as *HaeIII*, cut straight across the double helix and leave two blunt-ended DNA molecules; with others, such as *EcoRI* and *HindIII*, the cuts on each strand are staggered. These staggered cuts generate “sticky ends”—short, single-strand overhangs that help the cut DNA molecules join back together through complementary base-pairing. This rejoining of DNA molecules becomes important for DNA cloning, as we discuss later. Restriction nucleases are usually obtained from bacteria, and their names reflect their origins; for example, the enzyme *EcoRI* comes from *Escherichia coli*. Hundreds of different restriction enzymes are commercially available.

fragments will migrate more slowly because their progress is impeded to a greater extent by the gel matrix. In less than an hour, the DNA fragments become spread out across the gel according to size, forming a ladder of discrete bands, each composed of a collection of DNA molecules of identical length (**Figure 8-24A and B**). To separate DNA molecules longer than 500 nucleotide pairs, the gel is made of a diluted solution of agarose (a polysaccharide isolated from seaweed). For DNA fragments less than 500 nucleotides long, specially designed polyacrylamide gels allow the separation of molecules that differ in length by as little as a single nucleotide (**Figure 8-24C**).

A variation of agarose-gel electrophoresis, called *pulsed-field gel electrophoresis*, makes it possible to separate extremely long DNA molecules, even those found in whole chromosomes. Ordinary gel electrophoresis fails to separate very large DNA molecules because the steady electric field stretches them out so that they travel end-first through the gel in snake-like configurations at a rate that is independent of their length. In pulsed-field gel electrophoresis, by contrast, the direction of the electric field changes periodically, which forces the molecules to reorient before continuing to move snake-like through the gel. This reorientation takes much more time for larger molecules, so that longer molecules move more slowly than shorter ones. As a consequence, entire bacterial or yeast chromosomes separate into discrete bands in pulsed-field gels and so can be sorted and identified on the basis of their size (**Figure 8-24D**). Although a typical mammalian chromosome of 10^8 nucleotide pairs is still too long to be sorted even in this way, large segments of these chromosomes are readily separated and identified if the chromosomal

Figure 8-24 DNA molecules can be separated by size using gel electrophoresis.

(A) Schematic illustration comparing the results of cutting the same DNA molecule (in this case, the genome of a virus that infects wasps) with two different restriction nucleases, EcoRI (*middle*) and HindIII (*right*). The fragments are then separated by gel electrophoresis using a gel matrix of agarose. Because larger fragments migrate more slowly than smaller ones, the lowermost bands on the gel contain the smallest DNA fragments. The sizes of the fragments can be estimated by comparing them to a set of DNA fragments of known sizes (*left*). (B) Photograph of an actual agarose gel showing DNA bands that have been stained with ethidium bromide. (C) A polyacrylamide gel with small pores was used to separate short DNA molecules that differ by only a single nucleotide. Shown here are the results of a dideoxy sequencing reaction, explained later in this chapter. From left to right, the bands in the four lanes were produced by adding G, A, T, and C chain-terminating nucleotides (see Figure 8-42). The DNA molecules were labeled with ^{32}P , and the image shown was produced by laying a piece of photographic film over the gel and allowing the ^{32}P to expose the film, producing the dark bands observed when the film was developed. (D) The technique of pulsed-field agarose-gel electrophoresis was used to separate the 16 different chromosomes of the yeast species *Saccharomyces cerevisiae*, which range in size from 220,000 to 2.5 million nucleotide pairs. The DNA was stained as in panel B. DNA molecules as large as 10^7 nucleotide pairs can be separated in this way. (B, from U. Albrecht et al., *J. Gen. Virol.* 75:3353–3363, 1994; C, courtesy of Leander Lauffer and Peter Walter; D, from D. Vollrath and R.W. Davis, *Nucleic Acids Res.* 15:7865–7876, 1987. With permission from Oxford University Press.)



DNA is first cut with a restriction nuclease selected to recognize sequences that occur only rarely.

The DNA bands on agarose or polyacrylamide gels are invisible unless the DNA is labeled or stained in some way. A particularly sensitive method of staining DNA is to soak the gel in the dye *ethidium bromide*, which fluoresces under ultraviolet light when it is bound to DNA (see Figure 8-24B and D). Even more sensitive detection methods incorporate a radioisotope or chemical marker into the DNA molecules before electrophoresis, as we next describe.

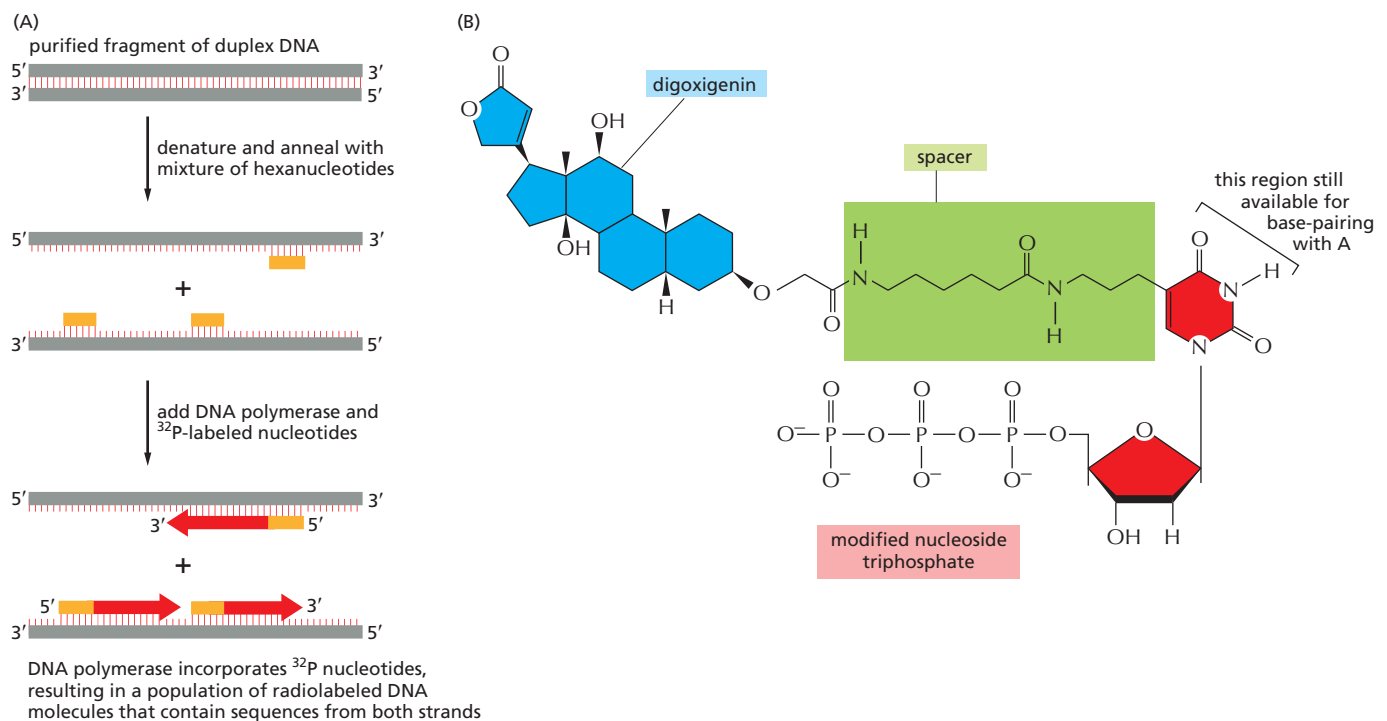
Purified DNA Molecules Can Be Specifically Labeled with Radioisotopes or Chemical Markers *in Vitro*

The DNA polymerases that synthesize and repair DNA (discussed in Chapter 5) have become important tools in experimentally manipulating DNA. They are often used in the test tube to create exact copies of existing DNA molecules. In some experiments, the copies can include specially modified nucleotides (Figure 8-25). To synthesize DNA in this way, the DNA polymerase is presented with a template and a pool of nucleotide precursors that contain the modification. As long as the polymerase can use these precursors, it automatically makes new, modified molecules that match the sequence of the template. Modified DNA molecules have many uses. DNA labeled with the radioisotope ^{32}P can be detected after gel electrophoresis by exposing the gel to photographic film or a radiation detector (see Figure 8-24C). Other types of modified DNA, such as that labeled by digoxigenin (see Figure 8-25B), are useful for visualizing DNA molecules in whole cells, a topic we discuss later in this chapter.

Genes Can Be Cloned Using Bacteria

Any DNA fragment can be cloned. In molecular biology, the term **DNA cloning** refers to the act of making many identical copies (typically billions) of a DNA molecule; that is, the amplification of a specific DNA sequence (often a particular gene) from the rest of the cell's genome. We note that elsewhere in the book, cloning can also refer to the generation of many genetically identical cells starting from a single cell or even to the generation of genetically identical organisms (see,

Figure 8-25 Methods for labeling DNA molecules *in vitro*. (A) A purified DNA polymerase enzyme can incorporate radiolabeled nucleotides as it synthesizes new DNA molecules. In this way, radiolabeled versions of any DNA sequence can be prepared in the laboratory. (B) The method in panel A is also used to produce nonradioactive DNA molecules that carry a specific chemical marker that can be detected with an appropriate antibody. The base on the nucleoside triphosphate shown is an analog of thymine, in which the methyl group on T has been replaced by a spacer arm linked to the plant steroid digoxigenin. An anti-digoxigenin antibody coupled to a visible marker such as a fluorescent dye is then used to visualize the DNA. Other chemical labels, such as biotin, can be attached to nucleotides and used in the same way. The only requirements are that the modified nucleotides properly base-pair and appear “normal” to the DNA polymerase.



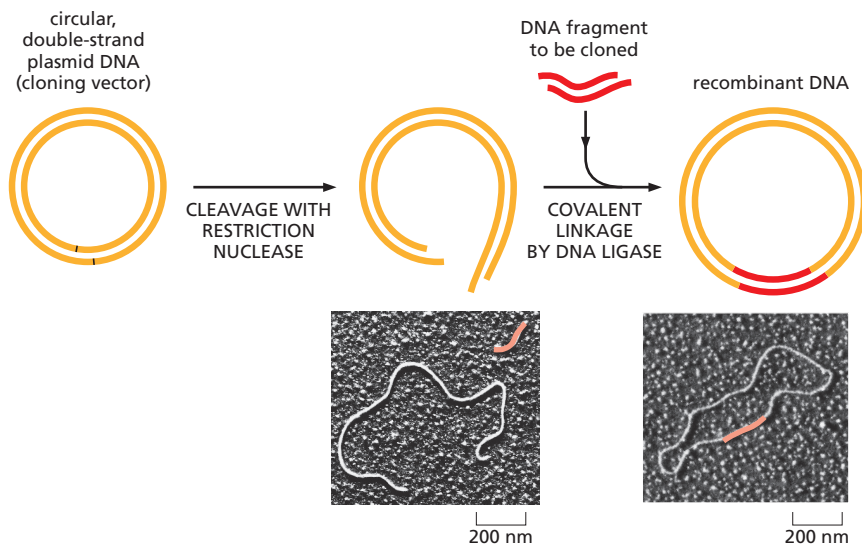


Figure 8–26 The insertion of a DNA fragment into a bacterial plasmid with the enzyme DNA ligase. The plasmid is cut open with a restriction nuclease (in this case, one that produces staggered ends) and is mixed with the DNA fragment to be cloned (which has been prepared with the same restriction nuclease). DNA ligase and ATP are added. The staggered ends base-pair, and DNA ligase seals the nicks in the DNA backbone, producing a complete recombinant DNA molecule. In the accompanying micrographs, the inserted DNA is colored red. (Micrographs courtesy of Huntington Potter and David Dressler.)

for example, Figure 7–2). In all cases, cloning refers to the act of making many identical copies, and here we use the term to refer to methods designed to generate many identical copies of a defined segment of nucleic acid.

DNA cloning can be accomplished in several ways. One of the simplest involves inserting a particular fragment of DNA into the purified DNA of a self-replicating genetic element—usually a plasmid. The **plasmid vectors** most widely used for gene cloning are small, circular molecules of double-stranded DNA derived from plasmids that occur naturally in bacterial cells. They generally account for only a minor fraction of the total host bacterial cell DNA, but owing to their small size, they can easily be separated from the much larger chromosomal DNA molecules. For use as cloning vectors, the purified plasmid DNA circles are first cut with a restriction nuclease to create linear DNA molecules. The DNA to be cloned is added to the cut plasmid and then covalently joined using the enzyme DNA ligase (Figure 8–26 and Figure 8–27). As discussed in Chapter 5, this enzyme is used by the cell to stitch together the Okazaki fragments produced during DNA replication. The recombinant DNA circle is introduced back into bacterial cells that have been made transiently permeable to DNA. As the cells grow and divide, doubling in number every 30 minutes, the recombinant plasmids also replicate to produce an enormous

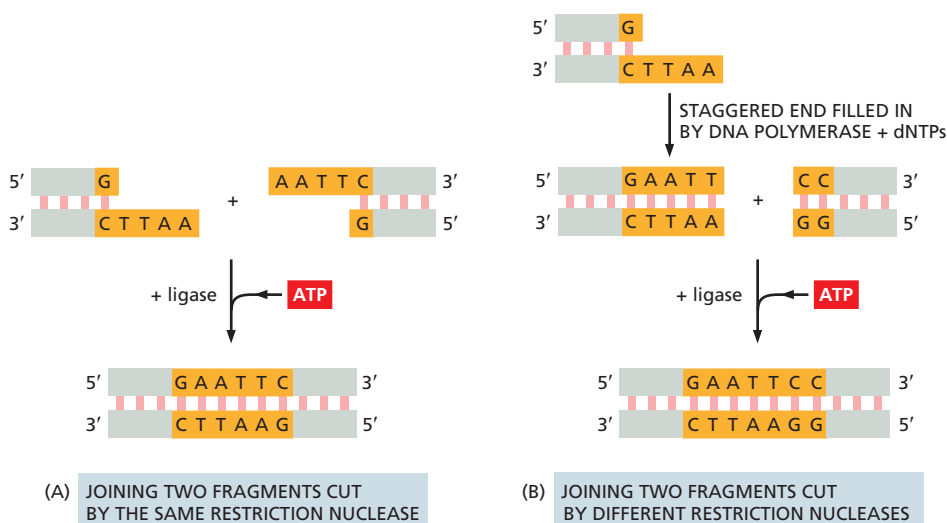


Figure 8–27 DNA ligase can join together any two DNA fragments *in vitro* to produce recombinant DNA molecules.

ATP provides the energy necessary to reseal the sugar–phosphate backbone of DNA (see Figure 5–12). (A) DNA ligase can readily join two DNA fragments produced by the same restriction nuclease, in this case EcoRI. Note that the staggered ends produced by this enzyme enable the ends of the two fragments to base-pair correctly with each other, greatly facilitating their rejoining. (B) DNA ligase can also be used to join DNA fragments produced by different restriction nucleases; for example, EcoRI and HaeIII. In this case, before the fragments undergo ligation, DNA polymerase plus a mixture of deoxyribonucleoside triphosphates (dNTPs) are used to fill in the staggered cut produced by EcoRI. Each DNA fragment shown in the figure is oriented so that its 5' ends are at the left end of the upper strand and the right end of the lower strand, as indicated.

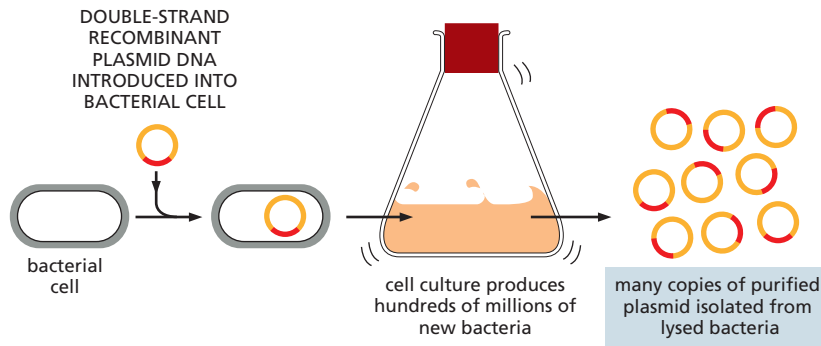


Figure 8–28 A DNA fragment can be replicated inside a bacterial cell. To clone a particular fragment of DNA, it is first inserted into a plasmid vector, as shown in Figure 8–26. The resulting recombinant plasmid DNA is then introduced into a bacterium, where it is replicated many millions of times as the bacterium multiplies. For simplicity, the genome of the bacterial cell is not shown.

number of copies of DNA circles containing the foreign DNA (**Figure 8–28**). Once the cells are lysed and the plasmid DNA isolated, the cloned DNA fragment can be readily recovered by cutting it out of the plasmid DNA with the same restriction nuclease that was used to insert it, and then separating it from the plasmid DNA by gel electrophoresis. Together, these steps allow the amplification and purification of any segment of DNA from the genome of any organism.

A particularly useful plasmid vector is based on the naturally occurring F plasmid of *E. coli*. Unlike smaller bacterial plasmids, the F plasmid—and its engineered derivative, the **bacterial artificial chromosome (BAC)**—is present in only one or two copies per *E. coli* cell. The fact that BACs are kept in such low numbers means that they can stably maintain very long DNA sequences, up to 1 million nucleotide pairs in length. With only a few BACs present per bacterium, it is less likely that the cloned DNA fragments will become scrambled by recombination with sequences carried on other copies of the plasmid. Because of their stability, ability to accept large DNA inserts, and ease of handling, BACs are the preferred vector for handling large fragments of foreign DNA.

An Entire Genome Can Be Represented in a DNA Library

Often it is useful to break up a genome into much smaller fragments and clone every fragment, separately, using a plasmid vector. This approach is useful because it allows scientists to work with easily managed, discrete pieces of a genome instead of whole, unwieldy chromosomes.

This strategy involves cleaving genomic DNA into small pieces using a restriction nuclease (or, in some cases, by mechanically shearing the DNA) and ligating the entire collection of DNA fragments into plasmid vectors, using conditions that favor the insertion of a single DNA fragment into each plasmid molecule. These recombinant plasmids are then introduced into *E. coli* at a concentration that ensures that no more than one plasmid molecule is taken up by each bacterium. The collection of cloned plasmid molecules is known as a **DNA library**. Because the DNA fragments were derived directly from the chromosomal DNA of the organism of interest, the resulting collection—called a **genomic library**—will represent the entire genome of that organism (**Figure 8–29**), spread out over tens of thousands of individual bacterial colonies.

An alternative strategy, one that enriches for protein-coding genes, is to begin the cloning process by selecting only those DNA sequences that are transcribed into mRNA and thus correspond to protein-encoding genes. This is done by extracting the mRNA from cells and then making a DNA copy of each mRNA



Figure 8–29 Human genomic libraries containing DNA fragments that represent the whole human genome can be constructed using restriction nucleases and DNA ligase. Such a genomic library consists of a set of bacteria, each carrying a different fragment of human DNA. For simplicity, only the colored DNA fragments are shown in the library; in reality, all of the different gray fragments will also be represented.

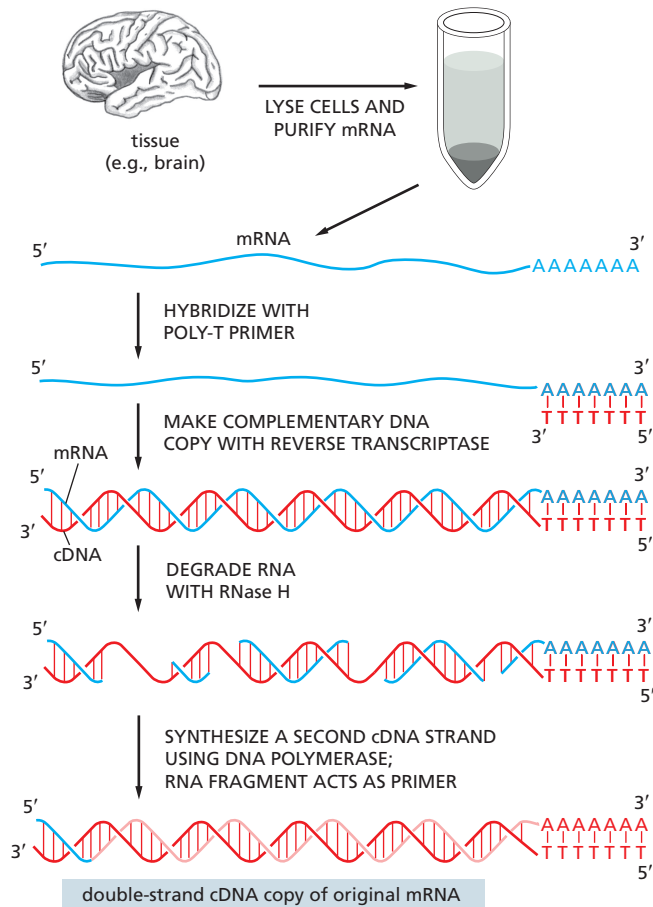


Figure 8–30 The synthesis of cDNA. Total mRNA is extracted from a particular tissue, and the enzyme reverse transcriptase (see Figure 5–61) is used to produce DNA copies (cDNA) of the mRNA molecules. For simplicity, the copying of just one of these mRNAs into cDNA is illustrated. A short oligonucleotide complementary to the poly-A tail at the 3' end of the mRNA is first hybridized to the RNA to act as a primer for the reverse transcriptase, which then copies the RNA into a complementary DNA chain, thereby forming a DNA–RNA hybrid helix. Treating the DNA–RNA hybrid with a specialized ribonuclease (RNase H) that attacks only the RNA produces nicks and gaps in the RNA strand. DNA polymerase then copies the remaining single-stranded cDNA into double-stranded cDNA. Because DNA polymerase can synthesize through the bound RNA molecules, the RNA fragment that is base-paired to the 3' end of the first DNA strand usually acts as the primer for the second strand synthesis, as shown. Any remaining RNA is eventually degraded during subsequent cloning steps. As a result, the nucleotide sequences at the extreme 5' ends of the original mRNA molecules are often absent from cDNA libraries.

molecule present—a so-called *complementary DNA*, or *cDNA*. The copying reaction is catalyzed by the reverse transcriptase enzyme of retroviruses, which synthesizes a complementary DNA chain on an RNA template. The single-strand cDNA molecules synthesized by the reverse transcriptase are converted by DNA polymerase into double-strand cDNA molecules, and these molecules are inserted into a plasmid or virus vector and cloned (**Figure 8–30**). Each clone obtained in this way is called a **cDNA clone**, and the entire collection of clones derived from one mRNA preparation constitutes a **cDNA library**.

The most important advantage of cDNA clones, over genomic clones, is that they contain the uninterrupted coding sequence of a gene. When the aim of the cloning is to produce the protein in large quantities by expressing the cloned gene in a bacterial or other cell type, it is best to start with cDNA.

Cloning DNA in bacteria revolutionized the study of genomes and is still in wide use today. However, there is an even simpler way to clone DNA and produce

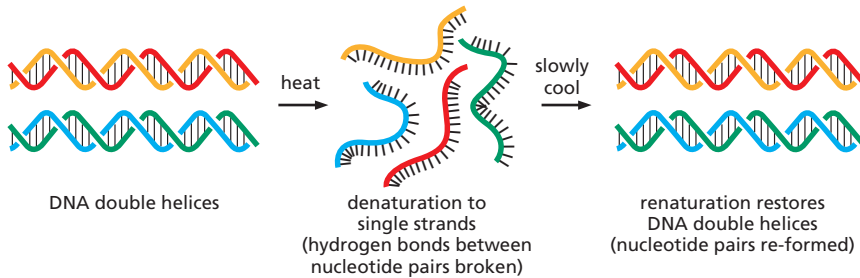


Figure 8–31 A molecule of DNA can undergo denaturation and renaturation (hybridization). For two single-strand molecules to hybridize, they must have complementary nucleotide sequences that allow base-pairing. In this example, the *red* and *orange* strands are complementary to each other, and the *blue* and *green* strands are complementary to each other. Although denaturation by heating is shown, DNA can also be renatured after being denatured by alkali treatment.

genomic libraries, entirely *in vitro*. We discuss this approach, called the *polymerase chain reaction*, shortly. However, first we need to review a fundamental, far-reaching property of DNA and RNA called *hybridization*.

Hybridization Provides a Powerful but Simple Way to Detect Specific Nucleotide Sequences

Under normal conditions, the two strands of a DNA double helix are held together by hydrogen bonds between the complementary base pairs (see Figure 4–3). These relatively weak, noncovalent bonds are easy to break. Such *DNA denaturation* releases the two strands from each other but does not break the covalent bonds that link together the nucleotides within each strand. The simplest way to achieve this separation is to heat the DNA to around 90°C. When the conditions are reversed—by slowly lowering the temperature—the complementary strands will readily come back together to re-form a double helix. This **hybridization**, or *DNA renaturation*, is driven by the re-formation of the hydrogen bonds between complementary base pairs (**Figure 8–31**).

This fundamental capacity of a single-strand nucleic acid molecule, either DNA or RNA, to form a double helix with a single-strand molecule of a complementary sequence provides a powerful and sensitive technique for detecting specific nucleotide sequences. Today, one simply designs a short, single-strand DNA molecule (often called a *DNA probe*) that is complementary to the nucleotide sequence of interest. Because the nucleotide sequences of so many genomes are known—and are stored in publicly accessible databases—designing a probe to hybridize anywhere in a genome is straightforward. Probes are single-stranded, typically 30 nucleotides in length, and are usually synthesized chemically by a commercial service for pennies per nucleotide. A DNA sequence of 30 nucleotides will occur by chance only once every 1×10^{18} nucleotides (4^{30}); so, even in the human genome of 3×10^9 nucleotide pairs, a DNA probe designed to match a unique 30-nucleotide sequence will be highly unlikely to hybridize—by chance—anywhere else on the genome. The hybridization conditions can be set so that even a single mismatch will prevent hybridization to “near-miss” sequences. The exquisite specificity of nucleic acid hybridization can be easily appreciated by the *in situ* (Latin for “in place”) *hybridization* experiment shown in **Figure 8–32**. As we will see throughout this chapter, nucleic acid hybridization has many uses in

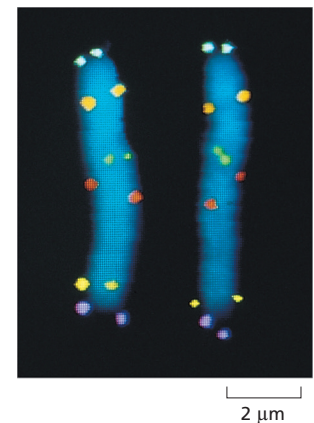


Figure 8–32 *In situ* hybridization can be used to locate genes on isolated chromosomes.

Here, six different DNA probes have been used to mark the locations of their complementary nucleotide sequences on human chromosome 5, isolated from a mitotic cell in metaphase. The DNA probes have been labeled with different chemical groups (see Figure 8–25B) and are detected using fluorescent antibodies specific for those groups. The chromosomal DNA has been partially denatured to allow the probes to base-pair with their complementary sequences. Both the maternal and paternal copies of chromosome 5 are shown, aligned side by side. Each probe produces two dots on each chromosome because chromosomes undergoing mitosis have already replicated their DNA; therefore, each chromosome contains two identical DNA helices. The technique employed here is nicknamed FISH, for *fluorescence in situ hybridization*. (Courtesy of David C. Ward.)

modern cell and molecular biology; one of the most powerful is in the cloning of DNA by the polymerase chain reaction, as we next discuss.

Genes Can Be Cloned *in Vitro* Using PCR

Genomic and cDNA libraries were once the only route to cloning genes. However, a powerful and versatile method for amplifying DNA, known as the **polymerase chain reaction (PCR)**, provides a more rapid and straightforward approach to DNA cloning, particularly in organisms whose complete genome sequence is known. Today, because genome sequences are abundant, cloning is often carried out by PCR.

Invented in the 1980s, PCR revolutionized the way that DNA and RNA are analyzed. The technique can amplify any nucleotide sequence selectively and is performed entirely in a test tube. Eliminating the need for bacteria makes PCR convenient and rapid—billions of copies of a nucleic acid sequence can be generated in a matter of hours. Starting with an entire genome, PCR allows DNA from a specified region—selected by the experimenter—to be greatly amplified, effectively purifying this DNA away from the remainder of the genome, which remains unamplified. Because of its power to greatly amplify nucleic acids, PCR is remarkably sensitive: the method can be used to detect the trace amounts of DNA in a drop of blood left at a crime scene or in a few copies of a viral genome in a sample of someone's saliva.

The success of PCR depends both on the selectivity of DNA hybridization and on the ability of DNA polymerase to copy a DNA template faithfully through repeated rounds of replication *in vitro*. As discussed in Chapter 5, this enzyme adds nucleotides to the 3' end of a growing strand of DNA (see Figure 5-4). To copy DNA, the polymerase requires a *DNA primer*—a short nucleotide sequence that provides a 3' end from which synthesis can begin. For PCR, the primers are designed by the experimenter, synthesized chemically, and, by hybridizing to genomic DNA, guide the polymerase to the part of the genome to copy. DNA primers can be designed to uniquely locate any position on a genome.

PCR is an iterative process in which the cycle of amplification is repeated many times. At the start of each cycle, the two strands of the double-strand DNA template are separated, and a different primer is annealed to each. These primers mark the right and left boundaries of the DNA to be amplified. DNA polymerase is then allowed to replicate each strand independently (**Figure 8-33**). In subsequent cycles, all the newly synthesized DNA molecules produced by the

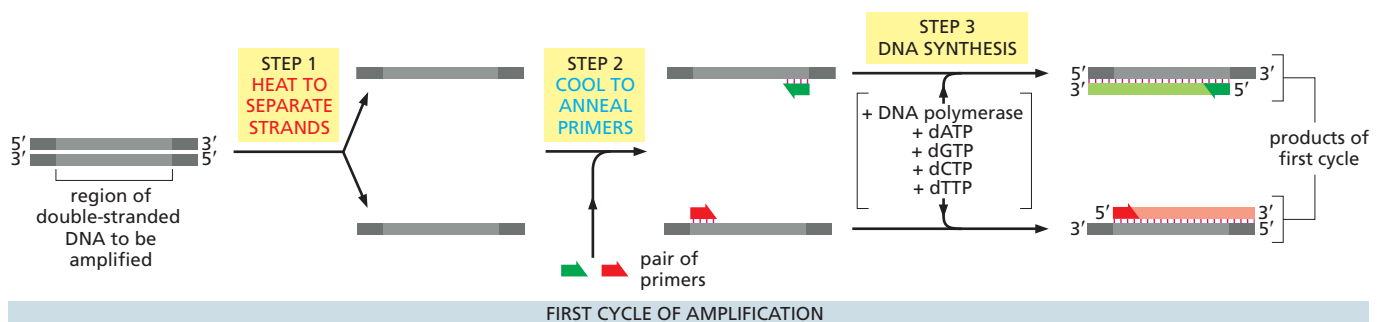


Figure 8-33 A pair of primers directs the synthesis of a desired segment of DNA in a test tube. Each cycle of PCR includes three steps: (1) The double-stranded DNA is heated briefly to separate the two strands. (2) The DNA is exposed to a large excess of a pair of specific primers—designed to bracket the region of DNA to be amplified—and the sample is cooled to allow the primers to hybridize to complementary sequences in the two DNA strands. (3) This mixture is incubated with DNA polymerase and the four deoxyribonucleoside triphosphates so that DNA can be synthesized, starting from the two primers. To amplify the DNA, the cycle is repeated many times by reheating the sample to separate the newly synthesized DNA strands (see Figure 8-34).

The technique depends on the use of a special DNA polymerase isolated from a thermophilic bacterium; this polymerase is stable at much higher temperatures than eukaryotic DNA polymerases, so it is not denatured by the heat treatment shown in step 1. The enzyme therefore does not have to be added again after each cycle.

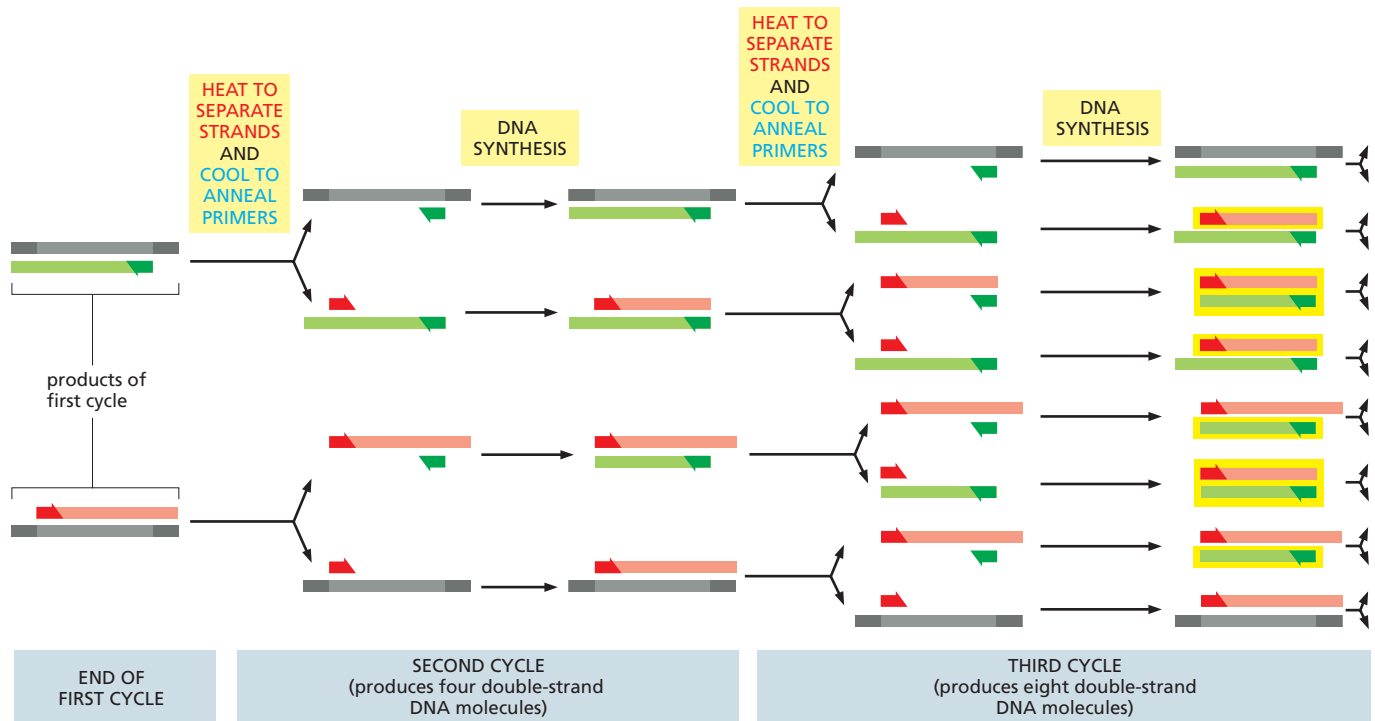


Figure 8–34 PCR uses repeated rounds of strand separation, hybridization, and synthesis to amplify DNA. As the procedure outlined in Figure 8–33 is repeated, all the newly synthesized fragments serve as templates in their turn. Because the polymerase and the primers remain in the sample after the first cycle, PCR involves simply heating and then cooling the same sample, in the same test tube, again and again. Each cycle doubles the amount of DNA synthesized in the previous cycle, so that within a few cycles, the predominant DNA is identical to the sequence bracketed by and including the two primers in the original template. In the example illustrated here, three cycles of reaction produce 16 DNA chains, 8 of which (boxed in yellow) correspond exactly to one or the other strand of the original bracketed sequence. After four more cycles, 240 of the 256 DNA chains will correspond exactly to the original sequence, and after several more cycles, essentially all of the DNA strands will be this length. Typically, 20–30 cycles are carried out to effectively clone a region of DNA starting from genomic DNA; the rest of the genome remains unamplified, and its concentration is therefore negligible compared with that of the amplified region ([Movie 8.2](#)).

polymerase serve as templates for the next round of replication ([Figure 8–34](#)). Through this iterative amplification process, many copies of the original sequence can be made—billions after about 20–30 cycles.

PCR is now the method of choice for cloning relatively short DNA fragments (say, under 10,000 nucleotide pairs). Each cycle takes only about 5 minutes, and automation of the whole procedure enables cell-free cloning of a DNA fragment in a few hours. The original template for PCR can be either DNA or RNA, so this method can be used to obtain either a genomic clone (complete with introns and exons) or a cDNA copy of an mRNA ([Figure 8–35](#)).

PCR Is Also Used for Diagnostic and Forensic Applications

The PCR method is extraordinarily sensitive; it can detect a single DNA molecule in a sample if at least part of the sequence of that molecule is known. Trace amounts of RNA can be analyzed in the same way by first transcribing them into DNA with reverse transcriptase. For these reasons, PCR is frequently employed for uses that go beyond simple cloning. For example, it can be used to detect invading pathogens at very early stages of infection. In this case, short sequences complementary to a segment of the infectious agent's genome are used as primers, and, after many cycles of amplification, even a few copies of an invading bacterial or viral genome in a human sample can be detected ([Figure 8–36](#)). For many infections, PCR has replaced the use of antibodies against microbial molecules to detect the presence of the invader. It is also used to verify the

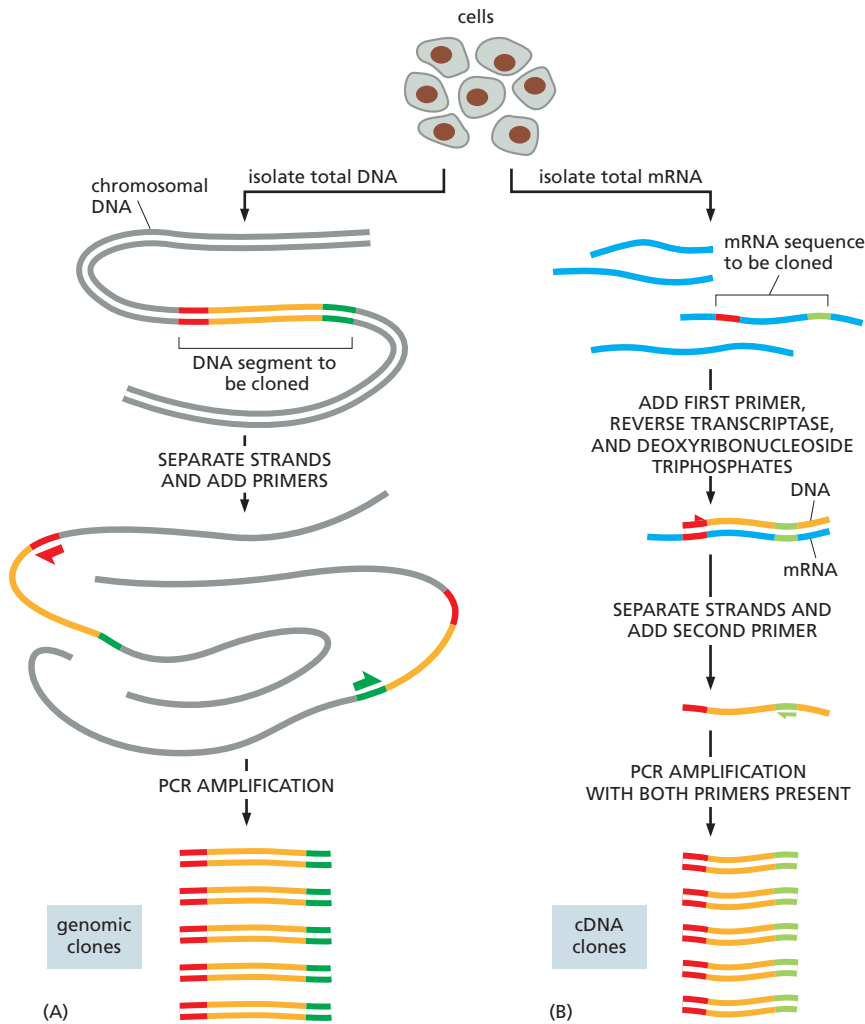


Figure 8-35 PCR can be used to obtain either genomic or cDNA clones.

(A) To use PCR to clone a segment of chromosomal DNA, total genomic DNA is first purified from cells. PCR primers that flank the stretch of DNA to be cloned are added, and many cycles of PCR are completed (see Figure 8-34). Because only the DNA between (and including) the primers is amplified, PCR provides a way to obtain selectively any short stretch of chromosomal DNA in an effectively pure form. (B) To use PCR to obtain a cDNA clone of a gene, total mRNA is first purified from cells. The first primer is added to the population of mRNAs, and reverse transcriptase is used to make a DNA strand complementary to the specific RNA sequence of interest. The second primer is then added, and the DNA molecule is amplified through many cycles of PCR.

authenticity of a food source; for example, whether a sample of beef actually came from a cow.

Finally, PCR is now widely used in forensics. The method's extreme sensitivity allows forensic investigators to isolate DNA from minute traces of human blood or other tissue to obtain a *DNA fingerprint* of the person who left the sample behind. With the possible exception of identical twins, the genome of each human differs in DNA sequence from that of every other person on Earth. Using primer pairs targeted at genome sequences that are known to be highly variable in the human population, PCR makes it possible to generate a distinctive DNA fingerprint for any individual (Figure 8-37). Such forensic analyses can be used not only to help

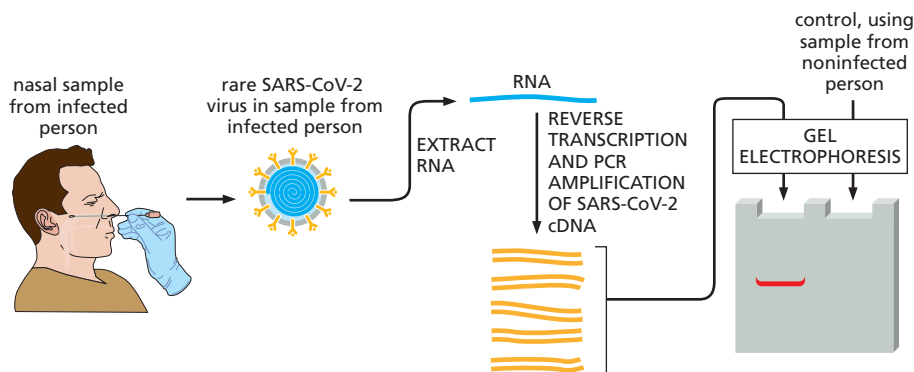


Figure 8-36 PCR can be used to detect the presence of a viral genome in a nasal sample.

Because of its ability to amplify enormously the signal from a single molecule of nucleic acid, PCR is an extraordinarily sensitive method for detecting trace amounts of virus in a sample of saliva, blood, or other tissue, without the need to purify the virus. For the coronavirus SARS-CoV-2, the virus that causes COVID-19, the genome is a single-strand molecule of RNA, as illustrated here. Typically, only a short segment of the viral genome (100–200 nucleotides) is amplified. Although it is possible to visualize the amplified DNA by gel electrophoresis as shown here, it is usually detected by rapid optical methods described later in the chapter (see Figure 8-64). Many other viruses that infect humans—such as HIV—are detected using this strategy.

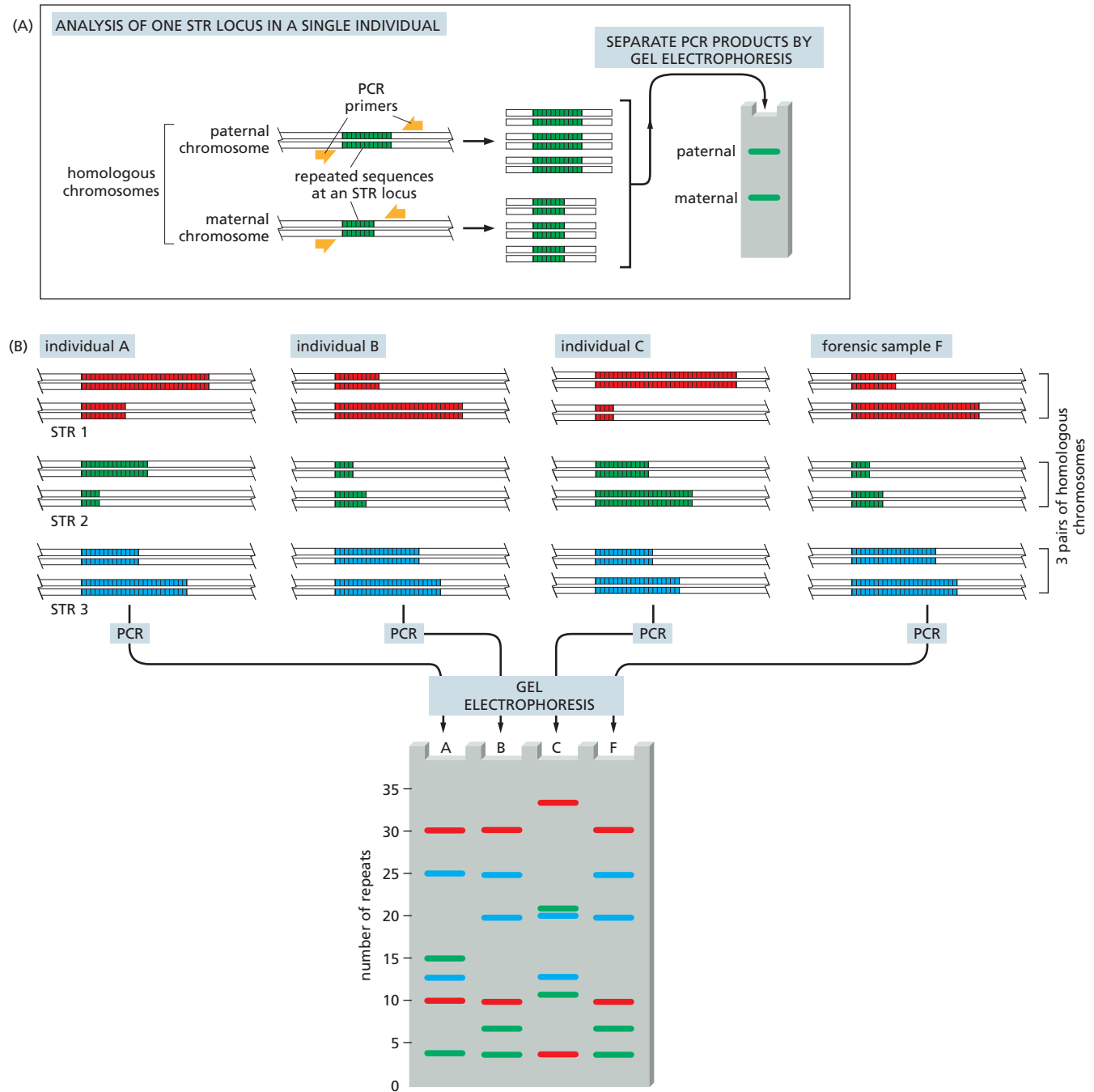


Figure 8–37 PCR is used in forensic science to distinguish one individual from another. The DNA sequences analyzed are short tandem repeats (STRs) composed of sequences such as CACACA... or GTGTGT.... STRs are found in various positions (loci) in the human genome. The number of repeats in each STR locus is highly variable in the population, ranging from 4 to 40 in different individuals. Because of the variability in these sequences, an individual will usually inherit a different number of repeats at each STR locus from his mother and from his father; two unrelated individuals, therefore, rarely contain the same pair of sequences at a given STR locus. (A) PCR using primers that recognize unique sequences on either side of one particular STR locus produces a pair of bands of amplified DNA from each individual, one band representing the maternal STR variant and the other representing the paternal STR variant. The length of the amplified DNA, and thus its position after gel electrophoresis, will depend on the exact number of repeats at the locus. (B) In the schematic example shown here, the same three STR loci are analyzed in samples from three suspects (individuals A, B, and C), producing six bands for each individual. Although different people can have several bands in common, the overall pattern is quite distinctive for each person. The band pattern can therefore serve as a DNA fingerprint to identify an individual nearly uniquely. The fourth lane (F) contains the products of the same PCR amplifications carried out on a hypothetical forensic DNA sample, which could have been obtained from a single hair or a tiny spot of blood left at a crime scene.

The more loci that are examined, the more confident one can be about the results. When examining the variability at 5–10 different STR loci, the odds that two random individuals would share the same fingerprint by chance are approximately 1 in 10 billion. In the case shown here, individuals A and C can be eliminated from inquiries, while B is a clear suspect. A similar approach is used routinely in paternity testing.

identify those who have done wrong, but also—equally important—to exonerate those who have been wrongfully accused.

PCR and Synthetic DNA Are Ideal Sources of Specific Gene Sequences for Cloning

PCR is a powerful technique for producing large amounts of specific DNA sequences *in vitro*. For many purposes—like those just described—there is no need to insert the PCR product in a bacterial plasmid for further analysis. Certain experiments, however, do require cloning of a specific DNA sequence in a plasmid for production in bacteria or other cells. Plasmids, for example, are important for the production and biochemical analysis of specific proteins, as we discuss later. In these cases, the desired DNA sequence can be readily produced by PCR amplification. Restriction nuclease cleavage sites in the PCR product are then cut to allow insertion into the appropriate plasmid as described earlier.

In recent years, a method called Gibson assembly (named for its inventor) has reduced the need for restriction nucleases in plasmid construction. A circular plasmid is cut open with a restriction nuclease that leaves blunt ends. In parallel, the desired DNA fragment is generated by PCR-mediated amplification from a genomic or cDNA source (see Figure 8–35). Importantly, the PCR primers are designed so that the ends of the PCR product contain 15–40 base pairs of the same sequence that surrounds the cut site in the plasmid. The PCR product and plasmid are then treated with a 5' exonuclease that degrades just one DNA strand at the ends of all DNA molecules, leaving single-strand sticky ends on the PCR product that will hybridize to complementary DNA on the plasmid (Figure 8–38). DNA polymerase then fills in the gaps, and DNA ligase joins the DNA strands. The resulting product is introduced into bacteria or other cell types as desired. This method can also be used to join a series of DNA fragments in a single plasmid. Each end of a DNA fragment must share sequence with the end of the fragment to which it will be joined. After treatment with 5' exonuclease, all the complementary sticky ends find each other and are thereby assembled in the correct order in the plasmid. Large genes can be assembled in this way from multiple subfragments.

Another key advance in recent years resulted from the rapid decline in the cost of methods for the chemical synthesis of DNA. Short single-strand DNA

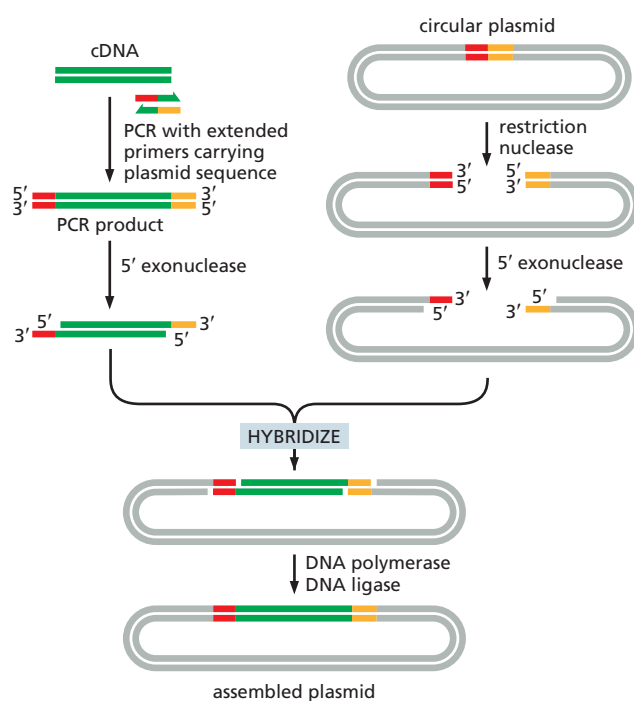


Figure 8–38 DNA cloning by Gibson assembly. It is often useful to insert a DNA fragment in a circular bacterial plasmid, as described earlier (see Figure 8–26). In Gibson assembly, as shown here, a plasmid is cleaved at a specific site with a restriction nuclease. In parallel, the DNA sequence to be cloned is amplified by PCR of a cDNA or other source. In addition to sequences complementary to the ends of the DNA to be amplified, the PCR primers include 15–40 nucleotides of sequence that matches the sequence on either side of the cut site in the linearized plasmid. The PCR product and plasmid are treated with a 5' exonuclease, which partially digests the DNA from the 5' end, resulting in single-strand 3' overhangs. These single-strand overhangs hybridize with their complementary sequence, neatly inserting the PCR product in the plasmid as shown. DNA polymerase is then added to fill in any gaps in the sequence, and DNA ligase seals the nicks to provide the fully assembled plasmid.

molecules, called *oligonucleotides*, are cheap to make and have been used for decades as probes and primers for PCR and other methods. Recently, it has become possible to cheaply synthesize much larger double-strand DNA molecules up to a few thousand base pairs in length. Thus, for many purposes, it is easier to order the desired DNA fragment from a DNA synthesis company than it is to produce it by PCR. If the ends of the synthetic DNA fragment are identical to those of a cut plasmid, then it is straightforward to insert the synthetic DNA in a plasmid by Gibson assembly (see Figure 8–38). Using multiple synthetic DNA fragments with interlocking ends, it is possible to assemble very large stretches of DNA (approaching the size of small genomes) from purely synthetic DNA.

We will see later in this chapter that the study of gene and protein function often requires methods to produce a mutant gene or protein that carries specific point mutations that alter its function. In molecular biology, the production of a mutant DNA sequence, called *site-directed mutagenesis*, is readily achieved through the clever application of PCR and synthetic DNA. If the goal is a single point mutation, then one of the two PCR primers can be designed to include the mutation while still having sufficient flanking sequence to hybridize to the non-mutant source DNA. PCR then generates a DNA product with the mutation near one end. This mutant DNA fragment can be assembled in a plasmid with other portions of the gene. If multiple mutations in a DNA sequence are needed, then the simplest approach is to purchase a synthetic DNA containing the mutations and insert that into the desired plasmid.

DNA Cloning Allows Any Protein to Be Produced in Large Amounts

Using the genetic code (and assuming all intron and exon boundaries are known), the amino acid sequence of any protein coded in a genome can be deduced. As was discussed earlier, this sequence can often provide an important clue to the protein's function if found to be similar to the amino acid sequence of a protein that has already been studied (see Figure 8–22). Although this strategy is often successful, it typically provides only the likely biochemical function of the protein; for example, whether the protein resembles a kinase or a protease. It usually remains for the experimenter to verify (or refute) this assignment and, most important, to discover the protein's biological function in the whole organism.

An important approach in determining gene function is to alter the gene (or in some cases, its expression pattern), place the altered copy back into the organism, and deduce the function of the normal gene by the changes caused by its alteration. Various techniques to implement this strategy are discussed in the next section of this chapter. But it is equally important to study the biochemical and structural properties of a gene product, as outlined earlier in this chapter. One of the most important contributions of DNA cloning to cell and molecular biology is the ability to produce any protein, even the rare ones, in nearly unlimited amounts. Such high-level production is usually carried out in living cells using expression vectors (Figure 8–39). These are generally plasmids that have been designed to produce a large amount of stable mRNA that can be efficiently translated into protein when the plasmid is introduced into bacterial, yeast, insect, or

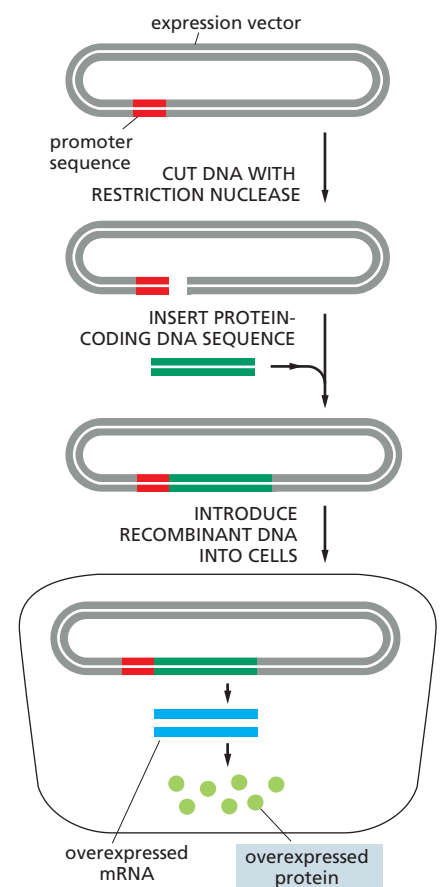
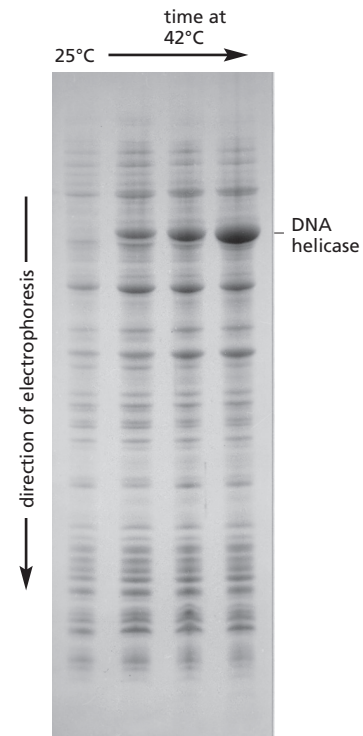


Figure 8–39 Production of large amounts of a protein from a protein-coding DNA sequence cloned into an expression vector and introduced into cells. A plasmid vector has been engineered to contain a highly active promoter, which causes unusually large amounts of mRNA to be produced from an adjacent protein-coding gene inserted into the plasmid vector. Depending on the characteristics of the cloning vector, the plasmid is introduced into bacterial, yeast, insect, or mammalian cells, where the inserted gene is efficiently transcribed and translated into protein. If the gene to be overexpressed has no introns (typical for genes from bacteria, archaea, and simple eukaryotes), it can simply be cloned from genomic DNA by PCR. For cloned animal and plant genes, it is often more convenient to obtain the gene as cDNA, either from a cDNA library (see Figure 8–30) or cloned directly by PCR from RNA isolated from the organism (see Figure 8–35). Alternatively, the DNA coding for the protein can be made by chemical synthesis.

Figure 8–40 Production of large amounts of a protein by using a plasmid expression vector. In this example, an expression vector that overproduces a DNA helicase has been introduced into bacteria. In this expression vector, transcription from this coding sequence is under the control of a viral promoter that becomes active only at a temperature of 37°C or higher. The total cell protein, either from bacteria grown at 25°C (no helicase protein made) or after a shift of the same bacteria to 42°C for up to 2 hours (helicase protein has become the most abundant protein species in the lysate), has been analyzed by SDS polyacrylamide-gel electrophoresis. (Courtesy of Kevin Hacker.)



mammalian cells. To prevent the high level of the foreign protein from interfering with the cell's growth, the expression vector is often designed to delay the synthesis of the foreign mRNA and protein until shortly before the cells are harvested and lysed (**Figure 8–40**).

Because the desired protein made from an expression vector is produced inside a cell, it must be purified away from the host-cell proteins by chromatography after cell lysis. However, because the protein is such a plentiful species in the cell (often 1–10% of the total cell protein), the purification is usually easy to accomplish in only a few steps. As we saw earlier, it is also possible to fuse a molecular tag—a cluster of histidine residues or a small marker protein—to the expressed protein to facilitate easy purification by affinity chromatography (see **Figure 8–11**). A variety of expression vectors is available, each engineered to function in the type of cell in which the protein is to be made.

This technology is also used to make large amounts of many medically useful proteins, including hormones (such as insulin and growth factors) used as pharmaceuticals, and viral proteins for use in vaccines. Expression vectors also allow scientists to produce many proteins of biological interest in large enough amounts for detailed structural studies. Nearly all three-dimensional protein structures depicted in this book are of proteins produced in this way. Recombinant DNA techniques thus allow scientists to move with ease from protein to gene, and vice versa, so that the functions of both can be explored on multiple fronts (**Figure 8–41**).

DNA Can Be Sequenced Rapidly by Dideoxy Sequencing

Most current methods of manipulating DNA, RNA, and proteins rely on prior knowledge of the nucleotide sequence of the genome of interest. But how are these sequences determined in the first place? In the late 1970s, researchers developed several strategies for determining the nucleotide sequence of any purified DNA fragment. The method that became the most widely used is called **dideoxy sequencing** or **Sanger sequencing** (named after the scientist who

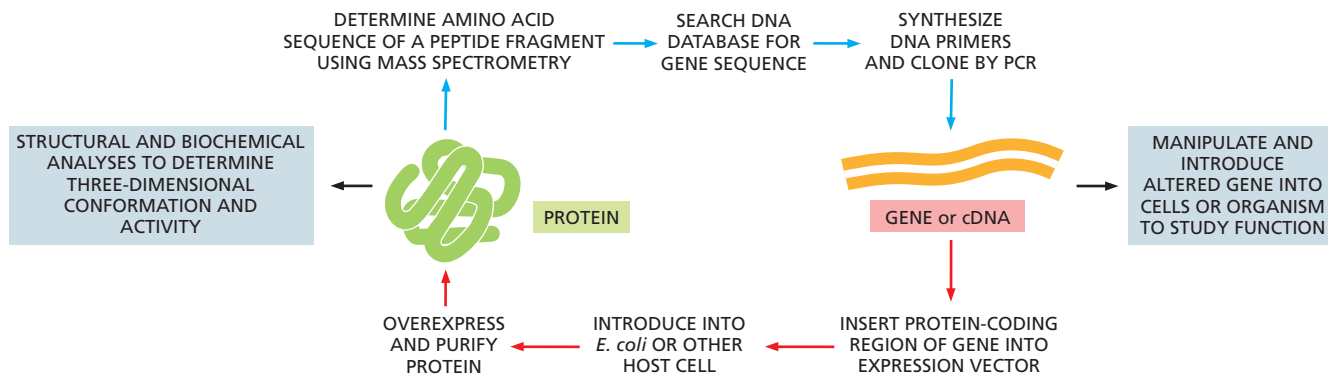


Figure 8–41 Recombinant DNA techniques make it possible to move experimentally from gene to protein and from protein to gene. If a gene has been identified (*right*), its protein-coding sequence can be inserted into an expression vector to produce large quantities of the protein (see **Figure 8–39**), which can then be studied biochemically or structurally. If a protein has been purified on the basis of its biochemical properties, mass spectrometry (see **Figure 8–18**) can be used to obtain a partial amino acid sequence, which is used to search a genome sequence for the corresponding nucleotide sequence. The complete gene can then be cloned by PCR from a sequenced genome (see **Figure 8–35**). The gene can also be manipulated and introduced into cells or organisms to study its function, a topic covered in the next section of this chapter.

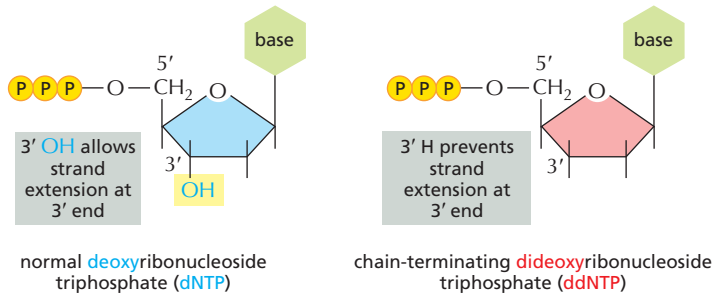


Figure 8-42 The dideoxy method of sequencing DNA relies on chain-terminating dideoxynucleoside triphosphates (ddNTPs). These ddNTPs are derivatives of the normal deoxyribonucleoside triphosphates (dNTPs) that lack the 3'-hydroxyl group. When incorporated into a growing DNA strand, they block further elongation of that strand.

invented it). This method uses DNA polymerase, along with special chain-terminating nucleotides called dideoxynucleoside triphosphates (**Figure 8-42**). Dideoxy sequencing reactions produce a collection of different DNA copies that terminate at every position in the original DNA sequence. In the original form of the method, four separate sequencing reactions were performed, each with a different dideoxynucleotide; the DNA copies were labeled with radioactivity and separated on polyacrylamide gels, which were then exposed to film to produce four ladders of bands that were read manually to reveal the sequence (see **Figure 8-24C**). This laborious method was replaced, beginning in the late 1980s, with technologies that are simpler, safer, and fully automated: robotic devices mix the reagents—including the four different chain-terminating dideoxynucleotides, each tagged with a different-colored fluorescent dye—and load the reaction samples onto long, thin capillary gels, which separate the reaction products into a series of distinct bands. A detector then records the color of each band, and a computer translates the information into a nucleotide sequence (**Figure 8-43**).

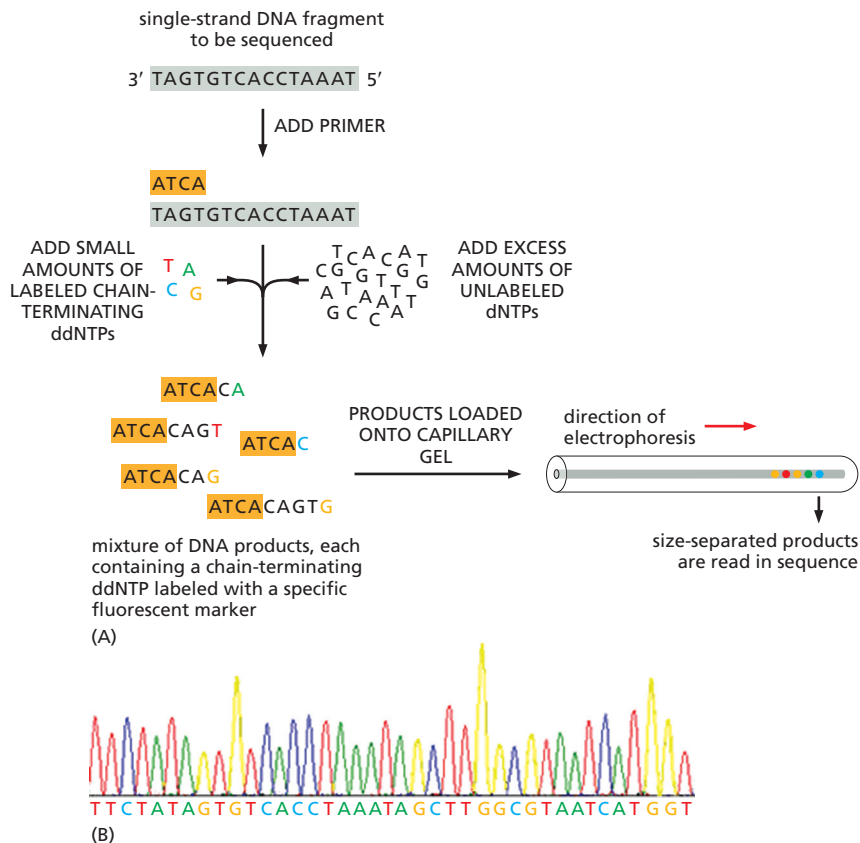


Figure 8-43 Automated dideoxy sequencing relies on a set of four ddNTPs, each bearing a uniquely colored fluorescent tag. (A) To determine the complete sequence of a single-strand fragment of DNA (gray), the DNA is first hybridized with a short DNA primer (orange). The DNA is then mixed with DNA polymerase (not shown), an excess amount of normal dNTPs, and a mixture containing small amounts of all four chain-terminating ddNTPs, each of which is labeled with a fluorescent tag of a different color. Because the chain-terminating ddNTPs will be incorporated only occasionally, each reaction produces a diverse set of DNA copies that terminate at different points in the sequence. The reaction products are loaded onto a long, thin capillary gel and separated by electrophoresis. A camera reads the color of each band on the gel and feeds the data to a computer that assembles the sequence (not shown). The sequence read from the gel will be complementary to the sequence of the original DNA molecule. (B) A tiny part of the data from such an automated sequencing run. Each colored peak represents a nucleotide in the DNA sequence.

Next-Generation Sequencing Methods Have Revolutionized DNA and RNA Analysis

Automated dideoxy sequencing was used in the late 1990s and early 2000s to determine the nucleotide sequences of many genomes, including those of *E. coli*, yeast, fruit flies, nematode worms, and humans. It continues to be used today as a low-cost approach to small-scale sequencing. But newer methods, developed since 2005, are now used for most large-scale genomic analysis. With these so-called *second-generation sequencing technologies*, the cost of sequencing DNA has decreased dramatically, and the number of sequenced genomes has increased enormously. These rapid methods allow multiple genomes to be sequenced in a matter of weeks, enabling investigators to examine thousands of individual human genomes, catalog the variation in nucleotide sequences from people around the world, and uncover the mutations that increase the risk of various diseases, from cancer to autism. These methods have also made it possible to determine the genome sequence of extinct species, including Neanderthal man and the woolly mammoth (**Movie 8.3**). By sequencing genomes from many closely related species, they have also helped us understand the molecular basis of key evolutionary events in the tree of life. The ability to rapidly sequence DNA has had major effects on all branches of biology, agriculture, and medicine; it is almost impossible to imagine where we would be without it.

Several second-generation sequencing methods are now in wide use. The most common is *Illumina sequencing*, named for the company that manufactures the equipment and reagents. This approach begins with the construction of libraries of small DNA fragments that represent the DNA of the entire genome. Instead of using bacterial cells to generate these libraries, they are made using PCR amplification of billions of DNA fragments, each attached to the glass surface of a flow cell. The amplification is carried out so that the PCR-generated copies of an original DNA fragment, instead of floating away in solution, remain bound in proximity to that original DNA fragment—resulting in a cluster of about 1000 identical copies of that small bit of the genome. These clusters—a billion of which can fit in a single flow cell—are then sequenced at the same time; that is, in parallel.

Sequencing is achieved using chain-terminating nucleotides with uniquely colored fluorescent tags. Unlike conventional dideoxy sequencing, however, the fluorescent tag and the chemical group that blocks elongation are both removable. Once DNA polymerase has added the fluorescent, chain-terminating nucleotide, a photo of the reaction records the color to reveal the identity of the nucleotide that was added. The colored label and the chain-terminating group are then removed, allowing DNA polymerase to add the next nucleotide (**Figure 8-44**). This cycle is repeated hundreds of times to provide the sequence of the DNA in each cluster. Billions of these clusters are sequenced in parallel. The full genome sequence is then reconstructed in the computer by stitching together the sequences of all fragments, using the overlaps between fragments as a guide.

Illumina sequencing provides short DNA sequences of a few hundred nucleotides, which can sometimes be difficult to assemble into a complete genome sequence because of the many repeated sequences that are often found in genomes. Recently developed *third-generation sequencing methods* are capable of sequencing much longer DNA molecules. Two methods are particularly promising. The first is *single-molecule real-time (SMRT)* sequencing, which is carried out in an array of tiny wells, each containing a single DNA polymerase anchored to its bottom surface. The key to SMRT sequencing is that it uses deoxyribonucleoside triphosphates in which the fluorescent dye is attached to the terminal phosphate. As the DNA polymerase copies the template DNA, the binding of a fluorescent nucleotide generates a color signal that reveals its identity. The signal disappears when the fluorescent terminal phosphate is released during incorporation of the nucleotide into the growing DNA chain (see **Figure 5-4**). The sequence of the DNA is thus revealed by the colors of the brief fluorescent pulses that appear as

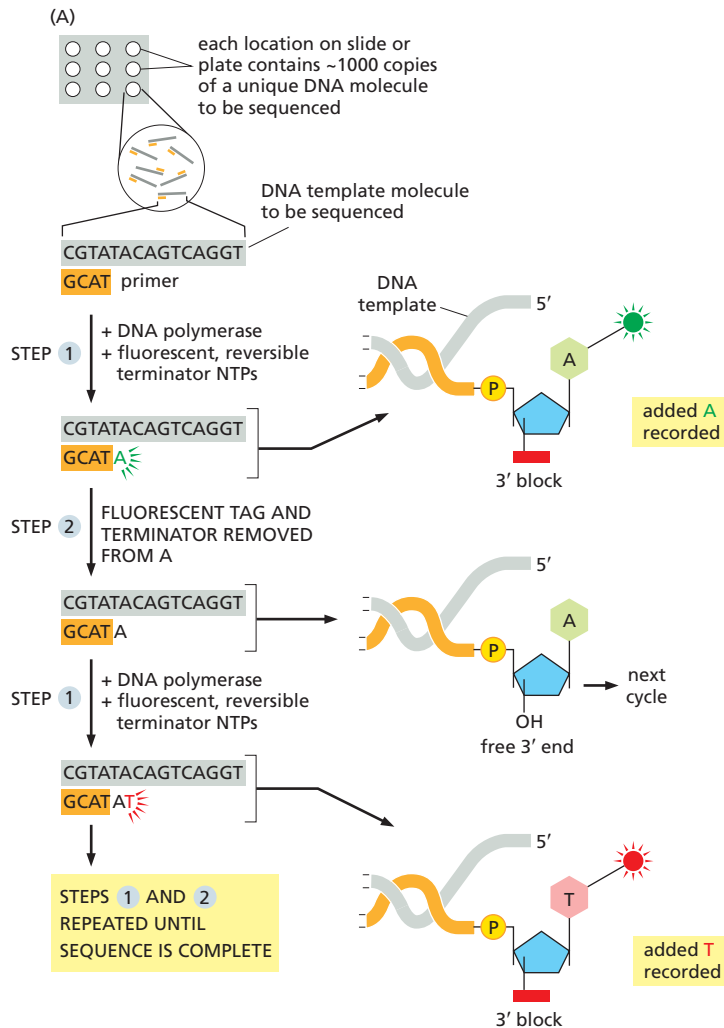


Figure 8–44 Principles of Illumina sequencing.

(A) A genome or other large DNA sample is broken into millions of short fragments. These fragments are attached to the surface of a flow cell and amplified by PCR to generate DNA clusters, each containing about a thousand copies of a single DNA fragment. The large number of clusters provides complete coverage of the genome. In the first step, the anchored DNA clusters are incubated with DNA polymerase and a special set of all four nucleoside triphosphates (NTPs), each with two reversible chemical modifications: a uniquely colored fluorescent marker and a 3' chemical group that terminates DNA synthesis. Normal dNTPs are not present. After a nucleotide is added by DNA polymerase, a high-resolution digital camera records the color of the fluorescence at each DNA cluster. In the second step, the DNA is chemically treated to remove the fluorescent markers and chemical blockers. A new batch of fluorescent, reversible terminator NTPs is then added to initiate another round of DNA synthesis. These steps are repeated until the sequence is complete. The snapshots of each round of synthesis are compiled by computer to yield the sequence of each DNA fragment. The sequence of the millions of overlapping DNA fragments can then be used to reconstruct the complete genome sequence. (B) An image of the surface of the Illumina flow cell, showing individual DNA clusters after a round of DNA synthesis with colored NTPs. (B, courtesy of Illumina, Inc.)

each nucleotide binds (Figure 8–45). Because very long DNA fragments (tens of kilobases) can be read by this method, and tens of thousands of reactions can be analyzed in parallel, complete genome sequences are well within reach in a short period of time. It is also possible to use circular DNA templates that are sequenced repeatedly on both strands, greatly improving the accuracy of the resulting sequence (see Figure 8–45C).

Another third-generation sequencing method, called *nanopore sequencing*, does not require DNA synthesis at all, but instead involves the transport of a single-strand DNA molecule through a tiny protein pore in a membrane. Voltage is applied across the membrane, resulting in current through the pore. The passage of nucleotides through the pore generates tiny shifts in electric current across the membrane, and the unique shape of each nucleotide base results in a slightly different disruption of the current. Measurement of these tiny current changes reveals the identity of each nucleotide as it passes through the pore. As in SMRT sequencing, extremely long DNAs can be sequenced in this manner. Another advantage is that modified nucleotides (such as 5-methylcytosine, depicted in Figure 7–46) can be identified because their effect on the current differs slightly from that of the unmodified nucleotide. Efforts are under way to allow direct sequencing of RNA by this approach as well. A major advantage of this method is that it can be performed with portable, handheld instruments that can be taken into the field, opening up exciting possibilities for DNA and RNA sequence analysis in global health and biology.

The development of cheaper and faster DNA sequencing methods has led to vast improvements in our ability to obtain and analyze genomic information. The

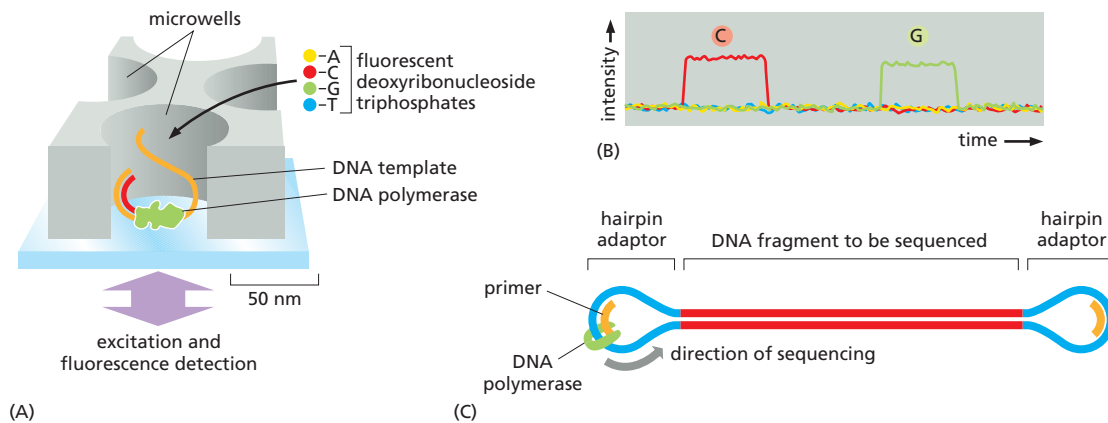


Figure 8-45 Single-molecule real-time (SMRT) sequencing. (A) SMRT sequencing uses a flow cell with thousands of tiny wells, each containing a single DNA polymerase, a single DNA template, and four fluorescently tagged deoxyribonucleoside triphosphates. Initial binding of a nucleotide to the template generates a local fluorescent signal that disappears when the terminal phosphates are removed during incorporation of the nucleotide into the DNA. To reduce background fluorescence from unbound nucleotides, only the bottom 30 nm of the well is illuminated, so that fluorescence is detected in a tiny volume (20 zeptoliters, or 20×10^{-21} liters). (B) Detection of fluorescent signals in the well reveals transient pulses of a single color, indicating the nucleotide that has been incorporated. (C) SMRT sequencing is often performed with a circular DNA template that is constructed by attaching hairpin adaptor DNAs to each end of the DNA to be sequenced. Using a primer that matches the adaptor, DNA polymerase can then replicate the template as shown in panel A. The enzyme used in this method is a *strand-displacing* polymerase that separates double-stranded DNA as it moves along the template, allowing it to continue around the entire circular molecule many times. Thus, both strands of the DNA are repeatedly sequenced, allowing the experimenter to eliminate sequence errors that arise from random mistakes made by the polymerase.

original “reference” sequence of the human genome, completed in 2003, cost more than \$1 billion and required many scientists from around the world working together for 13 years. The enormous progress made in the past 15 years has made it possible for a single person to complete the sequence of an individual human genome in less than a day, at a cost of less than \$1000.

As mentioned above, next-generation sequencing methods are being developed for the direct sequencing of RNA. Currently, however, RNA sequencing is typically carried out by converting the RNA to cDNA (using reverse transcriptase) and using one of the methods described above for DNA sequencing. It is important to keep in mind that although genomes remain the same from cell to cell and from tissue to tissue, the RNA produced from the genome can vary enormously. We will see later in this chapter that sequencing the entire repertoire of RNA from a cell or tissue (known as **deep RNA sequencing**, or **RNA-seq**) is a powerful way to understand how the information present in the genome is used by different cells under different circumstances. RNA-seq is also a valuable tool for annotating genomes, as we discuss next.

To Be Useful, Genome Sequences Must Be Annotated

Long strings of nucleotides, at first glance, reveal nothing about how this genetic information directs the development of a living organism—or even what types of DNA, protein, and RNA molecules are produced by a genome. The process of **genome annotation** attempts to mark out all the genes (both protein-coding and noncoding) in a genome and ascribe a role to each. It also seeks to understand more subtle types of genome information, such as the *cis*-regulatory sequences that specify the time and place that a given gene is expressed and whether its mRNA undergoes alternative splicing to produce different protein isotopes. Clearly, this is a daunting task, and we are far short of completing it for any form of life, even the simplest bacterium. For many organisms, we know the approximate number of genes, and, for very simple organisms, we understand the functions of about half their genes.

How does one begin to make sense of a genome sequence? The first step is usually to translate *in silico* the entire genome into protein. There are six different

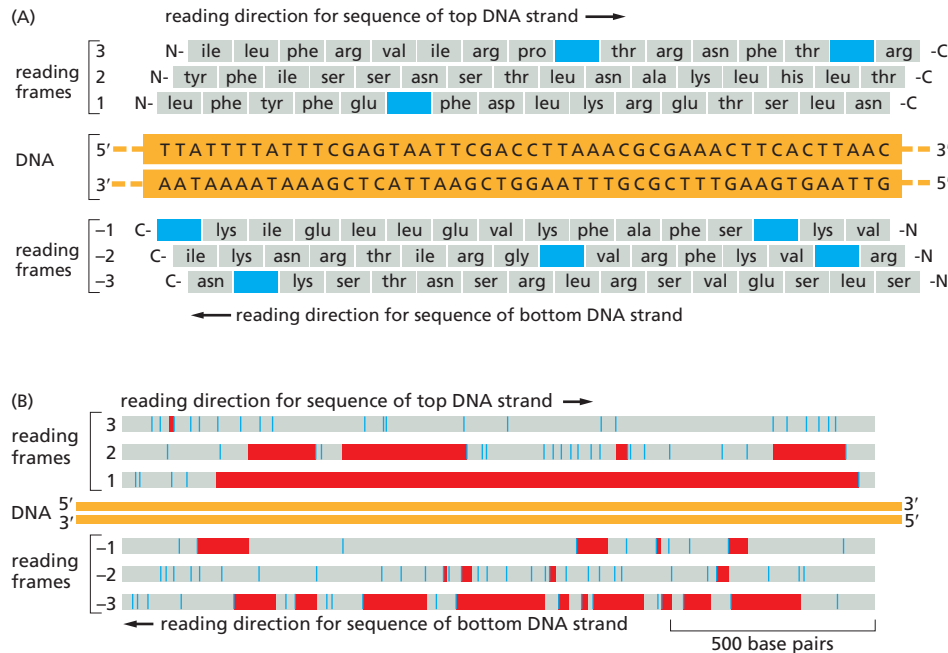


Figure 8-46 Finding the regions in a DNA sequence that encode a protein. (A) Any region of the DNA sequence can, in principle, code for six different amino acid sequences, because any one of three different reading frames can be used to interpret the nucleotide sequence on each strand. Note that a nucleotide sequence is always read in the 5'-to-3' direction and encodes a polypeptide from the N-terminus to the C-terminus. For a random nucleotide sequence read in a particular frame, a stop signal for protein synthesis is encountered, on average, about once every 20 amino acids. In this sample sequence of 48 base pairs, each such signal (*stop codon*) is colored *blue*, and only reading frame 2 lacks a stop signal. (B) Search of a 1700-base-pair DNA sequence for a possible protein-encoding sequence. The information is displayed as in panel A, with each stop signal for protein synthesis denoted by a *blue line*. In addition, all of the regions between possible start and stop signals for protein synthesis are displayed as *red bars*. Only reading frame 1 actually encodes a protein, which is 475 amino acid residues long.

reading frames for any piece of double-stranded DNA (three on each strand). We saw in Chapter 6 that a random sequence of nucleotides, read in frame, will contain a stop codon about every 20 amino acids. In contrast, protein-coding regions will usually contain much longer stretches without stop codons (**Figure 8-46**). Known as **open reading frames (ORFs)**, these usually signify bona fide protein-coding genes. This assignment is often “double-checked” by comparing the ORF amino acid sequence to the many databases of documented proteins from other species. If a match is found, even an imperfect one, it is very likely that the ORF will code for a functional protein (see **Figure 8-22**).

This strategy works very well for compact genomes, where intron sequences are rare and ORFs often extend for many hundreds of amino acids. However, in many animals and plants, the average exon size is 150–200 nucleotide pairs, and additional information is usually required to unambiguously locate all the exons of a gene. Although it is possible to search genomes for splicing signals and other features that help to identify exons (codon bias, for example), one of the most powerful methods is simply to sequence the total RNA produced from the genome in living cells. As can be seen in **Figure 7-3**, this RNA-seq information, when mapped onto the genome sequence, can be used to accurately locate all the introns and exons of even complex genes. By sequencing total RNA from different cell types, it is also possible to identify cases of alternative splicing.

RNA-seq also identifies noncoding RNAs produced by a genome. Although the function of some of them can be readily recognized (tRNAs or snoRNAs, for example), many have unknown functions and still others probably have no function at all. The existence of the many noncoding RNAs and our relative ignorance

of their function is the main reason that we know only the approximate number of genes in the human genome.

But even for protein-coding genes that have been unambiguously identified, we still have much to learn. Thousands of genomes have been sequenced, and we know from *comparative genomics* that many organisms share the same basic set of proteins. However, the functions of a very large number of identified proteins remain unknown. Depending on the organism, approximately one-third of the proteins encoded by a sequenced genome do not clearly resemble any protein that has been studied biochemically. This observation underscores a limitation of the emerging field of genomics: although comparative analysis of genomes reveals a great deal of information about the relationships between genes and organisms, it often does not provide immediate information about how these genes function or what roles they have in the physiology of an organism. Comparison of the full gene complement of several thermophilic bacteria, for example, does not reveal why these bacteria thrive at temperatures exceeding 70°C. And examination of the genome of the incredibly radiation-resistant bacterium *Deinococcus radiodurans* does not explain how this organism can survive a blast of radiation that can shatter glass. Further biochemical and genetic studies, like those described in the other sections of this chapter, are required to determine how genes, and the proteins they produce, function in the context of living organisms.

Summary

DNA cloning allows a copy of any specific part of a DNA or RNA sequence to be selected from the millions of other sequences in a cell and produced in unlimited amounts in pure form. DNA sequences can be amplified by inserting the desired DNA fragment into a self-replicating genetic element such as a bacterial plasmid. Bypassing cloning vectors and bacterial cells altogether, the polymerase chain reaction (PCR) allows DNA cloning to be performed directly with a DNA polymerase and DNA primers—provided that the DNA sequence of interest is already known.

The procedures used to obtain DNA clones that correspond in sequence to mRNA molecules are the same, except that a DNA copy of the mRNA sequence, called cDNA, is first made. Unlike genomic DNA clones, cDNA clones lack intron sequences, making them the clones of choice for analyzing the protein product of a gene.

Nucleic acid hybridization reactions provide a sensitive means of detecting any nucleotide sequence of interest. The enormous specificity of this hybridization reaction allows any single-strand sequence of nucleotides to be labeled with a radioisotope or chemical and used as a probe to find a complementary partner strand, even in a cell or cell extract that contains millions of different DNA and RNA sequences. DNA hybridization also makes it possible to use PCR to amplify any section of any genome once its sequence is known.

The nucleotide sequence of any genome can be determined rapidly and simply by using highly automated techniques that are based on several different strategies. Comparison of the genome sequences of different organisms allows us to trace the evolutionary relationships among genes and organisms, and it has proved valuable for discovering new genes and predicting their functions.

STUDYING GENE FUNCTION AND EXPRESSION

Ultimately, our goal is to understand how genes—and the proteins they encode—function in the intact organism. Although it may seem counterintuitive, one of the most direct ways to find out what a gene does is to see what happens to the organism when that gene is missing. Studying mutant organisms that have acquired changes or deletions in their nucleotide sequences is a time-honored practice in biology and forms the basis of the important field of **genetics**. Because mutations can disrupt cell processes, mutants often hold the key to understanding gene function. In the classical genetic approach, one begins by isolating mutants

that have an interesting or unusual appearance: fruit flies with white eyes or curly wings, for example. Working backward from the **phenotype**—the appearance or behavior of the individual—one then determines the organism's **genotype**, the form of the gene responsible for that characteristic (**Panel 8-1**).

Today, with numerous genome sequences available, the exploration of gene function often begins with a DNA sequence. Here, the challenge is to translate sequence into function. One approach, discussed earlier in the chapter, is to search databases for well-characterized proteins that have similar amino acid sequences to the protein encoded by a new gene. From there, the protein can be overexpressed and purified, and the methods described earlier in this chapter can be employed to study its biochemical properties and three-dimensional structure. But to determine directly a gene's function in a cell or organism, the most effective approach involves studying mutants that either lack the gene or express an altered version of it. Determining which cell processes have been disrupted or compromised in such mutants will usually shed light on a gene's biological role.

In this section, we describe several approaches to determining a gene's function, starting either from an individual with an interesting phenotype or from a DNA sequence. We begin with the classical genetic approach, which starts with a *genetic screen* for isolating mutants of interest and then proceeds toward identification of the gene or genes responsible for the observed phenotype. We then describe the set of techniques that are sometimes called reverse genetics, in which one begins with a gene or gene sequence and attempts to determine its function. This approach often involves some intelligent guesswork—searching for similar sequences in other organisms or determining when and where a gene is expressed—as well as generating mutant organisms and characterizing their phenotype.

Classical Genetic Screens Identify Random Mutants with Specific Abnormalities

Before the advent of gene cloning technology, most genes were identified by the abnormalities produced when the gene was mutated. Indeed, the very concept of the gene was deduced from the heritability of such abnormalities. This classical genetic approach—identifying the genes responsible for mutant phenotypes—is most easily performed in organisms that reproduce rapidly and are amenable to genetic manipulation, such as bacteria, yeasts, nematode worms, and fruit flies. Although spontaneous mutants can sometimes be found by examining extremely large populations—thousands or tens of thousands of individual organisms—isolating mutant individuals is much more efficient if one generates mutations with chemicals or radiation that damage DNA. By treating organisms with such mutagens, very large numbers of mutant individuals can be created quickly and then screened for a particular defect of interest.

An alternative approach to chemical or radiation mutagenesis is called *insertional mutagenesis*. This method relies on the fact that exogenous DNA inserted randomly into the genome can produce mutations if the inserted fragment interrupts a gene or its regulatory sequences. The inserted DNA, whose sequence is known, then serves as a molecular tag that aids in the subsequent identification and cloning of the disrupted gene (**Figure 8-47**). In *Drosophila*, the use of the transposable P element to inactivate genes has revolutionized the study of gene function in the fly. Transposable elements (see Table 5-4, p. 308) have also been used to generate mutations in bacteria, yeast, mice, and the flowering plant *Arabidopsis*.

Once a collection of mutants in a model organism has been produced, one generally must examine thousands of individuals to find the altered phenotype of interest. Such a search is called a **genetic screen**, and the larger the genome, the less likely it is that any particular gene will be mutated. Therefore, the larger the genome of an organism, the bigger the screening task becomes. The phenotype being screened for can be simple or complex. Simple phenotypes are easiest to detect: one can screen many organisms rapidly, for example, for mutations



Figure 8-47 Insertional mutant of the snapdragon, *Antirrhinum*. A mutation in a single gene coding for a regulatory protein causes leafy shoots (*left*) to develop in place of flowers, which occur in the normal plant (*right*). The mutation causes cells to adopt a character that would be appropriate to a different part of the normal plant, so instead of a flower, the cells produce a leafy shoot. (Courtesy of Enrico Coen and Rosemary Carpenter.)

GENES AND PHENOTYPES

Gene: a functional unit of inheritance, usually corresponding to the segment of DNA coding for a single protein.

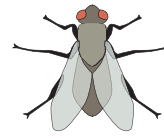
Genome: all of an organism's DNA sequences.

locus: the site of the gene in the genome

alleles: alternative forms of a gene

GENOTYPE: the specific set of alleles forming the genome of an individual

PHENOTYPE: the visible character of the individual



Wild type: the normal, naturally occurring type

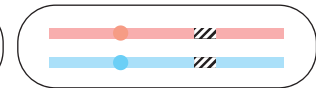
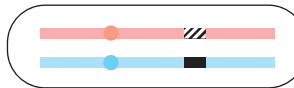


Mutant: differing from the wild type because of a genetic change (a mutation)

homozygous A/A

heterozygous a/A

homozygous a/a

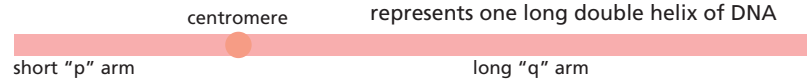


allele A is **dominant** (relative to a); allele a is **recessive** (relative to A)

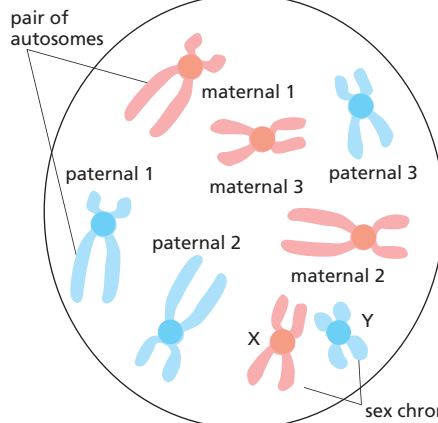
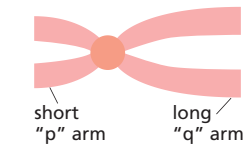
In the example above, the phenotype of the heterozygote is the same as that of one of the homozygotes; in cases where it is different from both, the two alleles are said to be codominant.

CHROMOSOMES

a chromosome at the beginning of the cell cycle, in G_1 phase; the single long bar represents one long double helix of DNA

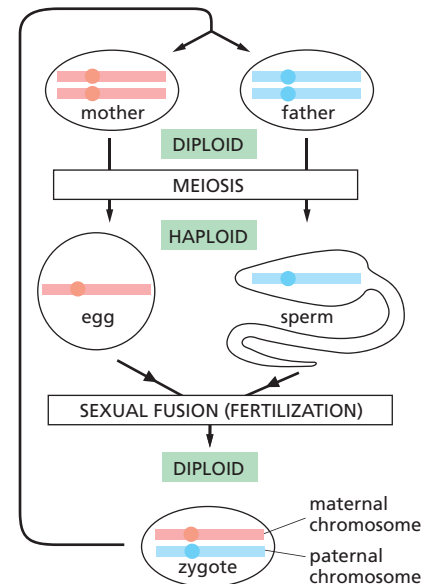


a chromosome near the end of the cell cycle, in metaphase; it is duplicated and condensed, consisting of two identical sister chromatids (each containing one DNA double helix) joined at the centromere.



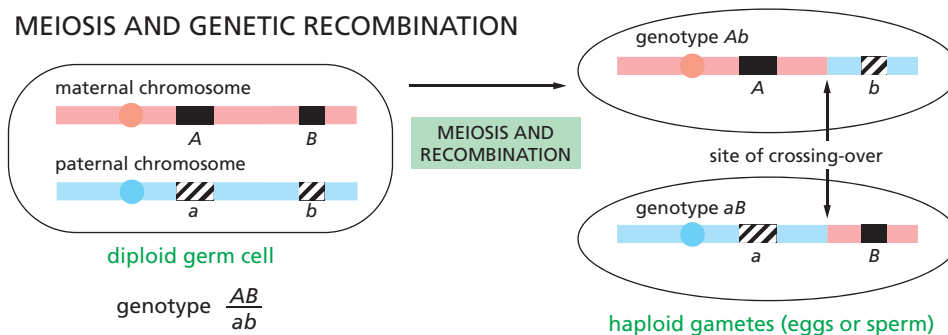
A normal diploid chromosome set, as seen in a metaphase spread, prepared by bursting open a cell at metaphase and staining the scattered chromosomes. In the example shown schematically here, there are three pairs of autosomes (chromosomes inherited symmetrically from both parents, regardless of sex) and two sex chromosomes—an X from the mother and a Y from the father. The numbers and types of sex chromosomes and their role in sex determination are variable from one class of organisms to another, as is the number of pairs of autosomes.

THE HAPLOID-DIPLOID CYCLE OF SEXUAL REPRODUCTION



For simplicity, the cycle is shown for only one chromosome/chromosome pair.

MEIOSIS AND GENETIC RECOMBINATION



The greater the distance between two loci on a single chromosome, the greater is the chance that they will be separated by crossing-over occurring at a site between them. If two genes are thus reassorted in $x\%$ of gametes, they are said to be separated on a chromosome by a **genetic map distance** of x map units (or x centimorgans).

TYPES OF MUTATIONS



POINT MUTATION: maps to a single site in the genome, corresponding to a single nucleotide pair or a very small part of a single gene



INVERSION: inverts a segment of a chromosome

lethal mutation: causes the developing organism to die prematurely.

conditional mutation: produces its phenotypic effect only under certain conditions, called the *restrictive* conditions. Under other conditions—the *permissive* conditions—the effect is not seen. For a *temperature-sensitive* mutation, the restrictive condition typically is high temperature, while the permissive condition is low temperature.

loss-of-function mutation: either reduces or abolishes the activity of the gene. These are the most common class of mutations. Loss-of-function mutations are usually *recessive*—the organism can usually function normally as long as it retains at least one normal copy of the affected gene.

null mutation: a loss-of-function mutation that completely abolishes the activity of the gene.



DELETION: deletes a segment of a chromosome



TRANSLOCATION: breaks off a segment from one chromosome and attaches it to another

gain-of-function mutation: increases the activity of the gene or makes it active in inappropriate circumstances; these mutations are usually *dominant*.

dominant-negative mutation: dominant-acting mutation that blocks gene activity, causing a loss-of-function phenotype even in the presence of a normal copy of the gene. This phenomenon occurs when the mutant gene product interferes with the function of the normal gene product.

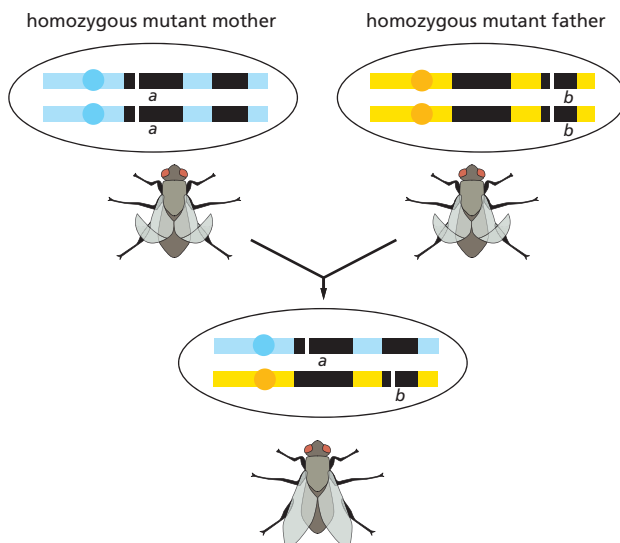
suppressor mutation: suppresses the phenotypic effect of another mutation, so that the double mutant seems normal.

An *intragenic* suppressor mutation lies within the gene affected by the first mutation; an *extragenic* suppressor mutation lies in a second gene—often one whose product interacts directly with the product of the first.

TWO GENES OR ONE?

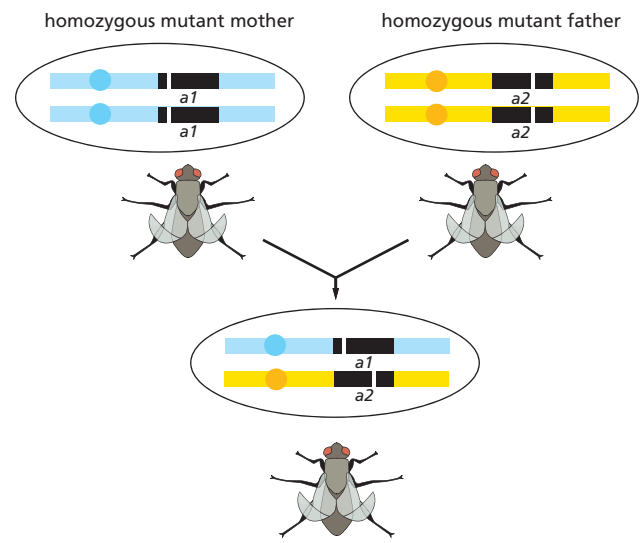
Given two mutations that produce the same phenotype, how can we tell whether they are mutations in the same gene? If the mutations are recessive (as they most often are), the answer can be found by a **complementation test**.

COMPLEMENTATION: MUTATIONS IN TWO DIFFERENT GENES



hybrid offspring shows normal phenotype:
one normal copy of each gene is present

NONCOMPLEMENTATION: TWO INDEPENDENT MUTATIONS IN THE SAME GENE



hybrid offspring shows mutant phenotype:
no normal copies of the mutated gene are present

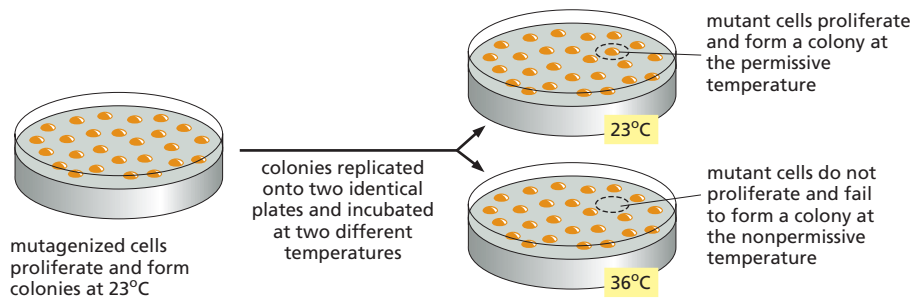


Figure 8–48 Screening for temperature-sensitive bacterial or yeast mutants.

Mutagenized cells are plated out at the permissive temperature. They divide and form colonies, which are transferred to two identical Petri dishes by replica plating. One of these plates is incubated at the permissive temperature, the other at the restrictive temperature. Cells containing a temperature-sensitive mutation in a gene essential for proliferation can divide at the normal (permissive) temperature but fail to divide at the elevated (restrictive) temperature.

that make it impossible for the organism to survive in the absence of a particular amino acid or nutrient.

Because defects in genes that are required for fundamental cell processes—RNA synthesis and processing or cell-cycle control, for example—are usually lethal, the functions of these genes are often studied in individuals with **conditional mutations**. The mutant individuals function normally as long as *permissive* conditions prevail but demonstrate abnormal gene function when subjected to *restrictive* (*nonpermissive*) conditions. In organisms with *temperature-sensitive mutations*, for example, the abnormality can be switched on and off experimentally simply by changing the ambient temperature; thus, a cell containing a temperature-sensitive mutation in a gene essential for survival will die at a restrictive temperature but proliferate normally at a permissive temperature (Figure 8–48). The temperature-sensitive gene in such a mutant usually contains a point mutation that causes a subtle change in its protein product; for example, the mutant protein may function normally at low temperatures but unfold at higher temperatures.

Temperature-sensitive mutations were crucial to find the bacterial genes that encode the proteins required for DNA replication. The mutants were identified by screening populations of mutagen-treated bacteria for cells that stop making DNA when they are warmed from 30°C to 42°C. These mutants were later used to identify and characterize the corresponding DNA replication proteins (discussed in Chapter 5). Similarly, screens for temperature-sensitive mutations in yeast led to the identification of many proteins involved in regulating the cell cycle, as well as many proteins involved in moving proteins through the secretory pathway. Related screening approaches demonstrated the function of enzymes involved in the principal metabolic pathways of bacteria and yeast (discussed in Chapter 2) and identified many of the gene products responsible for the orderly development of the *Drosophila* embryo (discussed in Chapter 21).

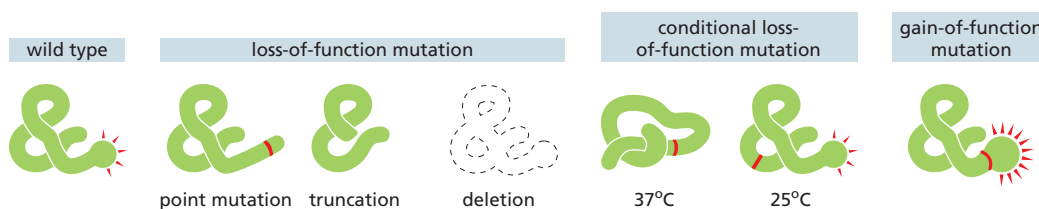
Mutations Can Cause Loss or Gain of Protein Function

Gene mutations are generally classed as *loss of function* or *gain of function*. A loss-of-function mutation results in a gene product that either does not work or works too little; thus, it can reveal the normal function of the gene. A gain-of-function mutation results in a gene product that works too much, works at the wrong time or place, or works in a new way (Figure 8–49).

An important early step in the genetic analysis of any mutant cell or organism is to determine whether the mutation causes a loss or a gain of function. A standard test is to determine whether the mutation is *dominant* or *recessive*. A

Figure 8–49 Gene mutations that affect their protein product in different ways.

In this example, the wild-type protein has a specific cell function denoted by the *red rays*. Mutations that eliminate this function or inactivate it at higher temperatures are shown. The conditional mutant protein carries an amino acid substitution (*red*) that prevents its proper folding at 37°C but allows the protein to fold and function normally at 25°C. Such temperature-sensitive conditional mutations are especially useful for studying essential genes (see Figure 8–48). In some cases, a mutation increases the activity of the mutant protein.



dominant mutation is one that still causes the mutant phenotype in the presence of a single copy of the wild-type gene. A recessive mutation is one that is no longer able to cause the mutant phenotype in the presence of a single wild-type copy of the gene. In the majority of cases, recessive mutations are loss of function and dominant mutations are gain of function—although cases have been described in which a loss-of-function mutation is dominant or a gain-of-function mutation is recessive. It is easy to determine if a mutation is dominant or recessive. One simply mates a mutant with a wild type to obtain diploid cells or organisms. The progeny from the mating will be heterozygous for the mutation. If the mutant phenotype is no longer observed, one can conclude that the mutation is recessive and is very likely to be a loss-of-function mutation (see Panel 8-1).

Complementation Tests Reveal Whether Two Mutations Are in the Same Gene or Different Genes

A large-scale genetic screen can turn up many different mutations that show the same phenotype. These defects might lie in different genes that function in the same process or they might represent different mutations in the same gene. Alternative forms of the same gene are known as **alleles**. The most common difference between alleles is a substitution of a single nucleotide pair, but different alleles can also bear deletions, substitutions, and duplications. How can we tell, then, whether two mutations that produce the same phenotype occur in the same gene or in different genes? If the mutations are recessive—if, for example, they represent a loss of function of a particular gene—a **complementation test** can be used to ascertain whether the mutations fall in the same gene or in different genes. To test complementation in a diploid organism, an individual that is homozygous for one mutation—that is, it possesses two identical alleles of the mutant gene in question—is mated with an individual that is homozygous for the other mutation. If the two mutations are in the same gene, the offspring show the mutant phenotype, because they still will have no normal copies of the gene in question. If, in contrast, the mutations fall in different genes, the resulting offspring show a normal phenotype, because they retain one normal copy (and one mutant copy) of each gene; the mutations thereby complement one another and restore a normal phenotype (**Figure 8-50**). Complementation testing of mutants identified during genetic screens has revealed, for example, that 5 different genes are required for yeast to digest the sugar galactose, 20 genes are needed for *E. coli* to build a functional flagellum, 48 genes are involved in assembling bacteriophage T4 viral particles, and hundreds of genes are involved in the development of an adult nematode worm from a fertilized egg.

Gene Products Can Be Ordered in Pathways by Epistasis Analysis

Once a set of genes involved in a particular biological process has been identified, it is helpful to determine the order in which the genes function. Gene order is perhaps easiest to explain for metabolic pathways, where, for example, enzyme A is necessary to produce the substrate for enzyme B. In this case, we would say that the gene encoding enzyme A acts before (upstream of) the gene encoding enzyme B in the pathway. Similarly, where one protein regulates the activity of another protein, we would say that the former gene acts before the latter. Gene order can, in many cases, be determined purely by genetic analysis without any knowledge of the mechanism of action of the gene products involved.

Suppose we have a biosynthetic process consisting of a sequence of steps, such that performance of step B requires completion of the preceding step A; and suppose gene A is required for step A, and gene B is required for step B. Then a null mutation (a mutation that abolishes function) in gene A will arrest the process at step A, regardless of whether gene B is functional or not, whereas a null mutation in gene B will cause arrest at step B only if gene A is still active. In such a case, gene A is said to be *epistatic* to gene B. By comparing the phenotypes of different combinations of mutations, we can therefore discover the order in which

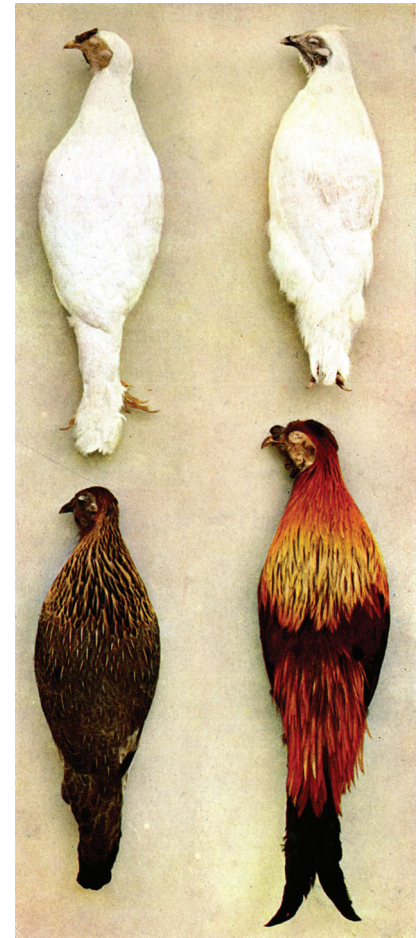


Figure 8-50 A complementation test can reveal that mutations in two different genes are responsible for the same abnormal phenotype. When an albino (white) bird from one strain is bred with an albino from a different strain, the resulting offspring (bottom) have normal coloration. This restoration of the wild-type plumage indicates that the two white breeds lack color because of recessive mutations in different genes. (From W. Bateson, *Mendel's Principles of Heredity*, 1st ed. Cambridge, UK: Cambridge University Press, 1913.)

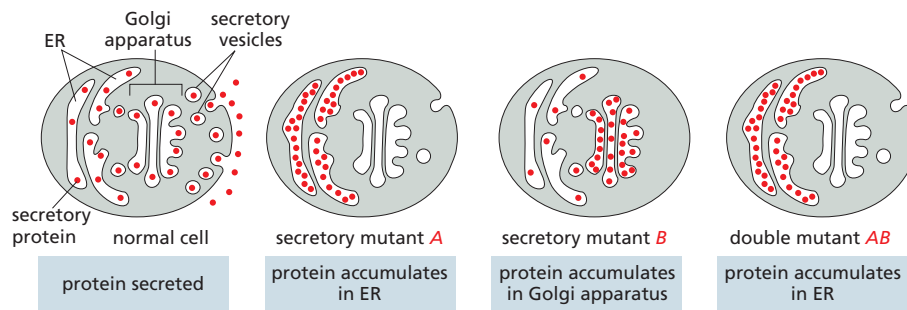


Figure 8–51 Using genetics to determine the order of function of genes. In normal cells, secretory proteins are loaded into vesicles, which fuse with the plasma membrane to secrete their contents into the extracellular medium. Two mutants, *A* and *B*, fail to secrete proteins. In mutant *A*, secretory proteins accumulate in the ER. In mutant *B*, secretory proteins accumulate in the Golgi. In the double mutant *AB*, proteins accumulate in the ER; this indicates that the gene defective in mutant *A* acts before the gene defective in mutant *B* in the secretory pathway.

the genes act. This type of analysis is called **epistasis analysis**. As an example, the pathway of protein secretion in yeast has been analyzed in this way. Different mutations in this pathway cause proteins to accumulate aberrantly in the endoplasmic reticulum (ER) or in the Golgi apparatus. When a yeast cell is engineered to carry both a mutation that blocks protein processing in the ER *and* a mutation that blocks processing in the Golgi apparatus, proteins accumulate in the ER. This indicates that proteins must pass through the ER before being sent to the Golgi before secretion (**Figure 8–51**). Strictly speaking, an epistasis analysis can only provide information about gene order in a pathway when both mutations are null alleles. When the mutations retain partial function, their epistasis interactions can be difficult to interpret.

Sometimes, a double mutant will show a new or more severe phenotype than either single mutant alone. This type of genetic interaction is called a *synthetic* phenotype, and if the phenotype is death of the organism, it is called *synthetic lethality*. In most cases, a synthetic phenotype indicates that the two genes act in two different parallel pathways, either of which is capable of mediating the same cell process. Thus, when both pathways are disrupted in the double mutant, the process fails altogether, and the synthetic phenotype is observed.

Mutations Responsible for a Phenotype Can Be Identified Through DNA Analysis

Once a collection of mutant organisms with interesting phenotypes has been obtained, the next task is to identify the gene or genes responsible for the altered phenotype. If the phenotype has been produced by insertional mutagenesis, locating the disrupted gene is fairly simple. DNA fragments containing the insertion (a transposon or a retrovirus, for example) are amplified by PCR, and the nucleotide sequence of the flanking DNA is determined. The gene affected by the insertion can then be identified by a computer-aided search of the complete genome sequence of the organism.

If a DNA-damaging chemical was used to generate the mutations, identifying the inactivated gene is often more laborious, but there are several powerful strategies available. With recent advances in DNA sequencing technology, it is possible to simply determine the genome sequence of the mutant organism and identify the affected gene by comparison with the wild-type sequence. Because of the continual accumulation of neutral mutations, there will probably be differences between the two genome sequences in addition to the mutation responsible for the phenotype. One way of proving that a mutation is causative is to introduce the putative mutation back into a normal organism and determine whether or not it causes the mutant phenotype. We will discuss how this is accomplished later in the chapter.

Rapid and Cheap DNA Sequencing Has Revolutionized Human Genetic Studies

Genetic screens in model experimental organisms have been spectacularly successful in identifying genes and relating them to various phenotypes, including many that are conserved between these organisms and humans. But how can we

study humans directly? They do not reproduce rapidly, cannot be treated with mutagens, and, if they have a defect in an essential process such as DNA replication, would die long before birth.

Despite their limitations compared to model organisms, humans are becoming increasingly attractive subjects for genetic studies. Because the human population is so large, spontaneous nonlethal mutations have arisen many times in all human genes. A substantial proportion of these mutations remains in the genomes of present-day humans. Deleterious mutations are discovered when the mutant individuals call attention to themselves by seeking medical help.

With the recent advances that have enabled the sequencing of entire human genomes cheaply and quickly, we can now identify such mutations and study their evolution and inheritance in ways that were impossible even a few years ago. By comparing the sequences of thousands of human genomes from all around the world, we can begin to identify directly the DNA differences that distinguish one individual from another. These differences hold clues to our evolutionary origins and can be used to explore the roots of disease.

Linked Blocks of Polymorphisms Have Been Passed Down from Our Ancestors

When we compare the sequences of multiple human genomes, we find that any two individuals will differ in roughly 1 nucleotide pair in 1000. As described in Chapter 4, most sequence variation results from substitution of a single nucleotide, called a *single-nucleotide variant* (SNV), while other variation is due to structural chromosome changes such as deletions and rearrangements. Human genetic studies have benefited greatly from a particularly common type of sequence variants, present in more than 1% of the population, that are called **polymorphisms**—most of which are **single-nucleotide polymorphisms**, or **SNPs** (Figure 8–52). Although these common variants can be found throughout the genome, they are not scattered randomly—or even independently. Instead, they tend to travel in groups called **haplotype blocks**—combinations of polymorphisms that are inherited as a unit.

To understand why such haplotype blocks exist, we need to consider our evolutionary history. It is thought that modern humans expanded from a relatively small population—perhaps around 10,000 individuals—that existed in Africa about 200,000 years ago. Among that small group of our ancestors, some individuals will have carried one set of genetic variants, others a different set. The chromosomes of a present-day human represent a shuffled combination of chromosome segments from different members of this small ancestral group of people. Because only about 2000 generations separate us from them, large segments of these ancestral chromosomes have passed from parent to child, unbroken by the crossover events that occur during meiosis. As described in Chapter 5, only a few crossovers occur between each set of homologous chromosomes during each meiosis (see Figure 5–52).

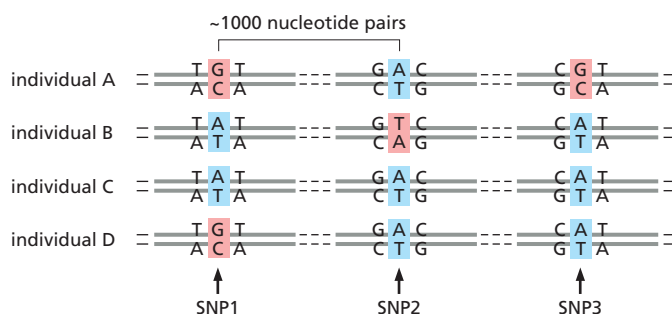


Figure 8–52 Single-nucleotide polymorphisms (SNPs) are sites in the genome where two or more alternative choices of a nucleotide are common in the population. Most such variations in the human genome occur at locations where they do not significantly affect a gene's function.

As a result, certain sets of DNA sequences—and their associated polymorphisms—have been inherited in linked groups, with little genetic rearrangement across the generations. These are the haplotype blocks. Like genes that exist in different allelic forms, haplotype blocks also come in a limited number of variants that are common in the human population, each representing a combination of DNA polymorphisms passed down from a particular ancestor long ago.

Sequence Variants Can Aid the Search for Mutations Associated with Disease

Mutations that give rise, in a reproducible way, to rare but clearly defined differences, such as albinism, hemophilia, or congenital deafness, can often be identified by studies of affected families. Such single-gene, or monogenic, disorders are referred to as *Mendelian* because their pattern of inheritance is easy to track. Moreover, individuals who inherit the causative mutation will exhibit the disorder irrespective of environmental factors such as diet or exercise. But for many common disorders, the genetic roots are more complex. Instead of a single allele of a single gene, such disorders stem from a combination of contributions from multiple genes. And often, environmental factors have strong influences on the severity of the disorder. For these *multigenic* conditions, such as diabetes or arthritis, population studies are often helpful in tracking down the genes that increase the risk of getting the disease.

In population studies, investigators collect DNA samples from a large number of people who have the disease and compare them to samples from a group of people who do not have the disease. They look for variants—SNPs, for example—that are more common among the people who have the disease. Because DNA sequences that are close together on a chromosome tend to be inherited together, the presence of such SNPs could indicate that an allele that increases the risk of the disease might lie nearby (Figure 8–53). Although, in principle, the disease could be caused by the SNP itself, the culprit is much more likely to be a change that is merely linked to the SNP as part of a haplotype block.

Such *genome-wide association studies* have been used to search for genes that predispose individuals to common diseases, including diabetes, coronary artery disease, rheumatoid arthritis, and even depression. For many of these conditions, the DNA polymorphisms identified increase the risk of disease only slightly. Moreover, environmental factors (diet and exercise, for example) play an important role in the onset and severity of the disease. Nonetheless, the identification of potential disease genes linked to polymorphisms is leading to a mechanistic understanding of some of our most common disorders.

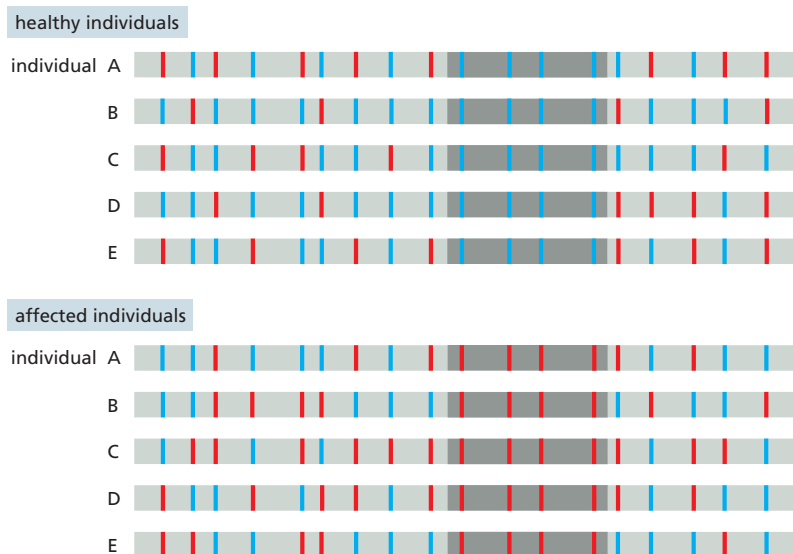


Figure 8–53 Genes that affect the risk of developing a common disease can often be tracked down through linkage to SNPs. Here, the patterns of SNPs are compared between two sets of individuals—a set of healthy controls and a set affected by a particular common disease. A segment of a typical chromosome is shown. For most polymorphic sites in this segment, it is a random matter whether an individual has one SNP variant (*red vertical bars*) or another (*blue vertical bars*); this same randomness is seen both for the control group and for the affected individuals. However, in the part of the chromosome that is shaded in *dark gray*, a bias is seen: most healthy individuals have the blue SNP variants, whereas most affected individuals have the red SNP variants. This suggests that this region contains or is close to a gene that is genetically linked to these red SNP variants and which predisposes individuals to the disease. Using carefully selected controls and thousands of affected individuals, this approach can help track down disease-related genes, even when they confer only a slight increase in the risk of developing the disease.

Genomics Is Accelerating the Discovery of Rare Mutations That Predispose Us to Serious Disease

The polymorphisms that have allowed us to identify some of the genes that increase our risk of disease are common. They arose long ago in our evolutionary past and are now present, in one form or another, in a substantial portion of the population. Such polymorphisms are thought to account for about 90% of the differences between one person's genome and another. But when we try to tie these common variants to differences in disease susceptibility or other heritable traits, such as height, we find that they do not have as much predictive power as we had anticipated: thus, for example, most confer relatively small increases—less than twofold—in the risk of developing a common disease.

Part of the problem is that many of the mutations that are directly responsible for complex human diseases appeared more recently in our evolutionary history—during a period when the human population expanded from the few million individuals who existed 10,000 years ago to the more than 7 billion who exist today. Because recent mutations occur more rarely than the ancient polymorphisms that are common in the human population, they could slip through the genome-wide association studies just described.

In contrast to polymorphisms, rare DNA variants—those much less frequent in humans than SNPs—can have large effects on the risk of developing some common diseases. For example, numerous loss-of-function mutations, each individually rare, have been found to increase greatly the predisposition to autism and schizophrenia. Many of these are *de novo* mutations, which arose spontaneously in the germ-line cells of one or the other parent. The fact that these mutations arise spontaneously with some frequency could help explain why these common disorders—each observed in about 1% of the population—remain with us, even though the affected individuals might leave few or no descendants. These rare mutations can arise in any one of hundreds of different genes, which could explain much of the clinical variability of autism and schizophrenia.

Now that DNA sequencing has become fast and inexpensive, the most efficient way to identify these rare, large-effect mutations is by comparing the genomes of large numbers of affected individuals with those of unaffected controls. When the key variants are identified, the major challenge is then to determine how they affect the individuals who carry them and how small variations in multiple genes produce the disease phenotype.

The Cellular Functions of a Known Gene Can Be Studied with Genome Engineering

As we have seen, classical genetics starts with a mutant phenotype and identifies the mutations, and consequently the genes, responsible for it. Recombinant DNA technology has made possible a different type of genetic approach that is used widely in a variety of species. Instead of beginning with a mutant organism and using it to identify a gene and its protein, an investigator can start with a particular gene and proceed to make mutations in it, creating mutant cells or organisms so as to analyze the gene's function. Because this approach reverses the traditional direction of genetic discovery—proceeding from genes to mutations, rather than vice versa—it is sometimes referred to as *reverse genetics*. And because the genome of the organism is deliberately altered in a particular way, this approach is also called *genome engineering* or *genome editing*. We shall see in this chapter that this approach can be scaled up so that whole collections of organisms can be created, each of which has a different gene altered.

There are several ways a gene of interest can be altered. In the simplest, the gene can simply be deleted from the genome, although in a diploid organism, this requires that both copies—one on each chromosome homolog—be deleted. Such *gene knockouts* are especially useful if the gene is not essential. The gene in question (even if it is essential) can also be replaced by one that is expressed in the wrong tissue or at the wrong time in development; this type of manipulation

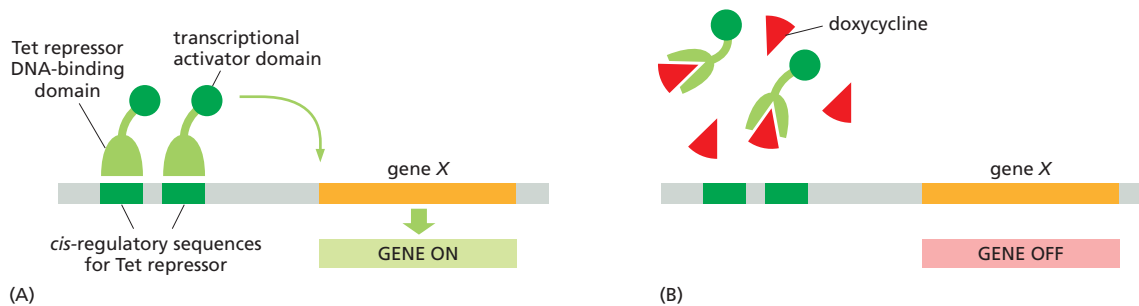


Figure 8-54 Engineered genes can be turned on and off with small molecules. Here, the DNA-binding portion of a bacterial protein (the tetracycline, Tet, repressor) has been fused to a portion of a mammalian transcriptional activator and expressed in cultured mammalian cells. The engineered gene X, present in place of the normal gene, has its usual gene control region replaced by *cis*-regulatory sequences recognized by the tetracycline repressor. In the absence of doxycycline (a particularly stable version of tetracycline), the engineered gene is expressed; in the presence of doxycycline, the gene is turned off because the drug causes the tetracycline repressor to dissociate from the DNA. This strategy can also be used in mice by incorporating the engineered genes into the germ line. In many tissues, the gene can be turned on and off simply by adding doxycycline to or removing it from the animal's water. If the tetracycline repressor construct is placed under the control of a tissue-specific gene control region, the engineered gene will be turned on and off only in that tissue.

often provides important clues to the gene's normal function. In a particularly powerful approach, a gene of interest can be modified to be expressed at will by the experimenter (Figure 8-54). Finally, genes can also be engineered so that they are expressed normally in most cell types and tissues but deleted in certain cell types or tissues selected by the experimenter (see Figure 5-66). This approach is especially useful when a gene has different roles in different tissues.

It is also possible to make subtler changes to a gene. It is sometimes useful to make slight changes in a protein's structure so that one can begin to dissect which portions of a protein are important for its function. The activity of an enzyme, for example, can be studied by changing a single amino acid in its active site. It is also possible, through genome engineering, to create new types of proteins in an animal. For example, a gene can be fused to the gene for a fluorescent protein. When this altered gene is introduced into the genome, the protein can be tracked in the living organism by monitoring its fluorescence.

Altered genes can be created in several ways. Perhaps the simplest is to chemically synthesize the DNA that makes up the gene. In this way, the investigator can specify any type of variant of the normal gene. It is also possible to construct altered genes using recombinant DNA technology, as described earlier in this chapter. Once obtained, altered genes can be introduced into cells in a variety of ways. DNA can be microinjected into mammalian cells with a glass micropipette or introduced by a virus that has been engineered to carry foreign genes. In plant cells, genes can be introduced by a technique called particle bombardment: DNA samples are painted onto tiny gold beads and then literally shot through the cell wall with a specially modified gun. *Electroporation* is sometimes used for introducing DNA into bacteria and some other cells. In this technique, a brief electric shock renders the cell membrane temporarily permeable, allowing foreign DNA to enter the cytoplasm.

To be most useful to experimenters, the altered gene, once it is introduced into a cell, must recombine with the cell's genome so that the normal gene is replaced. In simple organisms such as bacteria and yeasts, this process occurs with high frequency using the cell's own homologous recombination machinery, as described in Chapter 5. In more complex organisms that have elaborate developmental programs, the procedure is more complicated because the altered gene must be introduced into the germ line, as we next describe.

Animals and Plants Can Be Genetically Altered

Animals and plants that have been genetically engineered by gene deletion or gene replacement are called **transgenic organisms**, and any foreign or modified genes that are added are called **transgenes**. We discuss transgenic plants later

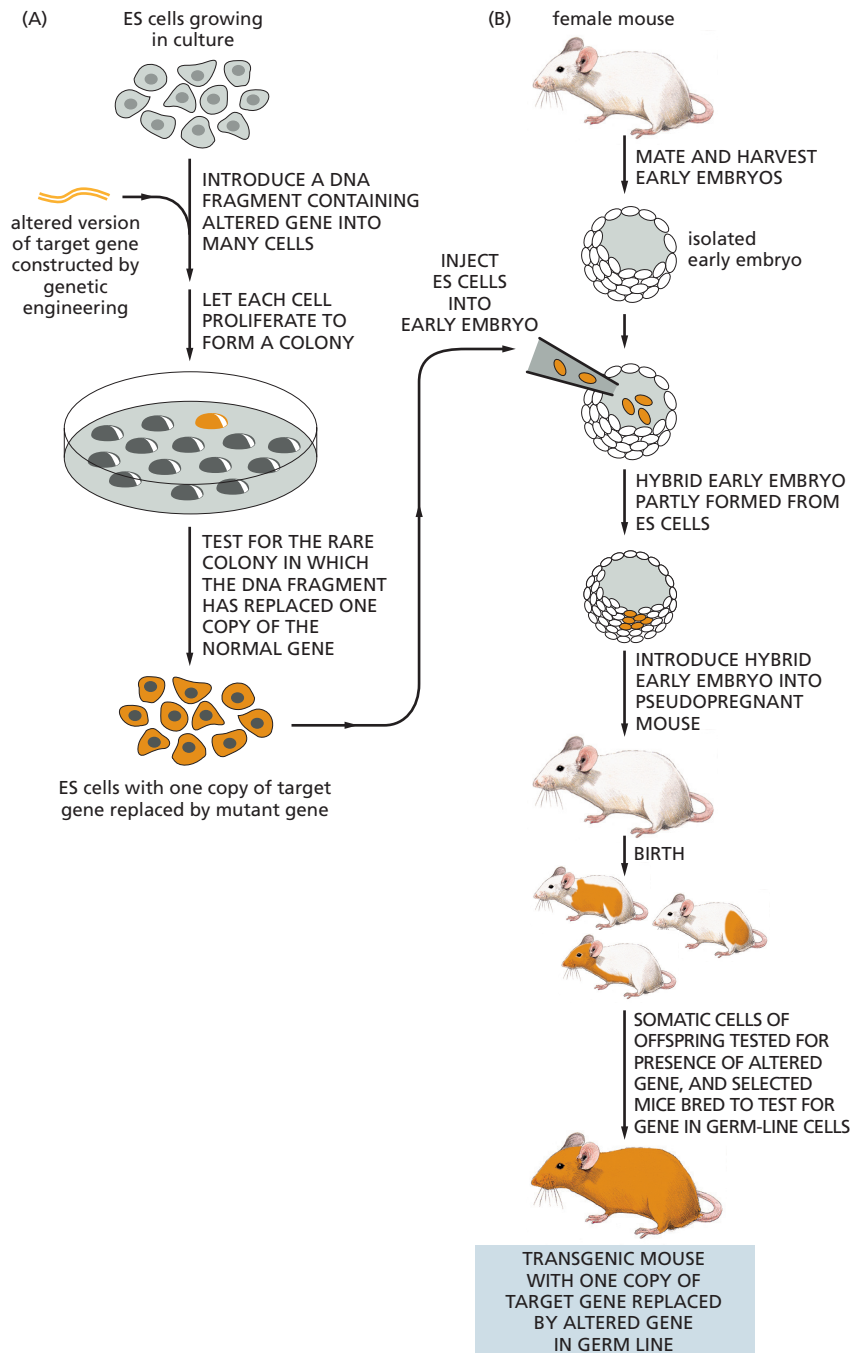


Figure 8-55 Summary of the procedures used for making gene replacements in mice. (A) In the first step, an altered version of the gene is introduced into cultured embryonic stem (ES) cells (described in Chapter 22). Only a few ES cells will have their corresponding normal genes replaced by the altered gene through a homologous recombination event. These cells can be identified by PCR and cultured to produce many descendants, each of which carries an altered gene in place of one of its two normal corresponding genes. (B) In the next step of the procedure, these altered ES cells are injected into a very early mouse embryo; the cells are incorporated into the growing embryo, and a mouse produced by such an embryo will contain some somatic cells (indicated by *orange*) that carry the altered gene. Some of these mice will also contain germ-line cells that contain the altered gene; when bred with a normal mouse, some of the progeny of these mice will contain one copy of the altered gene in all of their cells.

The mice with the transgene in their germ line are then bred to produce both a male and a female animal, each heterozygous for the gene replacement (that is, they have one normal and one mutant copy of the gene). When these two mice are mated (not shown), one-fourth of their progeny will be homozygous for the altered gene.

in this chapter and, for now, concentrate our discussion on transgenic mice. If a DNA molecule carrying a mutated mouse gene is transferred into a mouse cell, it is possible to direct the mutant gene to replace the normal gene by homologous recombination. By exploiting these gene-targeting events, any specific gene can be altered or inactivated in a mouse cell by a direct gene replacement. In the case in which both copies of the gene of interest are completely inactivated or deleted, the resulting animal is called a *knockout* mouse. The technique is summarized in **Figure 8-55**.

The ability to prepare transgenic mice lacking a known normal gene was a major advance, and the technique has been used to determine the functions of many mouse genes (**Figure 8-56**). If the gene functions in early development, a knockout mouse will usually die before it reaches adulthood. These lethal defects can be carefully analyzed to help determine the function of the missing gene.

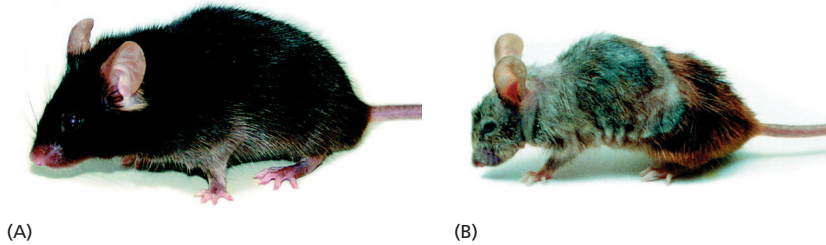


Figure 8-56 Transgenic mice engineered to express a mutant DNA helicase show premature aging. The helicase, encoded by the *Xpd* gene, is involved in both transcription and DNA repair. Compared with a wild-type mouse of the same age (A), a transgenic mouse that expresses a defective version of *Xpd* (B) exhibits many of the symptoms of premature aging, including osteoporosis, emaciation, early graying, infertility, and reduced life span. The mutation in *Xpd* used here impairs the activity of the helicase and mimics a mutation that in humans causes trichothiodystrophy, a disorder characterized by brittle hair, skeletal abnormalities, and a very reduced life expectancy. These results indicate that an accumulation of DNA damage can contribute to the aging process in both humans and mice. (From J. de Boer et al., *Science* 296:1276–1279, 2002. With permission from AAAS.)

As described in Chapter 5, an especially useful type of transgenic animal takes advantage of a site-specific recombination system to excise—and thus disable—the target gene in a particular place or at a particular time (see Figure 5-66). In this case, the target gene in embryonic stem (ES) cells is replaced by a fully functional version of the gene that is flanked by a pair of the short DNA sequences, called *lox sites*, that are recognized by the *Cre recombinase* protein. The transgenic mice that result are phenotypically normal. They are then mated with transgenic mice that express the *Cre recombinase* gene under the control of an inducible promoter. In the specific cells or tissues in which *Cre* is switched on, it catalyzes recombination between the *lox* sequences—excising a target gene and eliminating its activity (see Figure 22-7).

The Bacterial CRISPR System Has Been Adapted to Edit Genomes in a Wide Variety of Species

One of the difficulties in making transgenic mice by the procedure just described is that the introduced DNA molecule (bearing the experimentally altered gene) often inserts at random in the genome, and many ES cells must therefore be screened individually to find one that has the correct gene replacement.

Creative use of the CRISPR system, discovered in bacteria as a defense against viruses, has largely solved this problem. As described in Chapter 7, the CRISPR system uses a guide RNA sequence to target (through complementary base-pairing) double-stranded DNA, which it then cleaves (see Figure 7-81). The gene coding for the key component of this system, the bacterial Cas9 protein, has been transferred into a variety of organisms, where it greatly simplifies the process of making transgenic organisms (Figure 8-57A and B). The basic strategy is as follows: Cas9 protein is expressed in cultured cells along with a guide RNA designed by the experimenter to target a particular location on the genome. The Cas9 and guide RNA associate, the complex is brought to the matching sequence on the genome, and the Cas9 protein makes a double-strand break. As we saw in Chapter 5, these breaks are usually repaired by nonhomologous end joining, which often results in small sequence errors or deletions that disrupt gene function. In many cases, this repair process is sufficient to inactivate the gene, particularly if it produces a frameshift near the beginning of the coding sequence. If the goal is a precise gene knockout or replacement, then Cas9 and guide RNA can be co-expressed in ES cells with an altered homologous gene sequence, which the cell uses to repair the double-strand break by homologous recombination. In this way, the normal gene can be selectively damaged by the CRISPR system and replaced at high efficiency by an experimentally altered gene.

The CRISPR system has a variety of other uses. Its particular power lies with its ability to target Cas9 to thousands of different positions across a genome through the simple rules of complementary base-pairing. Thus, if a catalytically inactive Cas9 protein is fused to a transcription activator or repressor, it is possible, in principle, to turn any gene on or off by providing a guide RNA that matches a unique sequence in the gene promoter (Figure 8-57C and D; Movie 8.4).

The CRISPR system has several advantages over other strategies for experimentally manipulating gene expression. First, it is relatively easy for the experimenter to design the guide RNA: it simply follows standard base-pairing

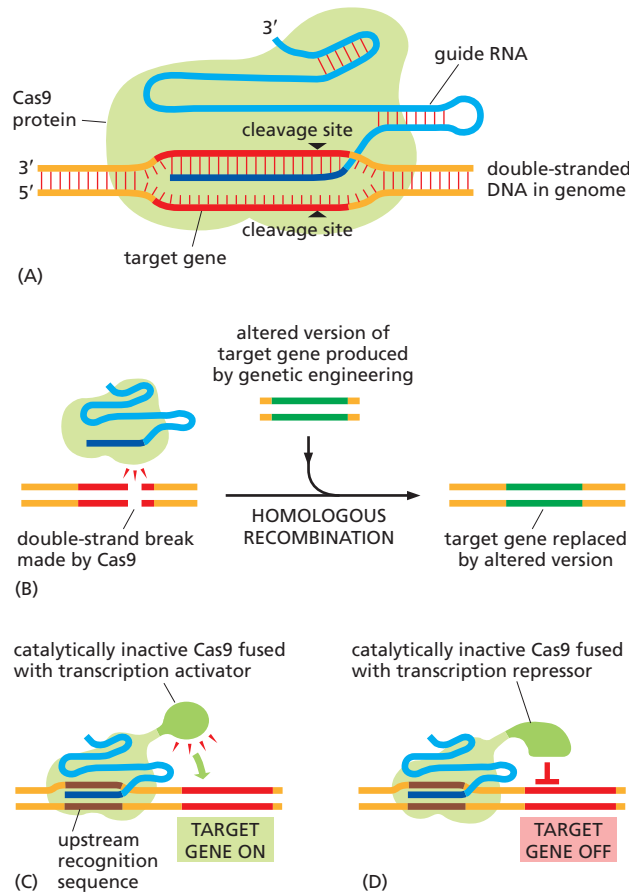


Figure 8-57 Use of CRISPR to study gene function in a wide variety of species. (A) The Cas9 protein (artificially expressed in the species of interest) binds to a guide RNA, designed by the experimenter and also expressed. The portion of RNA in *light blue* is needed for associations with Cas9; that in *dark blue* is specified by the experimenter to match a position on the genome. The only other requirement is that the adjacent genome sequence includes a short PAM (protospacer adjacent motif, not shown) that is needed for Cas9 to cleave. As described in Chapter 7, this sequence allows the CRISPR system in a bacterium to distinguish its own genome from that of invading viruses. (B) When Cas9 is directed to make a double-strand break in a gene, the break is generally repaired by nonhomologous end joining (not shown), which introduces local sequence errors that can disrupt gene function. A more precise mutation can be made as shown here, where the double-strand break is repaired by homologous recombination with an altered gene provided by the experimenter. (C, D) By using a mutant form of Cas9 that can no longer cleave DNA, Cas9 can be used to activate a normally dormant gene (C) or turn off an actively expressed gene (D). (Adapted from P. Mali et al., *Nat. Methods* 10:957–963, 2013.)

convention. Second, the gene to be controlled does not have to be modified; the CRISPR strategy exploits DNA sequences already present in the genome. Third, numerous genes can be controlled simultaneously. Cas9 has to be expressed only once, but many guide RNAs can be expressed in the same cell; this strategy allows the experimenter to turn on or off a whole set of genes at once.

The export of the CRISPR system from bacteria to virtually all other experimental organisms (including mice, zebrafish, worms, flies, rice, and wheat) has revolutionized the study of gene function. Like the earlier discovery of restriction nucleases, this breakthrough came from scientists studying a fascinating phenomenon in bacteria without—at first—realizing the enormous impact these discoveries would have on all aspects of biology.

Large Collections of Engineered Mutations Provide a Tool for Examining the Function of Every Gene in an Organism

Extensive collaborative efforts have produced comprehensive libraries of mutations in a variety of model organisms, including *S. cerevisiae*, *Caenorhabditis elegans*, *Drosophila*, *Arabidopsis*, and even the mouse. The ultimate aim in each case is to produce a collection of mutant strains in which every gene in the organism—one at a time—has been systematically deleted or altered in such a way that it can be conditionally disrupted. Collections of this type provide an invaluable resource for investigating gene function on a genomic scale. For example, a large collection of mutant organisms can be screened for a particular phenotype. Like the classic genetic approaches described earlier, this is one of the most powerful ways to identify the genes responsible for a particular phenotype. Unlike the classical genetic approach, however, the set of mutants is “pre-engineered,” so that there is no need to rely on chance events such as

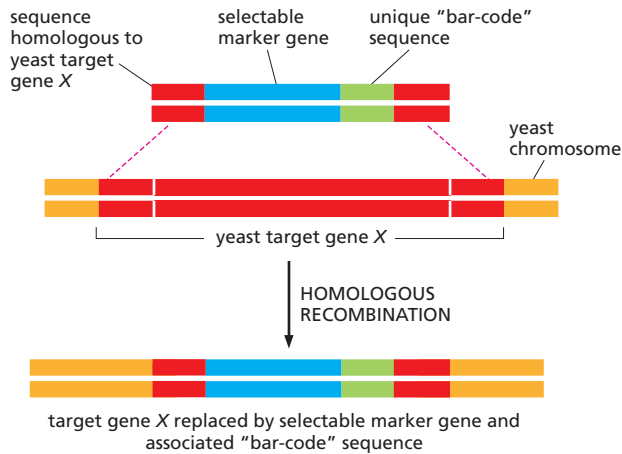


Figure 8–58 Making bar-coded collections of mutant organisms. A deletion construct for use in yeast contains DNA sequences (red) homologous to each end of a target gene X, a selectable marker gene (blue), and a unique “bar-code” sequence approximately 20 nucleotide pairs in length (green). This DNA is introduced into yeast cells, where it readily replaces the target gene by homologous recombination. Cells that carry a successful gene replacement are identified by expression of the selectable marker gene, typically a gene that provides resistance to a drug. By using a collection of such constructs, each specific for one gene, a library of yeast mutants was constructed containing a mutant for every gene. Essential genes cannot be studied this way, as their deletion from the genome causes the cells to die. In this case, the target gene is replaced by a version of the gene that can be regulated by the experimenter (see Figure 8–54). The gene can then be turned off, and the effect of this can be monitored before the cells die.

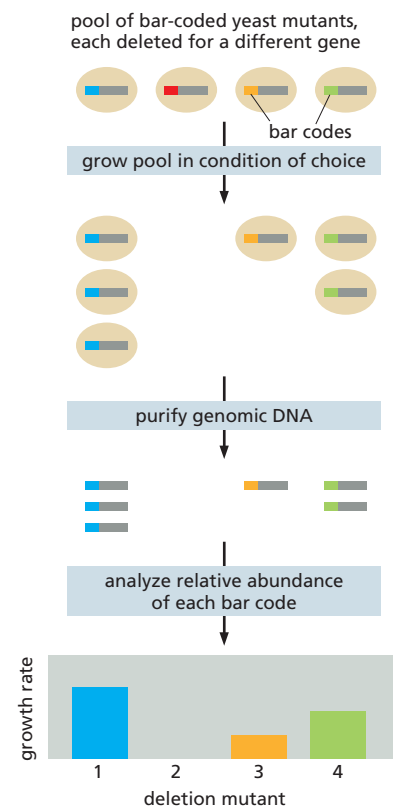
spontaneous mutations or transposon insertions. In addition, each of the individual mutations within the collection is often engineered to contain a distinct molecular “bar code”—in the form of a unique DNA sequence—designed to make identification of the altered gene rapid and routine (Figure 8–58).

In *S. cerevisiae*, the task of generating a complete set of 6000 mutants, each missing only one gene, was accomplished in the early 2000s. Because each mutant strain has an individual bar-code sequence embedded in its genome, a large mixture of engineered strains can be grown under various selective test conditions—such as nutritional deprivation, a temperature shift, or the presence of various drugs—and the cells that survive can be rapidly identified by the unique sequence tags present in their genomes. By assessing how well each mutant in the mixture fares, one can begin to discern which genes are essential, useful, or irrelevant for growth under the various conditions (Figure 8–59).

Similar methods can be applied to human cells using the CRISPR system described earlier. Using viral expression vectors, a large library of different guide RNAs can be expressed in a cell population, such that only one guide RNA is expressed in each cell, along with Cas9. After growth of the cells under various conditions, surviving cells are subjected to genomic sequencing to measure the abundance of guide RNAs in the population. Guide RNAs that target genes essential for survival will disappear from the population, whereas those that enhance survival will be enriched—providing important clues about the function of those genes under the conditions tested.

The insights generated by examining mutant libraries can be considerable. For example, studies of an extensive collection of mutants in *Mycoplasma genitalium*—the organism with the smallest known genome—have identified the minimum complement of genes essential for cellular life. Growth under laboratory conditions requires about three-quarters of the 480 protein-coding genes in *M. genitalium*. Approximately 100 of these essential genes are of unknown function, which suggests that a surprising number of the basic molecular mechanisms that underlie life have yet to be discovered.

Figure 8–59 Genome-wide screens for fitness using a large pool of bar-coded yeast deletion mutants. A large pool of yeast mutants, each with a different gene deleted and present in equal amounts, is grown under conditions selected by the experimenter. Some mutants (blue) grow normally, but others show reduced growth (orange and green) or no growth at all (red). The fitness of each mutant is experimentally determined in the following way. After the growth phase is completed, genomic DNA (isolated from the mixture of strains) is purified, and the relative abundance of each mutant is determined by quantifying the level of the DNA bar code matched to each deletion. This can be done by sequencing the pooled genomic DNA. In this way, the contribution of every gene to growth under the specified condition can be rapidly ascertained. This type of study has revealed that of the approximately 6000 coding genes in yeast, only about 1000 are essential under standard growth conditions.



Collections of mutant organisms are also available for many animal and plant species. For example, it is possible to “order,” by phone or e-mail from a consortium of investigators, a deletion or insertion mutant for almost all coding genes in *Drosophila*. Likewise, a nearly complete set of mutants exists for the model plant *Arabidopsis*. And the adaptation of the CRISPR system for use in mice means that, in the near future, we can expect to be able to turn on or off—at will—each gene in the mouse genome at different stages of development. Although we are still ignorant about the function of most genes in most organisms, these technologies allow an exploration of gene function on a scale that was unimaginable a decade ago.

RNA Interference Is a Simple and Rapid Way to Test Gene Function

Although knocking out (or conditionally expressing) a gene in an organism and studying the consequences is the most powerful approach for understanding the functions of the gene, *RNA interference* (RNAi, for short) is an alternative, particularly convenient approach. As discussed in Chapter 7, this method exploits a natural mechanism used in many plants, animals, and fungi to protect themselves against viruses and transposable elements. The technique introduces into a cell or organism a double-strand RNA molecule whose nucleotide sequence matches that of part of the gene to be inactivated. After the RNA is processed, it hybridizes with the target-gene RNA (either mRNA or noncoding RNA) and reduces its expression by the mechanisms shown in Figure 7–78.

RNAi is frequently used to inactivate genes in *Drosophila* and mammalian cell culture lines. Indeed, a set of 15,000 *Drosophila* RNAi molecules (one for every coding gene) allows researchers, in several months, to test the role of every fly gene in any process that can be monitored using cultured cells. RNAi has also been widely used to study gene function in whole organisms, including the nematode *C. elegans*. When working with worms, introducing the double-stranded RNA is quite simple: either the RNA can be injected directly into the intestine of the worm or the worm can be fed with *E. coli* engineered to produce the RNA (Figure 8–60). The RNA is amplified and distributed throughout the body of the worm, where it inhibits expression of the target gene in different tissue types. RNAi is being used to help in assigning functions to the entire complement of worm genes (Figure 8–61).

A related technique has also been applied to mice. In this case, the RNAi molecules are not injected or fed to the mouse; rather, recombinant DNA techniques are used to make transgenic animals that express the RNAi under the control of an inducible promoter. Often this is a specially designed RNA that can fold back on itself and, through base-pairing, produce a double-strand region that is recognized by the RNAi machinery. In the simplest cases, the process inactivates only the genes that exactly match the RNAi sequence. Depending on the

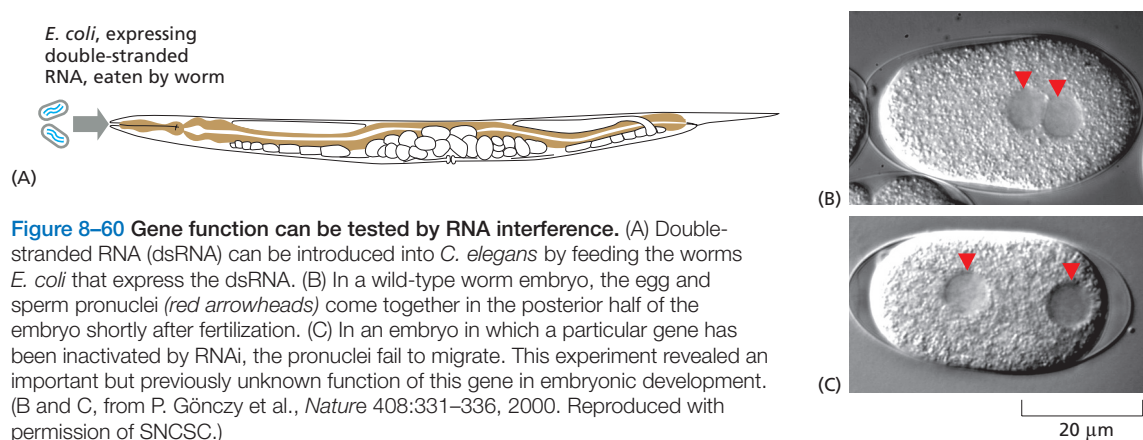


Figure 8–60 Gene function can be tested by RNA interference. (A) Double-stranded RNA (dsRNA) can be introduced into *C. elegans* by feeding the worms *E. coli* that express the dsRNA. (B) In a wild-type worm embryo, the egg and sperm pronuclei (red arrowheads) come together in the posterior half of the embryo shortly after fertilization. (C) In an embryo in which a particular gene has been inactivated by RNAi, the pronuclei fail to migrate. This experiment revealed an important but previously unknown function of this gene in embryonic development. (B and C, from P. Gönczy et al., *Nature* 408:331–336, 2000. Reproduced with permission of SNCSC.)

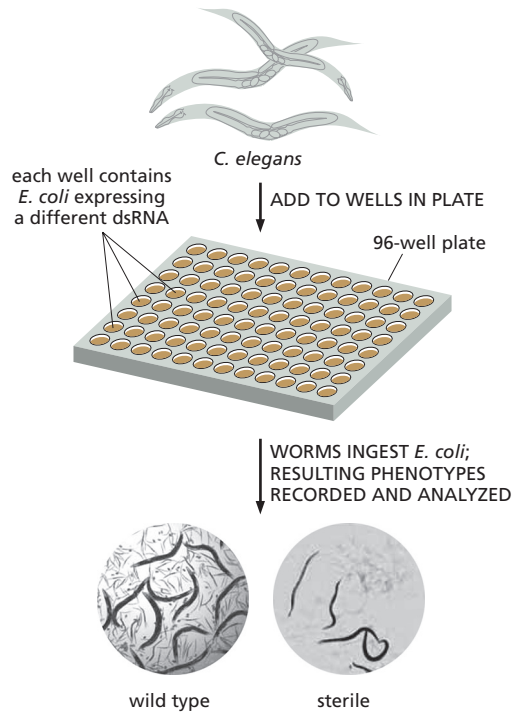


Figure 8–61 RNA interference provides a convenient method for conducting genome-wide genetic screens. In this experiment, each well in this 96-well plate is filled with *E. coli* that produce a different double-stranded RNA than that produced by *E. coli* in other wells. Each interfering RNA matches the nucleotide sequence of a single *C. elegans* gene, thereby inactivating it. About 10 worms are added to each well, where they ingest the genetically modified bacteria. The plate is incubated for several days, which gives the RNAs time to inactivate their target genes—and the worms time to grow, mate, and produce offspring. The plate is then examined in a microscope, which can be controlled robotically, to screen for genes that affect the worms' ability to survive, reproduce, develop, and behave. Shown here are normal worms alongside worms that show an impaired ability to reproduce because of inactivation of a particular "fertility" gene. (From B. Lehner et al., *Nat. Genet.* 38:896–903, 2006. With permission from Nature.)

inducible promoter used, the RNAi can be produced only in a specified tissue or only at a particular time in development, allowing the functions of the target genes to be analyzed in elaborate detail.

RNAi is a simple and efficient tool for analysis of gene function in many organisms, but it has several potential limitations compared with true genetic knockouts. For unknown reasons, RNAi does not efficiently inactivate all genes. Moreover, within whole organisms, certain tissues may be resistant to the action of RNAi (for example, neurons in nematodes). Another problem arises because many organisms contain large gene families, the members of which exhibit sequence similarity. RNAi therefore sometimes produces "off-target" effects, inactivating related genes in addition to the targeted gene. One strategy to avoid such problems is to use multiple small RNA molecules matched to different regions of the same gene. Ultimately, the results of any RNAi experiment must be viewed as a strong clue to, but not necessarily a proof of, normal gene function.

Reporter Genes Reveal When and Where a Gene Is Expressed

We have just discussed some of the approaches that can be used to assess a gene's function in cultured cells or, even better, in the intact organism. Although this information is crucial to understanding gene function, it does not generally reveal the molecular mechanisms through which the gene product works in the cell. For example, genetics on its own rarely tells us all the places in the organism where the gene is expressed or how its expression is controlled. It does not necessarily reveal whether the gene acts in the nucleus, the cytosol, on the cell surface, or in one of the numerous other compartments of the cell. And it does not reveal how a gene product might change its location or its expression pattern when the external environment of the cell changes. Key insights into gene function can be obtained by simply observing when and where a gene is expressed. A variety of approaches, most involving some form of genetic engineering, can easily provide this critical information.

As discussed in detail in Chapter 7, *cis*-regulatory DNA sequences, located upstream or downstream of the coding region, control gene transcription. These regulatory sequences, which determine precisely when and where the gene is

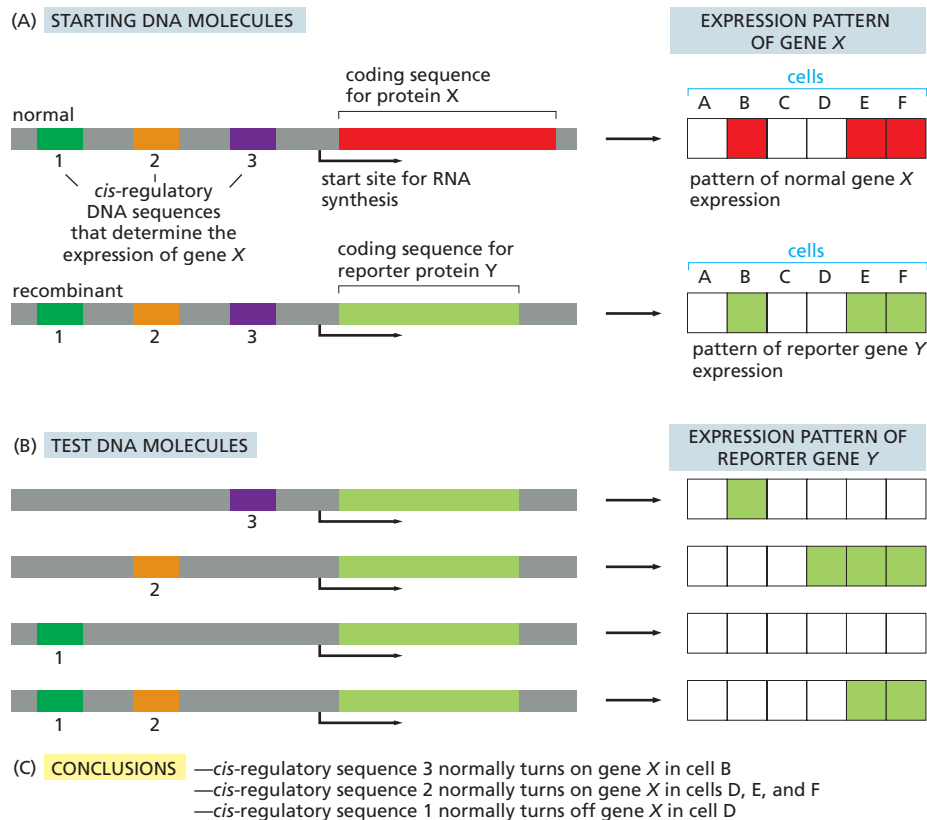


Figure 8–62 Using a reporter protein to determine the pattern of a gene's expression. (A) In this example, the coding sequence for protein X is replaced by the coding sequence for reporter protein Y. The expression patterns for X and Y are the same. (B) Various fragments of DNA containing candidate *cis*-regulatory sequences are added in combinations to produce test DNA molecules encoding reporter gene Y. These recombinant DNA molecules are then tested for expression after introducing them into a variety of different types of mammalian cells. The results are summarized in (C). For experiments in eukaryotic cells, two commonly used reporter proteins are the enzyme β -galactosidase (β -gal) (see Figure 7–30) and green fluorescent protein (GFP) (see Figure 9–16).

expressed, can be easily studied by placing a reporter gene under their control and introducing these recombinant DNA molecules into cells (Figure 8–62). In this way, the normal expression pattern of a gene can be determined, as well as the contribution of individual *cis*-regulatory sequences in establishing this pattern.

Reporter genes also allow any protein to be tracked over time in living cells. Here, the reporter gene typically encodes a fluorescent protein, often **green fluorescent protein (GFP)**, the molecule that gives luminescent jellyfish their greenish glow. The GFP is simply attached—in the coding frame—to the protein-coding gene of interest. The resulting *GFP fusion protein* often behaves in the same way the normal protein does, and its location can be monitored by fluorescence microscopy, a topic that is discussed in the next chapter (see Figure 9–10). GFP fusion has become a standard strategy for tracking not only the location but also the movement of specific proteins in living cells.

In Situ Hybridization Can Reveal the Location of mRNAs and Noncoding RNAs

It is also possible to directly observe the time and place that an RNA product of a gene is expressed using *in situ* hybridization. For protein-coding genes, this strategy often provides the same general information as the reporter gene approaches described above; however, it is crucial for genes whose final product is RNA rather than protein. We encountered *in situ* hybridization earlier in the chapter (see Figure 8–32); it relies on the basic principles of nucleic acid hybridization. Typically, tissues are gently fixed so that their RNA is retained in an exposed form that can hybridize with a labeled complementary DNA or RNA probe. In this way, the patterns of differential gene expression can be observed in tissues, and the location of specific RNAs can be determined (Figure 8–63). An advantage of *in situ* hybridization over other approaches is that genetic engineering is not required. Thus, it is often simpler and faster and can be used for genetically intractable species.

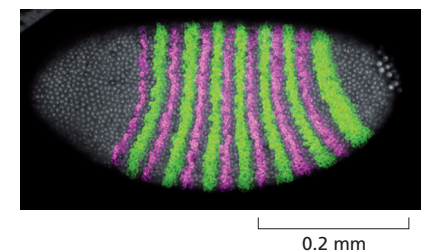
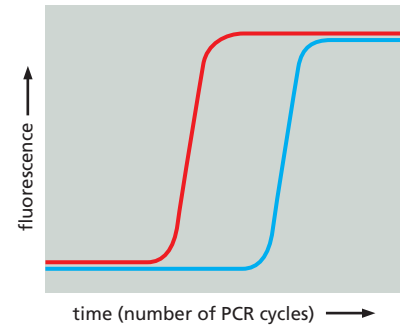


Figure 8–63 *In situ* hybridization to mRNAs reveals patterns of gene expression during development. Expression of specific regulatory genes is localized in the early *Drosophila* embryo, forming a series of stripes (see Figures 7–29 and 7–30). In this image, expression of the genes *eve* (magenta) and *ftz* (green) has been revealed by *in situ* hybridization with differently colored fluorescent probes. DNA is lightly stained with a white fluorescent dye to label all nuclei in the embryo. (Courtesy of Erik Clark and Angela DePace.)

Figure 8–64 RNA levels can be measured by quantitative RT-PCR. The fluorescence measured is generated by a dye that fluoresces only when bound to the double-strand DNA products of the RT-PCR (see Figure 8–34). The *red sample* has a higher concentration of the mRNA being measured than does the *blue sample*, because it requires fewer PCR cycles to reach the same half-maximal concentration of double-stranded DNA. On the basis of this difference, the relative amounts of the mRNA in the two samples can be precisely determined.

Expression of Individual Genes Can Be Measured Using Quantitative RT-PCR

Although reporter genes and *in situ* hybridization accurately reveal patterns of gene expression, they are not the most powerful methods for quantifying amounts of individual RNAs in cells. A more accurate method is based on the principles of PCR. Called **quantitative RT-PCR (reverse transcription–polymerase chain reaction)**, this method begins with the total population of RNA molecules purified from a tissue or a cell culture. It is important that no DNA be present in the preparation; it must be purified away or enzymatically degraded. Two DNA primers that specifically match the mRNA of interest are added, along with reverse transcriptase, DNA polymerase, and the four deoxyribonucleoside triphosphates needed for DNA synthesis. The first round of synthesis is the reverse transcription of the RNA into DNA using one of the primers. Next, a series of heating and cooling cycles allows the amplification of that DNA strand by PCR (see Figure 8–34). The quantitative part of this method relies on a direct relationship between the rate at which the PCR product is generated and the original concentration of the mRNA species of interest. By adding chemical dyes to the PCR that fluoresce only when bound to double-stranded DNA, a simple fluorescence measurement can be used to track the progress of the reaction and thereby accurately deduce the starting concentration of the mRNA that is amplified (**Figure 8–64**). This technique is relatively fast and simple to perform in the laboratory and is the preferred method for accurate measurement of mRNA levels from a specific gene or small group of genes. It is also used routinely to detect rare viral RNAs to determine if a person is infected.



Global Analysis of mRNAs by RNA-seq Provides a Snapshot of Gene Expression

As discussed in Chapter 7, a cell expresses only a subset of the many thousands of genes available in its genome; moreover, this subset differs from one cell type to another or, in the same cell, from one extracellular environment to the next. One way to determine which genes are being expressed by a population of cells or a tissue is to measure all of the RNAs that are being produced.

Global sequencing of all RNAs, or *RNA-seq*, provides the most direct approach for cataloging the RNAs produced by a cell or tissue. As mentioned earlier in this chapter, this approach uses reverse transcriptase to copy all RNAs into cDNAs, which are then fragmented and sequenced by next-generation sequencing methods such as Illumina sequencing (see Figure 8–44). More abundant RNAs will have more cDNA copies, resulting in higher numbers of “sequence reads” for those RNAs. Thus, as we have seen in Chapter 7 (see Figure 7–3), RNA-seq does not simply identify the RNAs in a sample but also provides information about their relative abundance. RNA-seq has other important benefits as well: it can detect alternative RNA splicing, RNA editing, and the many noncoding RNAs produced from a complex genome.

Comprehensive studies of gene expression often provide information that is useful for predicting gene function. Earlier in this chapter, we discussed how identifying a protein’s interaction partners can yield clues about that protein’s function. A similar principle holds true for genes: information about a gene’s function can be deduced by identifying genes that share its expression pattern. Using a computational approach called *cluster analysis*, one can identify sets of genes that are coordinately regulated. Genes that are turned on or turned off together under different circumstances are likely to work in concert in the

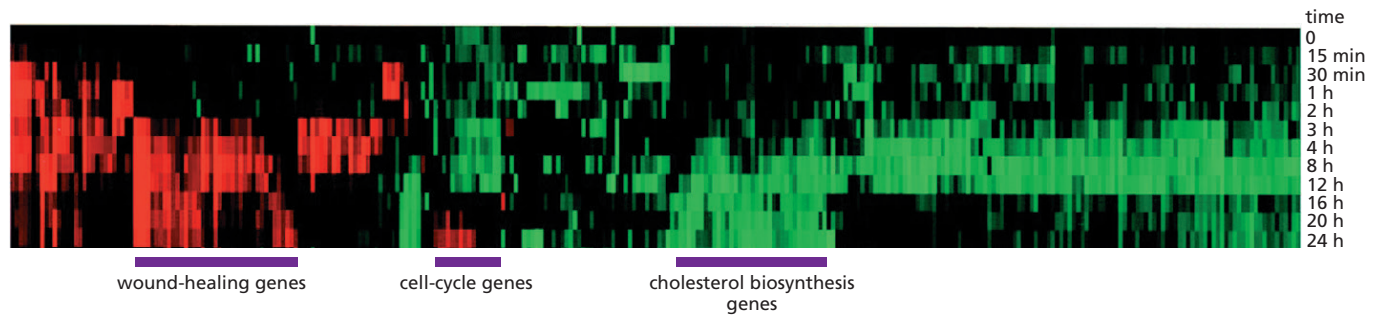


Figure 8–65 Using cluster analysis to identify sets of genes that are coordinately regulated. Genes that have the same expression pattern are likely to be involved in common pathways or processes. To perform a cluster analysis, gene expression data are obtained from cell samples exposed to a variety of different conditions, and genes that show coordinate changes in their expression pattern are grouped together. In this experiment, human fibroblasts were deprived of serum for 48 hours; serum was then added back to the cultures at time 0, and the cells were harvested for mRNA measurements at different time points. Of the 8600 genes depicted here (each represented by a thin vertical line), just over 300 showed threefold or greater variation in their expression patterns in response to serum reintroduction. Here, *red* indicates an increase in expression; *green* is a decrease in expression. On the basis of the results of many other experiments, the 8600 genes have been grouped in clusters according to similar patterns of expression. The results of this analysis show that genes involved in wound healing are turned on in response to serum, while genes involved in regulating cell-cycle progression and cholesterol biosynthesis are shut down. (From M.B. Eisen et al., *Proc. Natl. Acad. Sci. USA* 95:14863–14868, 1998. With permission from National Academy of Sciences.)

cell: they may encode proteins that are part of the same multiprotein machine or proteins that are involved in a complex coordinated activity, such as DNA replication or RNA splicing. Characterizing a gene of unknown function by grouping it with known genes that share its transcriptional behavior is sometimes called “guilt by association.” Cluster analyses have been used to analyze the gene expression profiles that underlie many interesting biological processes (Figure 8–65).

Major new insights into gene expression patterns have come from the development of sensitive methods for sequencing RNA from single cells. In this method, a complex tissue, tumor, embryo, or even organism is dissociated into single cells, after which microfluidics systems are used to separate the cell population into individual cells in single droplets that are each processed for RNA-seq. Sophisticated computational methods are then used to process the vast quantities of sequence data. These methods include cluster analysis algorithms that categorize cells into groups with similar gene expression patterns (Figure 8–66). The exciting outcome of these methods is a visualization of how individual cells fall into groups with similar expression patterns. In many cases, the cell groups identified in this fashion can be identified as previously known cell types, thereby revealing

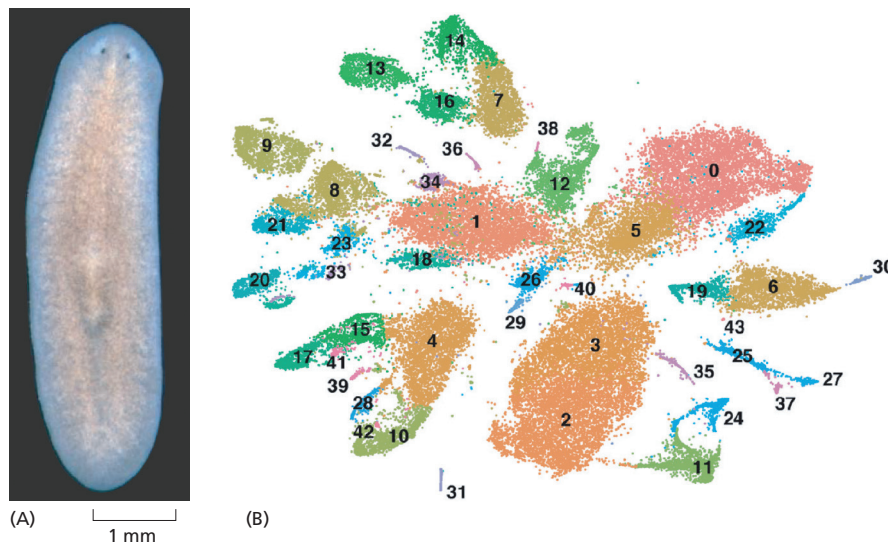


Figure 8–66 Using single-cell RNA-seq to identify the cell types in a whole animal. (A) The planarian flatworm is a small (3–5 mm) model organism used in studies of regeneration (see Figure 22–20). (B) In this experiment, flatworms were cut in small pieces and dissociated into individual cells. About 67,000 cells were analyzed by single-cell RNA sequencing, providing a complete list of all major mRNAs expressed in every cell. Sophisticated computational methods were then used to cluster cells into groups with similar gene expression patterns. These clusters are typically displayed in two dimensions on a plot as shown here, with each dot in a cluster representing a single cell. About 44 distinct cell types, highlighted with distinct colors, were identified in this cluster analysis. The pattern of gene expression in each cluster provides clues about the function of the cells in that cluster. Further insights can be obtained using *in situ* hybridization of intact flatworms, which reveals the precise location of cells and tissues expressing specific mRNAs from each cluster. Such analyses reveal that cell clusters 2 and 3, for example, represent cells from the epidermis. The end result is a comprehensive catalog of the genes expressed in every major cell type of the animal's body. (From C.T. Fincher et al., *Science* 360:eaq1736, 2018. With permission from AAAS.)

new clues about the genes expressed in those cells and even the function of those cells. In other cases, this approach can reveal the existence of cell types that were not previously known. Similar methods can be used to analyze gene expression in distinct cell types during multiple stages of embryonic development, providing comprehensive insights into cell lineage and gene expression patterns during development.

Genome-wide Chromatin Immunoprecipitation Identifies Sites on the Genome Occupied by Transcription Regulators

We have discussed several strategies to measure the levels of individual RNAs in a cell and to monitor changes in their levels in response to external signals. But this information does not tell us how such changes are brought about. We saw in Chapter 7 that transcription regulators, by binding to *cis*-regulatory sequences in DNA, are responsible for establishing and changing patterns of transcription. Typically, these proteins do not occupy all of their potential *cis*-regulatory sequences in the genome under all conditions. For example, in some cell types, the regulatory protein may not be expressed, or it may be present but lack an obligatory partner protein, or it may be excluded from the nucleus until an appropriate signal is received from the cell's environment. Even if the protein is present in the nucleus and is competent to bind DNA, other transcription regulators or components of chromatin can occupy overlapping DNA sequences and thereby occlude some of its *cis*-regulatory sequences in the genome.

Chromatin immunoprecipitation provides a way to experimentally determine all the *cis*-regulatory sequences in a genome that are occupied by a given transcription regulator under a particular set of conditions (Figure 8-67). In this approach, proteins are covalently cross-linked to DNA in living cells, the cells are broken open, and the DNA is mechanically sheared into small fragments. Antibodies directed against a given transcription regulator are then used to purify the DNA that became covalently cross-linked to that protein in the cell. This DNA is then sequenced using the rapid methods discussed earlier; the precise location of each precipitated DNA fragment along the genome is determined by comparing its DNA sequence to that of the whole genome sequence (Figure 8-68). In this way, all of the sites occupied by the transcription regulator in the cell sample can be mapped across the cell's genome. In combination with RNA-seq information, chromatin immunoprecipitation can identify the key transcriptional regulator responsible for specifying a particular pattern of gene expression.

Chromatin immunoprecipitation can also be used to deduce the *cis*-regulatory sequences recognized by a given transcription regulator. Here, all the DNA sequences precipitated by the regulator are lined up (by computer), and features in common are tabulated to produce the spectrum of *cis*-regulatory sequences recognized by the protein (see Figure 7-10A). Chromatin immunoprecipitation is also used routinely to identify the positions along a genome that are bound by the various types of modified histones discussed in Chapter 4. In this case, antibodies specific to the particular histone modification are employed (see Figure 8-68). A variation of the technique can also be used to map positions of chromosomes that are in physical proximity (see Figure 4-53).

Ribosome Profiling Reveals Which mRNAs Are Being Translated in the Cell

We have learned that there are several ways that RNA levels in the cell can be monitored. But for mRNAs, this represents only one step in gene expression, and we are often more interested in the final level of the protein produced by the gene. As described earlier in this chapter, mass spectrometry can be used to monitor the levels of all proteins in the cell, including modified forms of the proteins. However, if we want to understand *how* synthesis of proteins is controlled by the cell, we need to consider the translation step of gene expression.

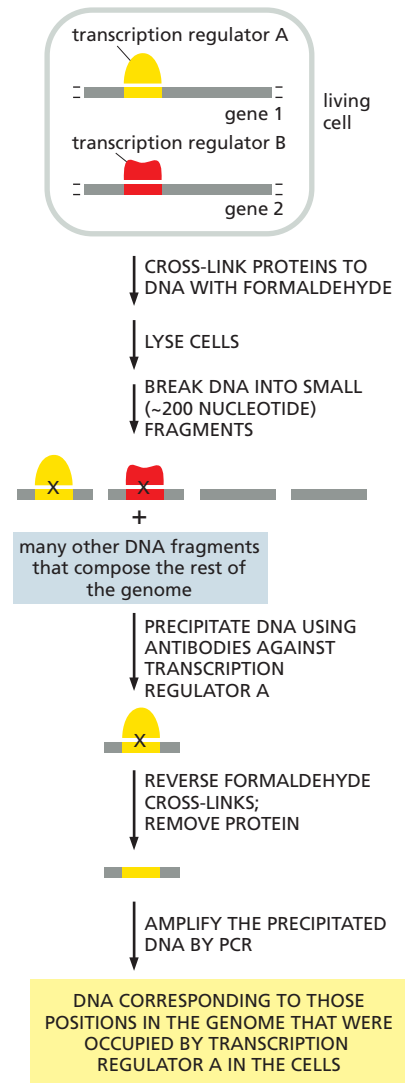


Figure 8-67 Chromatin immunoprecipitation.

This method allows the identification of all the sites in a genome that a transcription regulator occupies *in vivo*. The identities of the precipitated, amplified DNA fragments are determined by DNA sequencing.

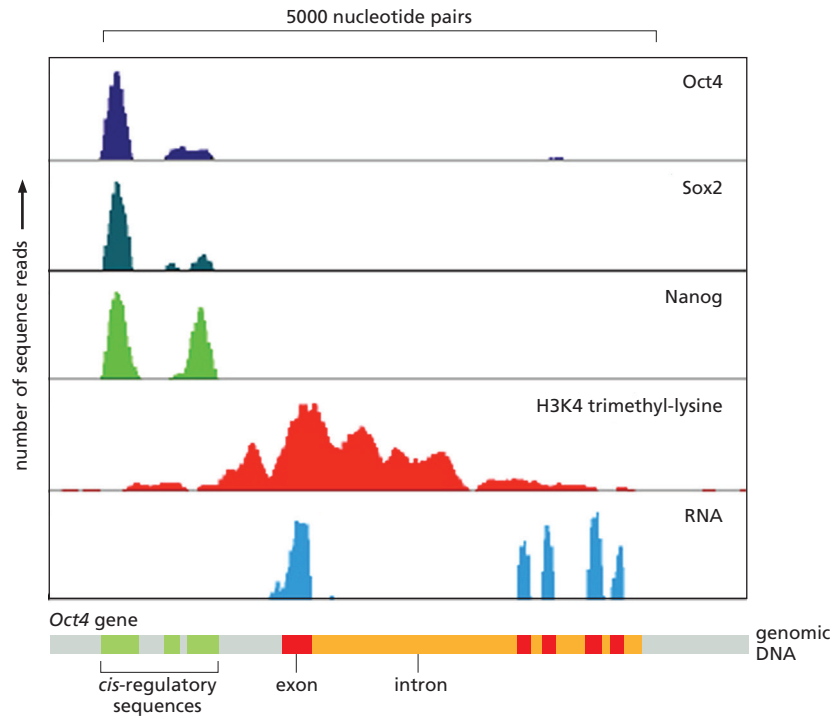


Figure 8–68 Chromatin immunoprecipitations showing proteins bound to the genomic region that controls expression of the *Oct4* gene. In this series of chromatin immunoprecipitation experiments, antibodies directed against a transcription regulator (first three panels) or a particular histone modification (fourth panel) were used to precipitate bound, cross-linked DNA. Precipitated DNA was sequenced, and the positions across the genome were mapped. (Only the small part of the mouse genome containing the *Oct4* gene is shown.) The results show that, in the embryonic stem cells analyzed in these experiments, Oct4 binds upstream of its own gene, and Sox2 and Nanog are bound in close proximity. Oct4, Sox2, and Nanog are key regulators in embryonic stem cells (discussed in Chapter 22), and this experiment reveals the position on the genome through which they exert their effects on *Oct4* expression. In the fourth panel, the positions of a histone modification associated with actively transcribed genes are shown (see Figure 4–35). Finally, the bottom panel shows the RNA produced from the *Oct4* gene under the same conditions used for the chromatin immunoprecipitations. Note that the introns and exons are relatively easy to identify from these RNA-seq data.

An approach called *ribosome profiling* provides an instantaneous map of the positions of ribosomes on each mRNA in the cell and thereby identifies those mRNAs that are being actively translated. To accomplish this, total RNA from a cell line or tissue is exposed to ribonucleases under conditions where only those RNA sequences covered by ribosomes are spared. The protected RNAs are released from ribosomes, converted to DNA, and the nucleotide sequence of each is determined (Figure 8–69). When these sequences are mapped on the genome, the positions of ribosomes across each mRNA species can be ascertained.

Ribosome profiling has revealed many cases where mRNAs are abundant but are not translated until the cell receives an external signal. It has also shown that many open reading frames (ORFs) that were too short to be annotated as genes are actively translated and probably encode functional, albeit very small, proteins (Figure 8–70). Finally, ribosome profiling has revealed the ways that cells rapidly and globally change their translation patterns in response to sudden changes in temperature, nutrient availability, or chemical stress.

Recombinant DNA Methods Have Revolutionized Human Health

We have seen that nucleic acid methodologies developed in the past 40 years have completely changed the study of cell biology. But they have also had a profound effect on our day-to-day lives. Many pharmaceuticals in routine use (insulin, human growth hormone, blood-clotting factors, and interferon, for example) are based on cloning human genes and expressing the encoded proteins in large amounts. As DNA sequencing continues to drop in cost, more and more individuals will elect to have their genome sequenced; this information can be used to predict susceptibility to diseases (often with the option of minimizing this possibility by appropriate behavior) or to predict the way an individual will respond to a given drug. The genomes of tumor cells from an individual can be sequenced to determine the best type of anticancer treatment. And mutations that cause or greatly increase the risk of disease continue to be identified at an unprecedented pace. Using the recombinant DNA technologies discussed in this chapter, these mutations can then be introduced into animals, such as mice, that can be studied in the laboratory. The resulting transgenic animals, which often mimic some of

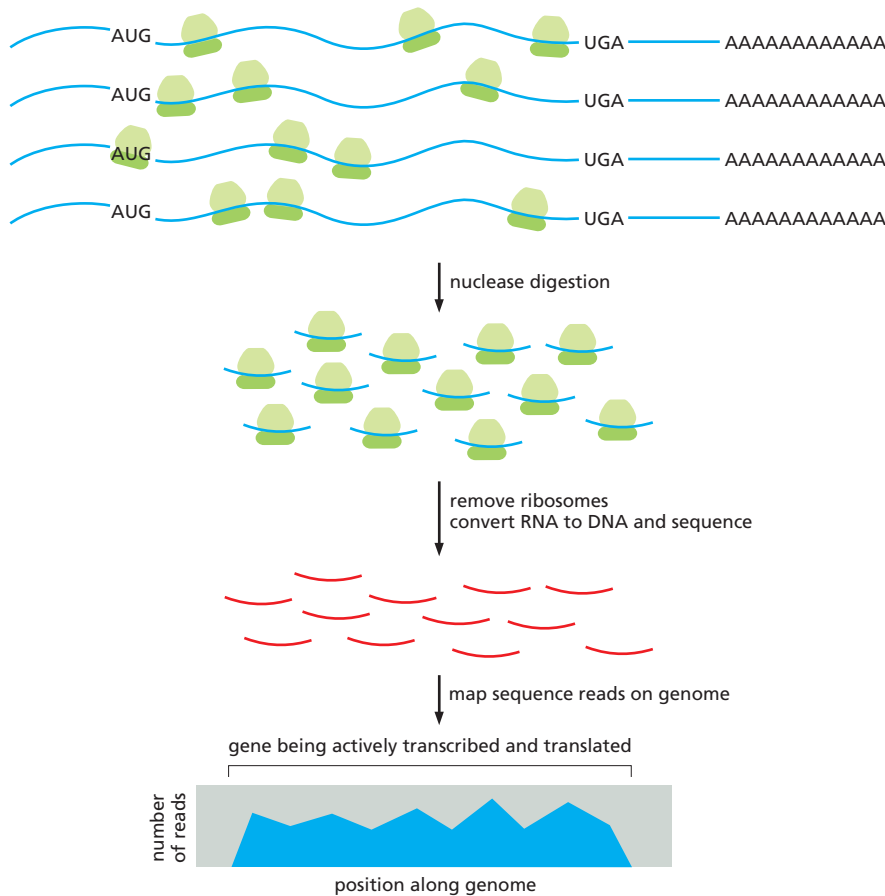


Figure 8–69 Ribosome profiling. RNA is purified from cells and digested with a ribonuclease to leave only those portions of the mRNAs that are protected by a bound ribosome. These short pieces of protected RNA (approximately 20 nucleotides in length) are converted to DNA and sequenced. The resulting information is displayed as the number of sequence reads along each position of the genome. In the diagram here, the data for only one gene, whose mRNA is being efficiently translated, are shown. Ribosome profiling provides this type of information for every mRNA produced by the cell.

the phenotypic abnormalities associated with the condition in humans, can be used to explore the cellular and molecular basis of the disease and to screen for drugs that could potentially be used for human therapy.

Transgenic Plants Are Important for Agriculture

Recombinant DNA technology has had a profound impact on the study of plants. In fact, certain features of plants make them especially amenable to recombinant DNA methods.

When a piece of plant tissue is cultured in a sterile medium containing nutrients and appropriate growth regulators, some of the cells are stimulated to proliferate indefinitely in a disorganized manner, producing a mass of relatively undifferentiated cells called a *callus*. If the nutrients and growth regulators are carefully manipulated, one can induce the formation of a shoot within the callus, and in many species a whole new plant can be regenerated from such shoots. In a

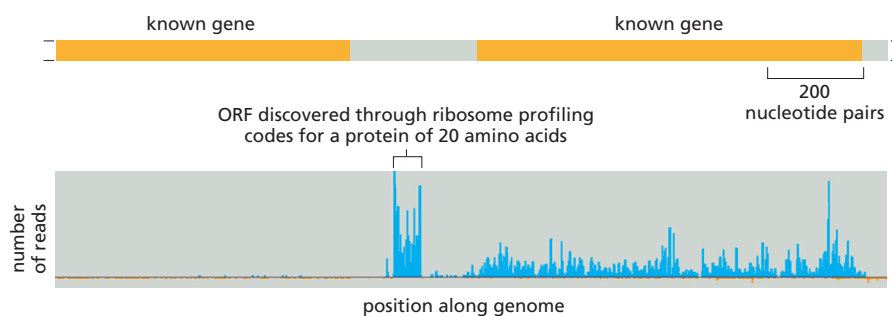


Figure 8–70 Ribosome profiling can identify new genes. This experiment shows the discovery of a previously unrecognized gene—one that encodes a protein of only 20 amino acids. At the top is shown a portion of a viral genome with two previously annotated genes. Below are the results of a ribosome profiling experiment, displayed across the same section of the genome, after human cells were infected by the virus. The results show that the left-hand gene is not expressed under these conditions, the right-hand gene is expressed at low levels, and a previously unrecognized gene that lies between them is expressed at high levels.

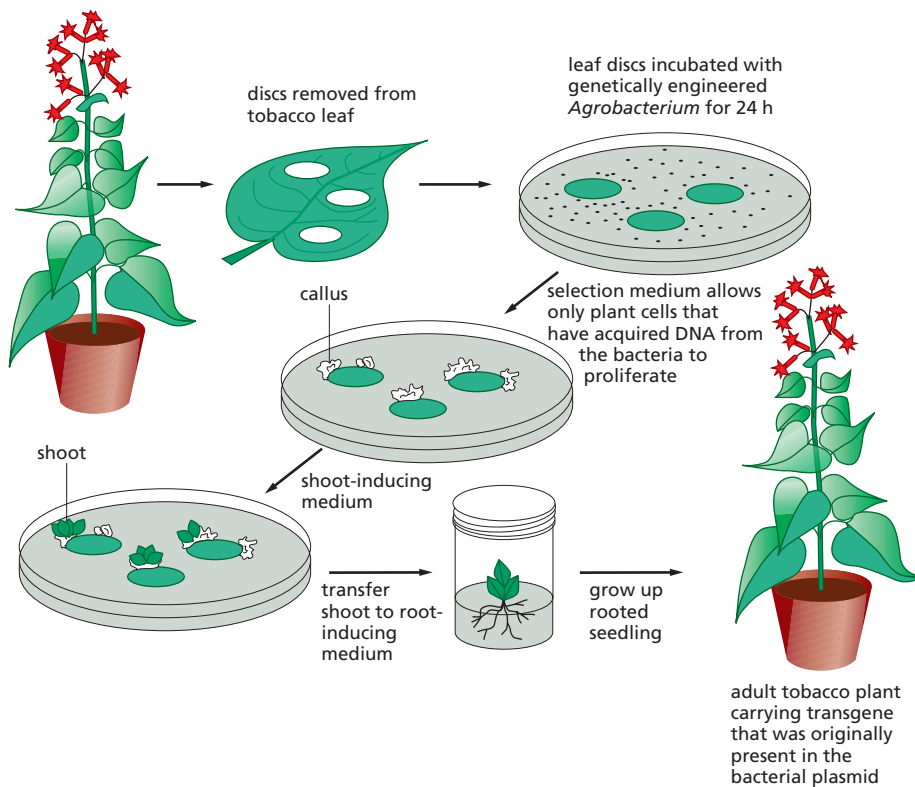


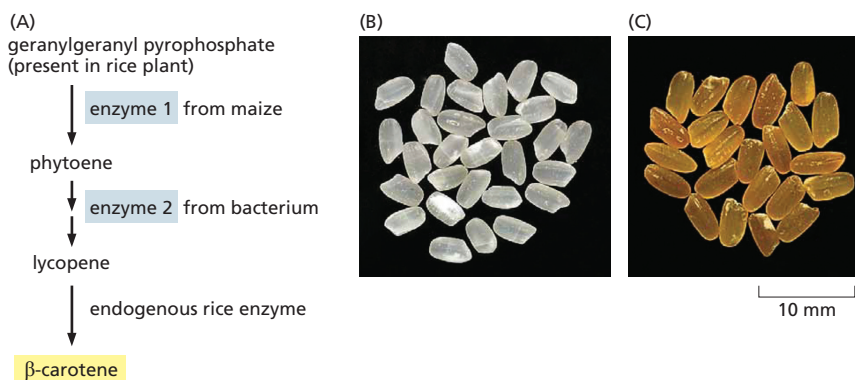
Figure 8-71 Transgenic plants can be made using recombinant DNA techniques optimized for plants. A disc is cut out of a leaf and incubated in a culture of *Agrobacterium* that carries a recombinant plasmid with both a selectable marker and a desired genetically engineered gene. The wounded plant cells at the edge of the disc release substances that attract the bacteria, which inject their DNA into the plant cells. Only those plant cells that take up the appropriate DNA and express the selectable marker gene survive and proliferate and form a callus. The manipulation of growth factors supplied to the callus induces it to form shoots, which subsequently root and grow into adult plants carrying the engineered gene.

number of plants—including tobacco, petunia, carrot, potato, and *Arabidopsis*—a single cell from such a callus (known as a *totipotent cell*) can be grown into a small clump of cells from which a whole plant can be regenerated (see Figure 7-2B). Just as mutant mice can be derived by the genetic manipulation of embryonic stem cells in culture, so transgenic plants can be created from plant cells transfected with DNA in culture (Figure 8-71).

The ability to produce transgenic plants has greatly accelerated progress in many areas of plant cell biology. It has played an important part, for example, in isolating receptors for growth regulators and in analyzing the mechanisms of morphogenesis and of gene expression. These techniques have also opened up many new possibilities in agriculture that could benefit both the farmer and the consumer. They have made it possible, for example, to modify the ratio of lipid, starch, and protein in seeds, to impart pest and virus resistance to plants, and to create modified plants that tolerate extreme habitats such as salt marshes or water-stressed soil. One variety of rice has been genetically engineered to produce β -carotene, the precursor of vitamin A (Figure 8-72). If it replaced conventional rice, this “golden rice”—so called because of its yellow color—could help to

Figure 8-72 DNA technology allows the production of rice grains with high levels of β -carotene.

To help reduce vitamin A deficiency in the developing world, a strain of rice, called “golden rice,” was developed in which the edible part of the grain (called the endosperm) contains large amounts of β -carotene, which is converted in the human gut to vitamin A. (A) Rice plants, like most other plants, can synthesize β -carotene in their leaves from an abundant precursor (geranylgeranyl pyrophosphate) found in all plant tissues. However, the genes that code for two of the enzymes that act early in this biosynthetic pathway are turned off in the endosperm, preventing the production of β -carotene in rice grains. To produce golden rice, the genes for these two enzymes were obtained from organisms that produce large amounts of β -carotene: one from maize and the other from a bacterium. Using DNA technology, these genes were connected to a promoter that drives gene expression in rice endosperm. Using the method outlined in Figure 8-71, this engineered DNA was then used to generate a transgenic rice plant that expresses these enzymes in endosperm, resulting in rice grains that contain high levels of β -carotene. Compared to the milled grains of wild-type rice (B), the grains of the transgenic rice are a deep yellow/orange due to the presence of β -carotene (C). (B and C, from J.A. Paine et al., *Nat. Biotechnol.* 23:482–487, 2005. Reproduced with permission of SNCSC.)



alleviate severe vitamin A deficiency, which causes blindness in hundreds of thousands of children in the developing world each year.

Summary

Genetics and genetic engineering provide powerful tools for understanding the function of individual genes in cells and organisms. In the classical genetic approach, random mutagenesis is coupled with screening to identify mutants that are deficient in a particular biological process. These mutants are then used to locate and study the genes responsible for that process.

Gene function can also be ascertained by reverse genetic techniques. DNA engineering methods can be used to alter genes and to reinsert them into a cell's chromosomes so that they become a permanent part of the genome. If the cell used for this gene transfer is a fertilized egg (for an animal) or a totipotent plant cell in culture, transgenic organisms can be produced that express the mutant gene and pass it on to their progeny. Especially important for cell and molecular biology is the ability to alter cells and organisms in highly specific ways—allowing one to discern the effect on the cell or the organism of a designed change in a single protein or RNA molecule. For example, genomes can be altered so that the expression of any gene can be switched on or off by the experimenter.

Many of these methods are being expanded to investigate gene function on a genome-wide scale. The generation of mutant libraries in which every gene in an organism has been systematically deleted, disrupted, or made controllable by the experimenter provides invaluable tools for exploring the role of each gene in the elaborate molecular collaboration that gives rise to life. Technologies such as RNA-seq can monitor the expression of tens of thousands of genes simultaneously, providing detailed, comprehensive snapshots of the dynamic patterns of gene expression that underlie complex cell processes.

MATHEMATICAL ANALYSIS OF CELL FUNCTION

Quantitative experiments combined with mathematical theory mark the beginning of modern science. Galileo, Kepler, Newton, and their contemporaries did more than set out some rules of mechanics and offer an explanation of the movements of the planets around the Sun: they showed how a quantitative mathematical approach could provide a depth and precision of understanding, at least for physical systems, that had never before been dreamed to be possible.

What is it that gives mathematics this almost magical power to explain the natural world, and why has mathematics played so much more important a part in physical sciences than in biology? What do biologists need to know about mathematics?

Mathematics can be viewed as a tool for deriving logical consequences from propositions. It differs from ordinary intuitive reasoning in its insistence on rigorous, accurate logic and the precise treatment of quantitative information. If the initial propositions are correct, then the deductions drawn from them by mathematics will be true. The surprising power of mathematics comes from the length of the chains of reasoning that rigorous logic and mathematical arguments make possible, and from the unexpectedness of the conclusions that can be reached, often revealing connections that one would not otherwise have guessed at. Reversing the argument, mathematics provides a way to test experimental hypotheses: if mathematical reasoning from a given hypothesis leads to a prediction that is not true, then the hypothesis is not true.

Clearly, mathematics is not much use unless we can frame our ideas—our initial hypotheses—about the given system in a precise, quantitative form. A mathematical edifice raised on a rickety or—even worse—a vague or overcomplicated set of propositions is likely to lead us astray. For mathematics to be useful, we must focus our analysis on simple subsystems in which we can pick out key quantitative parameters and frame well-defined hypotheses. This approach has been used with great success in physics for centuries, but it has been less common

in biology. But times are changing, and more and more it is becoming possible for biologists to exploit the power of quantitative mathematical analysis.

In this final section of our methods chapter, we do not attempt to teach readers every way in which mathematics can be fruitfully applied to biological problems. Rather, we simply aim to give a sense of what mathematics and quantitative approaches can do for us in modern biology. We focus primarily on the important principles that mathematics teaches us about the dynamics of molecular interactions and how mathematics can unveil surprising and useful features of complex systems containing feedback. We will illustrate these principles using the regulation of gene expression by transcription regulators like those discussed in Chapter 7. The same principles apply to the post-transcriptional regulatory systems that govern cell signaling (see Chapter 15), cell-cycle control (see Chapter 17), and essentially all cell processes.

Regulatory Networks Depend on Molecular Interactions

Cell function and regulation depend on transient interactions among thousands of different macromolecules in the cell. We often summarize these interactions in this book with schematic cartoons. These diagrams are useful, but a complete picture requires a deeper, more *quantitative* level of understanding. To meaningfully assess the biological impact of any interaction in the cell, we need to know in precise terms how the molecules interact, how they catalyze reactions, and, most important, how the behaviors of the molecules change over time. If a cartoon shows that protein A activates protein B, for example, we cannot judge the importance of this relationship without quantitative details about the concentrations, affinities, and kinetic behaviors of proteins A and B.

Let us begin by defining two different types of regulatory interaction in our cartoons: one designating inhibition and the other designating activation. If the protein product of gene X is a transcription repressor that inhibits the expression of gene Z, we depict the relationship as a *red bar-headed line* ($\dashv$) drawn between genes X and Z (**Figure 8-73**). If the protein product of gene Y is a transcription activator that induces the expression of gene Z, then a *green arrow* ($\rightarrow$) is drawn between genes Y and Z.

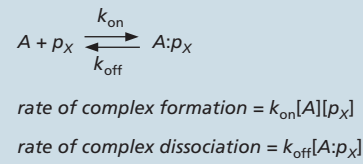
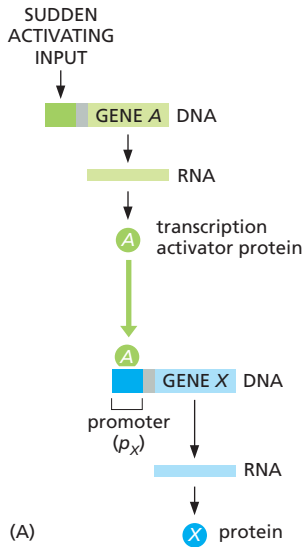
The regulation of one gene's expression by another is more complicated than a single arrow connecting them, and a complete understanding of this regulation requires that we tease apart the underlying biochemical processes. **Figure 8-74A** sketches some of the biochemical steps in the activation of gene expression by a transcription activator. A gene encoding the activator, designated as gene A, will produce its product, protein A, via an RNA intermediate. This protein A will then bind to p_X , the regulatory *promoter* of gene X, to form the complex $A:p_X$. Once the $A:p_X$ complex forms, it stimulates the production of an RNA transcript that is subsequently translated to produce protein X.

We will focus here on the binding interaction that lies at the heart of this regulatory system: the interaction between protein A and the promoter p_X . This interaction is reversible: any molecule of protein A that is bound to p_X can also dissociate from it. The steps represented by the green activation arrow in **Figure 8-74A** include both the binding of A to p_X and the dissociation of the complex $A:p_X$ to re-form A and p_X , as illustrated by the notation in **Figure 8-74B**. This reaction notation is more informative than the diagrams in our figures but has its own limitations. Suppose that the concentration of A increases by a factor of 10 as a response to an environmental input. If A increases, we intuitively know that $A:p_X$ should increase too, but we cannot determine the amount of the increase without additional information. We need to know the affinity of the binding interaction and the concentrations of the two binding partners. With this information in hand, we can rigorously derive the answer.

As discussed earlier and in Chapter 3 (see **Figure 3-42**), we know that the formation of a complex between two binding partners, such as A and p_X , depends on a rate constant k_{on} , which describes how many productive collisions occur per unit time per protein at a given concentration of p_X . The rate of complex



Figure 8-73 Diagrams that summarize biochemical relationships. Here, a simple cartoon indicates that gene X represses gene Z (left) whereas gene Y activates gene Z (right).



(B)

at steady state:

$$k_{\text{on}}[A][p_X] = k_{\text{off}}[A:p_X]$$

$$[A:p_X] = \frac{k_{\text{on}}}{k_{\text{off}}} [A][p_X] = K[A][p_X] \quad \text{Equation 8-1}$$

(C)

$$[p_X^T] = [p_X] + [A:p_X]$$

substituting $[p_X]$ from the above equation into Equation 8-1 yields:

$$[A:p_X] = K[A]([p_X^T] - [A:p_X])$$

$$[A:p_X](1 + K[A]) = K[A][p_X^T]$$

$$[A:p_X] = \frac{K[A]}{1 + K[A]} [p_X^T] \quad \text{Equation 8-2}$$

(D)

$$\text{bound fraction} = \frac{[A:p_X]}{[p_X^T]} = \frac{K[A]}{1 + K[A]} \quad \text{Equation 8-3}$$

(E)

Figure 8-74 A simple transcriptional interaction.

(A) Genes A and X each produce a protein, with the product of gene A serving as a transcription activator to stimulate expression of gene X. As indicated by the *green arrow*, stimulation depends in part on the binding of protein A to the promoter region of gene X, designated as p_X . (B) The binding of protein A to the gene promoter is determined by the concentrations of the two binding partners (denoted as $[A]$ and $[p_X]$, in units of mol/liter, or M), the association rate constant k_{on} (in units of $\text{M}^{-1} \text{sec}^{-1}$), and the dissociation rate constant k_{off} (in units of sec^{-1}). (C) At steady state, the rates of association and dissociation are equal, and the concentration of the bound complex is determined by Equation 8-1, in which the two rate constants are combined in the equilibrium constant K . (D) Equation 8-2 can be derived to calculate the steady-state concentration of bound complex at a known total concentration of the promoter $[p_X^T]$. (E) Rearrangement of Equation 8-2 yields Equation 8-3, which allows calculation of the fraction of promoter p_X that is occupied by protein A.

formation equals the product of this rate constant k_{on} and the concentrations of A and p_X (see Figure 8-74B). Complex dissociation occurs at a rate k_{off} multiplied by the concentration of the complex. The rate constant k_{off} can differ by orders of magnitude for different DNA sequences because it depends on the strength of the noncovalent bonds formed between A and p_X .

We are primarily interested in understanding the amount of bound promoter complex at equilibrium or *steady state*, where the rate of complex formation equals the rate of complex dissociation. Under these conditions, the concentration of the promoter complex is specified by a simple equation that combines the two rate constants into a single equilibrium constant $K = k_{\text{on}}/k_{\text{off}}$ (Equation 8-1; Figure 8-74C). K is sometimes called the association constant, K_a . The larger this constant K , the stronger the interaction between A and p_X . The reciprocal of K is the dissociation constant, K_d .

To calculate the steady-state concentration of the promoter complex using Equation 8-1, we need to account for another complication: both A and p_X exist in two forms—free in solution and bound to each other. In most cases, we know the total concentration of p_X and not the free or bound concentrations, so we must find a way to use the total concentration in our calculations. To do this, we first specify that the total concentration of p_X ($[p_X^T]$) is the sum of the concentrations of free ($[p_X]$) and bound ($[A:p_X]$) forms (Figure 8-74D). This leads to a new equation that allows us to use $[p_X^T]$ to calculate the steady-state concentration of the promoter complex ($[A:p_X]$) (Equation 8-2; Figure 8-74D).

Protein A also exists in two forms: free ($[A]$) and bound to p_X ($[A:p_X]$). In a cell, there are typically one or two copies of p_X (assuming there is only one gene X per haploid genome) and multiple copies of A . As a result, we can safely assume that from the viewpoint of A , $[A:p_X]$ is negligible relative to the total $[A^T]$. This means $[A] \approx [A^T]$, and we can just plug in the values of total $[A^T]$ in Equation 8-2 without incurring appreciable error in the calculation of $[A:p_X]$.

Now, we are ready to determine the effects of increasing the concentration of A . Suppose that $K = 10^8 \text{ M}^{-1}$, which is a typical value for many such interactions. The starting concentration of A is $[A^T] = 10^{-9} \text{ M}$, and $[p_X^T] = 10^{-10} \text{ M}$ (assuming there is one copy of gene X in a haploid yeast cell, for example, with a volume of around $2 \times 10^{-14} \text{ L}$). Using Equation 8-2, we find that a tenfold increase in the concentration of A causes the amount of promoter complex $[A:p_X]$ to increase 5.5-fold, from $0.09 \times 10^{-10} \text{ M}$ to $0.5 \times 10^{-10} \text{ M}$ at steady state. The effects of a tenfold increase in the concentration of A will vary dramatically depending on its starting concentration relative to the equilibrium constant. Only through this mathematical approach can we achieve a thorough understanding of what these effects will be and what impact they will have on the biological response.

To assess the biological impact of a change in transcription activator levels, it is also important in many cases to determine the fraction of the target gene promoter that is bound by the activator, because this number will be directly proportional to the activity of the gene's promoter. In our case, we can calculate the fraction of the gene X promoter, p_X , that has protein A bound to it by rearranging Equation 8-2 (Equation 8-3; [Figure 8-74E](#)). This fraction can be viewed as the probability that promoter p_X is occupied, averaged over time. It is also equal to the average occupancy across a large population of cells at any instant in time. When there is no protein A present, p_X is always free, the bound fraction is zero, and transcription is off. When $[A] = 1/K$, the promoter p_X has a 50% chance of being occupied. When $[A]$ greatly exceeds $1/K$, the bound fraction is almost equal to one, meaning that p_X is fully occupied and transcription is maximal.

Differential Equations Help Us Predict Transient Behavior

The most important and basic insights for which we, as biologists, depend on mathematics concern the behavior of regulatory systems over time. This is the central theme of dynamics, and it was for the solution of problems in dynamics that the techniques of calculus were developed, by Newton and Leibniz, in the seventeenth century. Briefly, the general problem is this: if we are given the rates of change of a set of variables that characterize the system at any instant, how can we compute its future state? The problem becomes especially interesting, and the predictions often remarkable, when the rates of change themselves depend on the values of the state variables, as in systems with feedback.

Let us return to Equation 8-2 ([Figure 8-74D](#)), which tells us that when $[A]$ changes, $[A:p_X]$ at steady state will also change to a new concentration that we can calculate with precision. However, $[A:p_X]$ does not change instantaneously to this value. If we hope to understand the behavior of this system in detail, we must also ask how long it takes $[A:p_X]$ to get to its new steady-state value inside the cell. Equation 8-2 cannot answer this question. We need calculus.

The most common strategy for solving this problem is to use ordinary differential equations. The equations that describe biochemical reactions have a simple premise: the rate of change in the concentration of any molecular species X (that is, $d[X]/dt$) is given by the balance of the rate of its appearance with that of its disappearance. For our example, the rate of change in the concentration of the bound promoter complex, $[A:p_X]$, is determined by the rates of complex assembly and disassembly. We can incorporate these rates into the differential equation shown in [Figure 8-75A](#) (Equation 8-4). When $[A]$ changes, Equation 8-4 can be solved to generate the concentration of $[A:p_X]$ as a function of time. Notice that when $k_{\text{on}}[A][p_X] = k_{\text{off}}[A:p_X]$, then $d[A:p_X]/dt = 0$ and $[A:p_X]$ stops changing. At this point, the system has reached the steady state.

Calculation of all $[A:p_X]$ values as a function of time, using Equation 8-4, allows us to determine the rate at which $[A:p_X]$ reaches its steady-state value. Because this value is attained asymptotically, it is often most useful to compare the times needed to get to 50%, 90%, or 99% of this new steady state. The simplest way to determine these values is to solve Equation 8-4 with a method called numerical integration, which involves plugging in values for all of the parameters (k_{on} , k_{off} ,

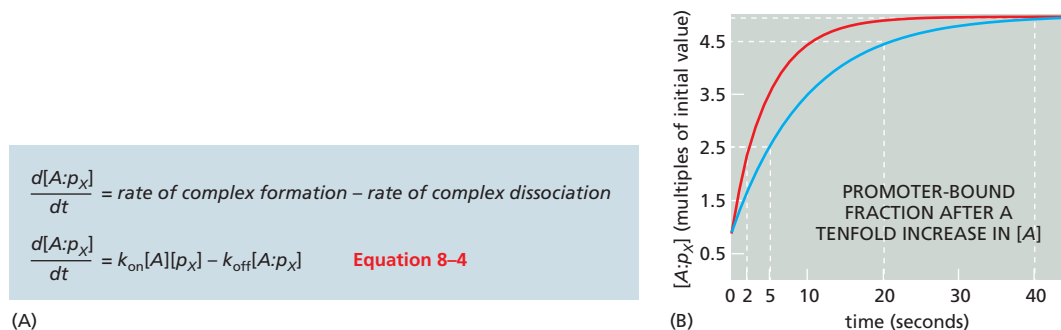


Figure 8-75 Using differential equations to study the dynamics and steady-state behavior of a biological system. (A) Equation 8-4 is an ordinary differential equation for calculating the rate of change in the formation of bound promoter complex in response to a change in other components. (B) Formation of $[A:p_X]$ after a tenfold increase in $[A]$, as determined by solving Equation 8-4. In blue is the solution corresponding to $k_{\text{on}} = 0.5 \times 10^7 \text{ M}^{-1} \text{ sec}^{-1}$ and $k_{\text{off}} = 0.5 \times 10^{-1} \text{ sec}^{-1}$. In this case, it takes $[A:p_X]$ about 5, 20, and 40 seconds to reach 50%, 90%, and 99% of the new steady-state value. For the red curve, the k_{on} and k_{off} values are doubled, and the system reaches the same steady state more rapidly.

etc.) and then using a computer to determine the values of $[A:p_X]$ over time, starting from given initial concentrations of $[A]$ and $[p_X]$. For $k_{\text{on}} = 0.5 \times 10^7 \text{ M}^{-1} \text{ sec}^{-1}$, $k_{\text{off}} = 0.5 \times 10^{-1} \text{ sec}^{-1}$ ($K = 10^8 \text{ M}^{-1}$ as above), and $[p_X^T] = 10^{-10} \text{ M}$, it takes $[A:p_X]$ about 5, 20, and 40 seconds to reach 50%, 90%, and 99% of the new steady-state value after a sudden tenfold change in $[A]$ (Figure 8-75B). Thus, a sudden jump in $[A]$ does not have instantaneous effects, as we might have assumed from looking at the cartoon in Figure 8-74A.

Differential equations therefore allow us to understand the transient dynamics of biochemical reactions. This tool is critical for achieving a deep understanding of cell behavior, in part because it allows us to determine the dependence of the dynamics inside cells on parameters that are specific to the particular molecules involved. For example, if we double the values of both k_{on} and k_{off} , then Equation 8-1 (Figure 8-74C) indicates that the steady-state value of $[A:p_X]$ does not change. However, the time it takes to reach 50% of this steady state after a tenfold change in $[A]$ in our example changes from about 5 seconds to 2 seconds (see Figure 8-75B). These insights are not accessible from either cartoons or equilibrium equations. This is an unusually simple example; mathematical descriptions such as differential equations become more indispensable for understanding biological interactions as the number of interactions increases.

Promoter Activity and Protein Degradation Affect the Rate of Change of Protein Concentration

To understand our gene regulatory system further, we also need to describe the dynamics of protein X production in response to changes in the amount of transcription activator protein A . Here again, we use an ordinary differential equation for the rate of change of protein X concentration—determined by the balance of the rate of production of protein X through expression of gene X and the protein's rate of degradation.

Let us begin with the rate of protein X production, which is determined primarily by the occupancy of the promoter of gene X by protein A . The binding and dissociation of a transcription regulator at a promoter generally occur on a much faster time scale than transcription initiation, causing many binding and unbinding events to occur before transcription proceeds. As a result, we can assume that the binding reaction is at equilibrium on the time scale of transcription, and we can calculate promoter occupancy by protein A using the equilibrium equation discussed earlier (Equation 8-3; Figure 8-74E). To determine the transcription rate, we simply multiply the occupied promoter fraction by a *transcription rate constant*, β , that represents the binding of RNA polymerase and the subsequent

$$\begin{aligned} \text{transcription rate} &= \beta \frac{K[A]}{1 + K[A]} \\ \text{protein production rate} &= \beta \cdot m \frac{K[A]}{1 + K[A]} \\ \text{protein degradation rate} &= \frac{[X]}{\tau_X} \end{aligned}$$

(A)

$$\begin{aligned} \frac{d[X]}{dt} &= \text{protein production rate} - \text{protein degradation rate} \\ \frac{d[X]}{dt} &= \beta \cdot m \frac{K[A]}{1 + K[A]} - \frac{[X]}{\tau_X} \end{aligned} \quad \text{Equation 8-5}$$

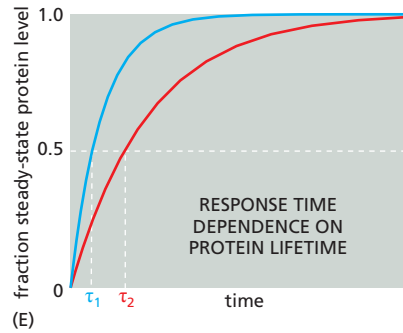
(B)

$$\begin{aligned} \text{at steady state:} \\ [X_{st}] &= \beta \cdot m \frac{K[A]}{1 + K[A]} \cdot \tau_X \end{aligned} \quad \text{Equation 8-6}$$

(C)

$$[X](t) = [X_{st}](1 - \exp(-t/\tau_X))$$

(D)



(E)

Figure 8-76 Effect of protein lifetime on the timing of the response.

(A) Equations for calculation of the rates of gene X transcription, protein X production, and protein X degradation, as explained in the text. (B) Equation 8-5 is an ordinary differential equation for calculating the rate of change in protein X in response to changes in other components. (C) When the rate of change in protein X is zero (steady state), its concentration can be calculated with Equation 8-6, revealing a direct relationship with protein lifetime (τ). (D) The solution of Equation 8-5 specifies the concentration of protein X over time as it approaches its steady-state concentration. (E) Response time depends on protein lifetime. As described in the text, the time that it takes a protein to reach a new steady state is greater when the protein is more stable. Here, the *blue line* corresponds to a protein with a lifetime that is 2.5-fold shorter than the lifetime of the protein in *red*.

steps that lead to production of mRNA and protein (**Figure 8-76A**). If each mRNA molecule produces, on average, m molecules of protein product, then we can determine the protein production rate by multiplying the transcription rate by m (**Figure 8-76A**).

Now let us consider the factors that influence protein X degradation and its dilution due to cell growth. Degradation generally results in an exponential decline in protein levels, and the average time required for a specific protein to be degraded is defined as its mean lifetime, τ . In our current example, the rate of degradation of protein X depends on its mean lifetime τ_X , which takes into account active degradation as well as its dilution as the cell grows. The degradation rate depends on the concentration of protein X and is calculated by dividing this concentration by the lifetime (see **Figure 8-76A**).

With equations for rates of production and degradation in hand, we can now generate a differential equation to determine the rate of change of protein X as a function of time (**Equation 8-5**; **Figure 8-76B**). This equation can be solved by the numerical methods mentioned earlier. According to the solution of this equation, when transcription begins, the concentration of protein X rises to a steady-state level at which the concentration of X is not changing anymore; that is, its rate of change is zero. When this occurs, rearrangement of **Equation 8-5** yields an equation that can be used to determine the steady-state value of X, $[X_{st}]$ (**Equation 8-6**; **Figure 8-76C**). An important concept emerges from the mathematics: the steady-state concentration of a gene product is directly proportional to its lifetime. If lifetime doubles, protein concentration doubles as well.

The Time Required to Reach Steady State Depends on Protein Lifetime

We can see from **Equation 8-6** (see **Figure 8-76C**) that when the concentration of protein A rises, protein X increases to a new steady-state value, $[X_{st}]$. But this cannot happen instantaneously. Instead, X changes dynamically according to the solution of its differential rate equation (**Equation 8-5**). The solution of this equation reveals that the concentration of X over time is related to its steady-state

concentration according to the equation in [Figure 8-76D](#). Once again, mathematics uncovers a simple but important concept that is not intuitively obvious: after a sudden increase in $[A]$, $[X]$ rises to a new steady state at an exponential rate that is inversely related to its lifetime; the faster X is degraded, the less time it takes it to reach its new steady-state value ([Figure 8-76E](#)). The faster response time comes at a higher metabolic cost, however, because proteins with a rapid response time must be produced and degraded at a high rate. For proteins that are not rapidly turned over, the response time is very long, and protein concentration is determined primarily by the dilution that results from cell growth and division.

Quantitative Methods Are Similar for Transcription Repressors and Activators

Positive control is not the only mechanism that cells use to regulate the expression of their genes. As we discussed in Chapter 7, cells also actively shut off genes, often by employing transcription repressor proteins that bind to specific sites on target genes, thereby blocking access to RNA polymerase. We can analyze the function of these repressors by the same quantitative methods described above for transcription activators. If a repressor protein R binds to the regulatory region of gene X and represses its transcription, then the fraction of gene-binding sites occupied by the repressor is specified by the same equation we used earlier for the transcription activator ([Figure 8-77A](#)). In this case, however, it is only when the DNA is free that RNA polymerase can bind to the promoter and transcribe the gene. Thus, the quantity of interest is the unbound fraction, which can be viewed as the probability that the site is free, averaged over multiple binding and unbinding events. When the repressor concentration is zero, the unbound fraction is 1 and the promoter is fully active; when the repressor concentration greatly exceeds $1/K$, the unbound fraction approaches zero. [Figure 8-77B](#) and [Figure 8-77C](#) compare these relationships for a transcription activator and a transcription repressor.

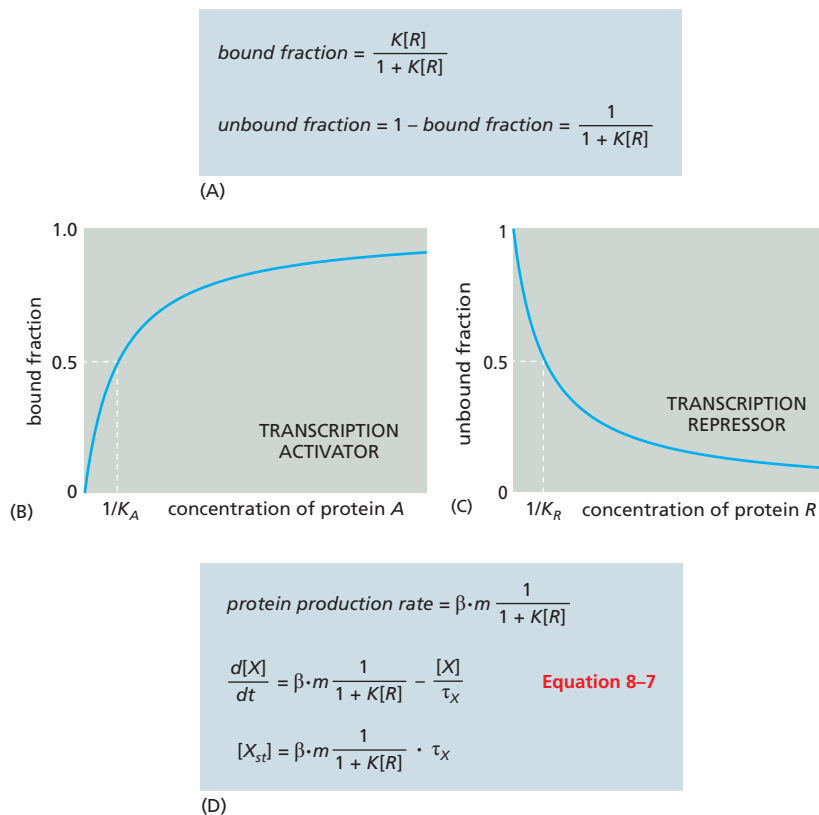


Figure 8-77 How promoter occupancy depends on the binding affinity of a transcription regulator protein. (A) The fraction of a binding site that is occupied by a transcription repressor R is determined by an equation that is similar to the one we used for a transcription activator (see [Figure 8-74E](#)), except that in the case of a repressor we are interested primarily in the unbound fraction. (B) For a transcription activator A , half of the promoters are occupied when $[A] = 1/K_A$. Gene activity is proportional to this bound fraction. (C) For a transcription repressor R , gene activity is proportional to the unbound fraction of promoters. As indicated, this fraction is reduced to half of its maximal value when $[R] = 1/K_R$. (D) As in the case of the transcription activator A (see [Figure 8-76](#)), we can derive equations to assess the timing of protein X production as a function of repressor concentrations.

We can create a differential equation that provides the rate of change in protein X when repressor concentrations change (Equation 8-7; **Figure 8-77D**). As in the case of the transcription activator, the steady-state concentration of protein X increases as its lifetime increases, but it decreases as the concentration of the transcription repressor increases.

Negative Feedback Is a Powerful Strategy in Cell Regulation

Thus far, we have considered simple regulatory systems of just a few components. In most of the complex regulatory systems that govern cell behaviors, multiple modules are linked to produce larger circuits that we call *network motifs*, which can produce surprisingly complex and biologically useful responses whose properties become apparent only through mathematical analysis. A particularly common and important network motif is the negative feedback loop, which can have dramatically different functions depending on how it is structured.

We take as a first example a network motif consisting of two linked modules (**Figure 8-78A**). Here, an input signal initiates the transcription of gene A , which produces a transcription activator protein A . This activates gene R , which synthesizes a transcription repressor protein R . Protein R in turn binds to the promoter of gene A to inhibit its expression. This cyclical organization creates a negative feedback loop that one can intuitively understand as a mechanism to prevent proteins from accumulating to high levels. But what can we learn about negative feedback loops, and their value in biology, by using mathematics to model them?

The negative feedback loop in **Figure 8-78A** can be modeled using Equation 8-7 (see **Figure 8-77D**) for the repression of gene A and Equation 8-5 (see **Figure 8-76B**) for the activation of gene R . Thus, for proteins A and R , we use the set of differential equations (Equation set 8-8) shown in **Figure 8-78B**. The two equations in this set are coupled, which means that they must be solved together to describe the behavior of A and R over time for any value of the input. As before, we plug in values for the parameters (β_R , τ_R , etc.) and then use a computer to determine the values of $[A]$ and $[R]$ as a function of time after a sudden input activates gene A .

The results reveal several important properties of negative feedback. First, rather surprisingly, negative feedback increases the speed of the response to the activating input. As shown in **Figure 8-78C**, the system with negative feedback reaches its new steady state faster than the system with no feedback.

Second, negative feedback is useful for protecting cells from perturbations that continually arise in the cell's internal environment—due either to random variations in the birth and death of molecules or to fluctuations in environmental variables such as temperature and nutritional supplies. Let us imagine, for example, that β_A , the transcription rate constant for gene A , fluctuates by 25% of its value and ask whether and how much the levels of protein R are affected. The results, shown in **Figure 8-79**, reveal that a change in β_A causes a smaller change in the steady-state value of R when the network has negative feedback.

Delayed Negative Feedback Can Induce Oscillations

A beautiful thing happens when a negative feedback loop contains some delay mechanism that slows the feedback signal through the loop: rather than generating a new stable state as in a rapid negative feedback loop, a delayed loop

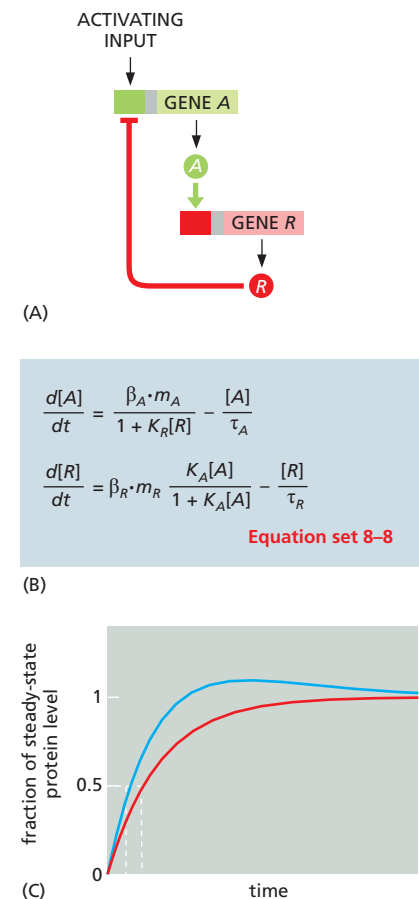


Figure 8-78 A simple negative feedback motif. (A) Gene A negatively regulates its own expression by activating gene R . The product of gene R is a transcription repressor that inhibits gene A . (B) Equation set 8-8 can be solved to determine the dynamics of system components over time. (C) A system with negative feedback (*blue*) reaches its steady state faster than a system with no feedback (*red*). The plots indicate the levels of protein A , expressed as a fraction of the steady-state level. The *blue* line reflects the solution of Equation set 8-8, which includes negative feedback of gene A by the repressor R . The *red* line represents the solution when the rate of synthesis of A was set to a constant value that is unaffected by the repressor R .

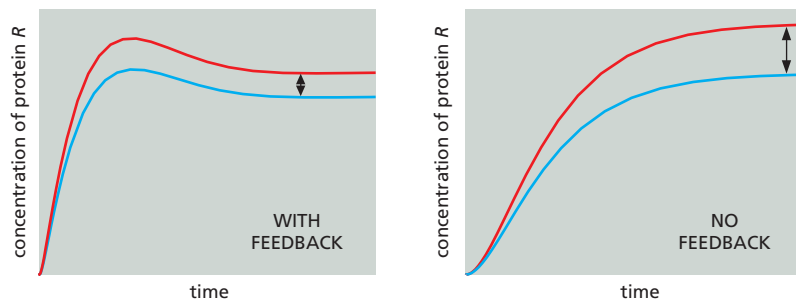


Figure 8-79 The effect of fluctuations in kinetic rate constants on a system with negative feedback compared to one without feedback. The plot at left represents the levels of protein *R* after a sudden activating stimulus, according to the regulatory scheme in Figure 8-78A and determined by the solution of Equation set 8-8 (see Figure 8-78B). A perturbation was induced by changing β_A from 4 M/min (red line) to 3 M/min (blue line). The plot at right shows the results when negative feedback was removed. The system with negative feedback deviates less from its normal operation as β changes than does the system with no feedback. Notice that, as in Figure 8-78C, the system with negative feedback also reaches its steady state more rapidly.

generates pulses, or *oscillations*, in the levels of its components. This can be seen, for example, if the number of components in a negative feedback loop increases, which leads to delays in the amount of time required for the cycle of signals to be completed. **Figure 8-80** compares the behavior of two network motifs—one with a three-stage and one with a five-stage negative feedback loop. Using the same kinetic parameters at each stage in the two loops, one finds that stable oscillations arise in the longer loop, while in the shorter loop the same parameters lead to relatively rapid convergence to a stable steady state.

Changes in the parameters of a delayed negative feedback loop—binding affinities, transcription rates, or protein stabilities, for example—can change the amplitude and period of the oscillations, providing a remarkably versatile mechanism for generating all sorts of oscillators that can be used for various purposes in the cell. Indeed, many naturally occurring oscillators, including the calcium oscillators described in Chapter 15 and the cell-cycle network described in Chapter 17, use delayed negative feedback as the basis for biologically important oscillations. Not all of the oscillations observed in cells are thought to have a function, however. Oscillations become inevitable in a highly complex, multicomponent biochemical pathway such as glycolysis, due simply to the large number of feedback loops that appear to be required for its regulation.

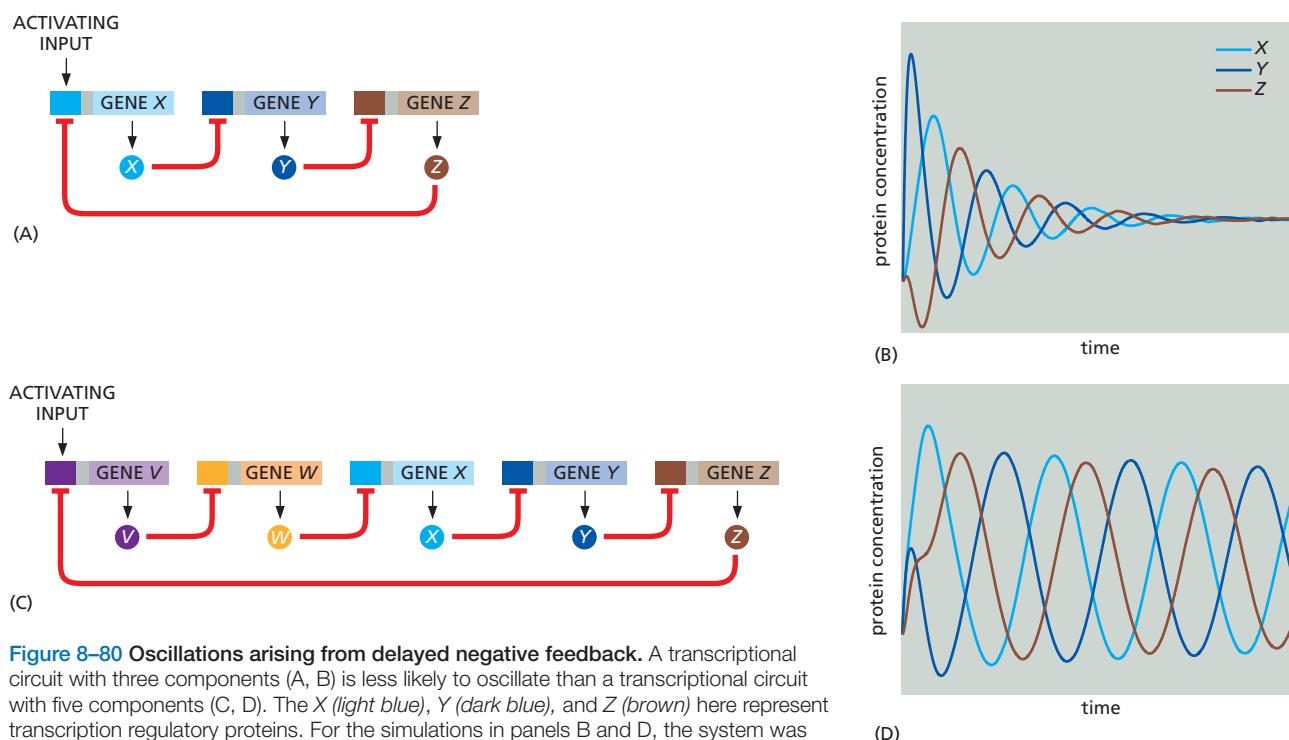


Figure 8-80 Oscillations arising from delayed negative feedback. A transcriptional circuit with three components (A, B) is less likely to oscillate than a transcriptional circuit with five components (C, D). The *X* (light blue), *Y* (dark blue), and *Z* (brown) here represent transcription regulatory proteins. For the simulations in panels B and D, the system was initiated from random initial conditions for *X*, *Y*, and *Z*. Oscillations are produced by a delay induced as the signal propagates through the loop.

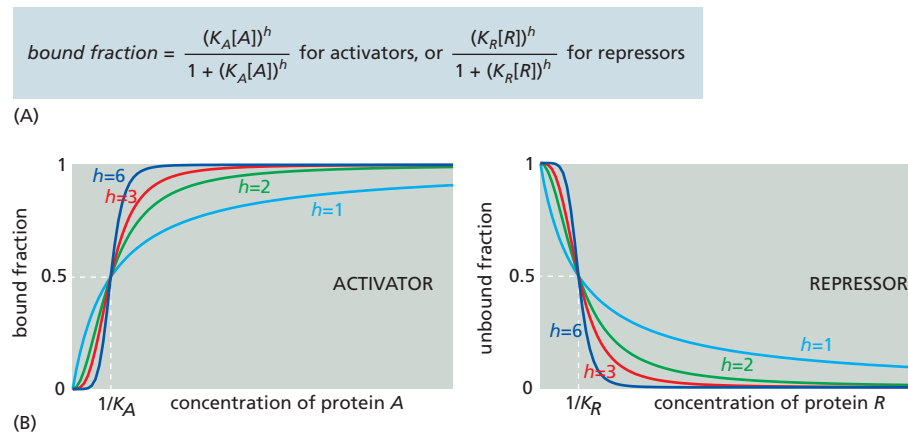


Figure 8-81 How the cooperative binding of transcription regulatory proteins affects the fraction of promoters bound. (A) Cooperativity is incorporated into our mathematical models by including a Hill coefficient (h) in the equations used previously to determine the fraction of bound promoter (see Figures 8-74E and 8-77A). When h is 1, the equations shown here become identical to the equations used previously, and there is no cooperativity. (B) The left panel depicts a cooperatively bound transcription activator, and the right panel depicts a cooperatively bound transcription repressor. Recall from Figure 8-77B that gene activity is proportional to bound activator (left panel) or unbound repressor (right panel). Note that the plots get steeper as the Hill coefficient increases.

DNA Binding by a Repressor or an Activator Can Be Cooperative

We have focused thus far on the binding of a single transcription regulator to a single site in a gene promoter. Many promoters, however, contain multiple adjacent binding sites for the same transcription regulator, and it is not uncommon for these regulators to interact with each other on the DNA to form dimers or larger oligomers. These interactions can result in a *cooperative* form of DNA binding, such that DNA-binding affinity increases at higher concentrations of the transcription regulator. Cooperativity produces a steeper transcriptional response to increasing regulator concentration than the response that can be generated by the binding of a monomeric protein to a single site. A steep transcriptional response of this sort, when present in conjunction with positive feedback, is an important ingredient for producing systems with the ability to switch between different discrete phenotypic states. To begin to understand how this occurs, we need to modify our equations to include cooperativity.

Cooperative binding events can produce steep S-shaped (or *sigmoidal*) relationships between the concentration of regulatory protein and the amount bound on the DNA (see Figure 7-11 and Figure 15-17). In this case, a number called the *Hill coefficient* (h) describes the degree of cooperativity, and we can include this coefficient in our equations for calculating the bound fraction of promoter (Figure 8-81A). As the Hill coefficient increases, the dependence of binding on protein concentration becomes steeper (Figure 8-81B). In principle, the Hill coefficient is similar to the number of molecules that must come together to generate a reaction. In practice, however, cooperativity is rarely complete, and the Hill coefficient does not reach this number.

Positive Feedback Is Important for Switchlike Responses and Bistability

We turn now to positive feedback and its very important consequences. First and foremost, positive feedback can make a system *bistable*, enabling it to persist in either of two (or more) alternative steady states. The idea is simple and can be conveyed by drawing an analogy with a candle, which can exist either in a burning state or in an unlit state. The burning state is maintained by positive feedback: the heat generated by burning keeps the flame alight. The unlit state is maintained by the absence of this feedback signal: so long as sufficient heat has never been applied, the candle will stay unlit.

For the biological system, as for the candle, bistability has an important corollary: it means that the system has a memory, such that its present state depends on its history. If we start with the system in an Off state and gradually rack up the concentration of the activator protein, there will come a point where autostimulation becomes self-sustaining (the candle lights), and the system moves rapidly to an On state. If we now intervene to decrease the level of activator, there will come a point where the same thing happens in reverse, and the system moves rapidly

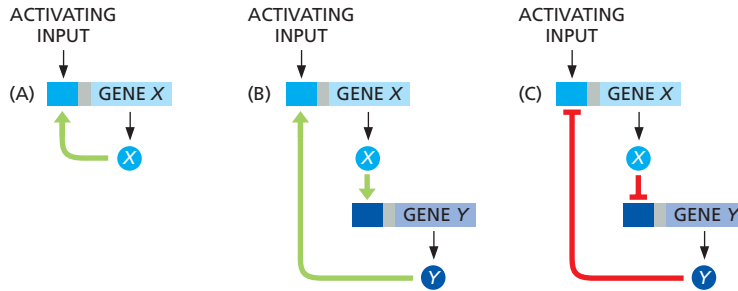


Figure 8-82 Positive feedback of a gene onto itself through serially connected interactions. A sequence of activators and repressors of any length can be connected to produce a positive feedback loop, as long as the overall sign is positive. Because the negative of a negative is positive, not only circuit (A) and (B) but also circuit (C) create positive feedback.

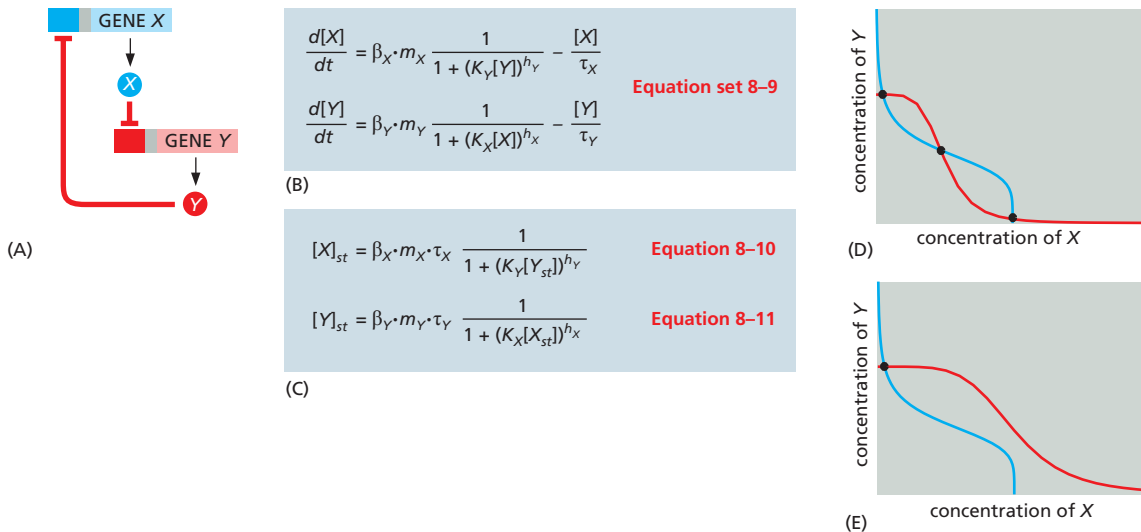
back to an Off state. But the transition points for switching on and switching off are different, and so the current state of the system depends on the route by which it has been taken in the past—a phenomenon called *hysteresis*.

A simple case of positive feedback can be seen in a regulatory system in which a transcription regulator activates (directly or indirectly) its own expression, as in **Figure 8-82A**. Positive feedback can also arise in a circuit with many intervening repressors or activators, so long as the net overall effect of the interactions is activation (**Figure 8-82B and C**).

To illustrate how positive feedback can generate stable states, let us focus on a simple positive feedback loop containing two repressors, X and Y , each of which inhibits expression of the other (**Figure 8-83A**). As we saw with Equation set 8-8 (**Figure 8-78B**) earlier, we can create differential equations describing the rate of change of $[X]$ and $[Y]$ (Equation set 8-9; **Figure 8-83B**). We can further modify these equations to include cooperativity by adding Hill coefficients. As we did earlier, we can then create equations for calculating the concentrations of $[X]$ and $[Y]$ when the system reaches a steady state—that is, when $(d[X]/dt) = 0$ and $(d[Y]/dt) = 0$ (Equations 8-10 and 8-11, **Figure 8-83C**).

Equations 8-10 and 8-11 can be used to carry out an intriguing mathematical procedure called a *nullcline* analysis. These equations define the relationships between the concentration of X at steady state, $[X_{st}]$, and the concentration of Y at steady state, $[Y_{st}]$, which must be simultaneously satisfied. We can plug in different values for $[Y_{st}]$ in Equation 8-10 and calculate the corresponding $[X_{st}]$ for each of these values. We can then graph $[X_{st}]$ as a function of $[Y_{st}]$. Next, we repeat the process by varying $[X_{st}]$ in Equation 8-11 to graph the resulting $[Y_{st}]$. The intersections of these two graphs determine the theoretically possible steady states of the system. For systems in which the Hill coefficients h_X and h_Y are much larger than 1, the lines in the two graphs intersect at three locations (**Figure 8-83D**). In other systems that have the same arrangement of regulators but different parameters, there might only be one intersection, indicating the presence of only a single

Figure 8-83 A graphical nullcline analysis. (A) X inhibits Y and Y inhibits X , resulting in a positive feedback loop. (B) Equation set 8-9 can be used to determine the rate of change in the concentrations of proteins X and Y . (C) Equations 8-10 and 8-11 provide the concentrations of proteins X and Y , respectively, when these concentrations reach a steady state. (D, E) *Blue curves* (called nullclines) are plots of $[X_{st}]$ calculated from Equation 8-10 over a range of concentrations of $[Y_{st}]$. *Red curves* are nullclines that indicate values of $[Y_{st}]$ calculated from Equation 8-11 over a range of concentrations of $[X_{st}]$. At an intersection of the two lines, both $[X]$ and $[Y]$ are at steady state. For plot D, the binding of both proteins to their target gene promoters was cooperative (h_X and h_Y much larger than 1), resulting in the presence of multiple intersections of the nullclines—suggesting that the system can assume multiple discrete steady states. In plot E, the binding of protein X to the promoter of gene Y was not cooperative (h_X close to 1), resulting in only one nullcline intersection and thus just one likely steady state.



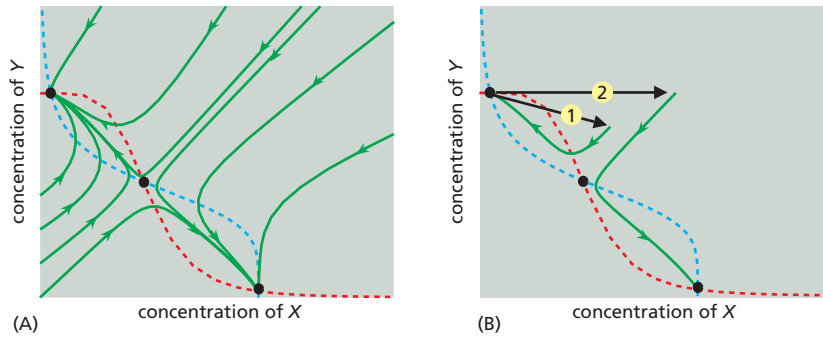


Figure 8-84 Analysis of the stability of a system's steady states. (A) The dotted lines are the nullclines for the system shown in Figure 8-83. Also shown are dynamic trajectories (green) that show the changes over time in $[X]$ and $[Y]$, starting at a variety of different initial concentrations (determined by solution of Equation set 8-9; see Figure 8-83B). By plotting $[X]$ versus $[Y]$ at each time point, we find that, although there are three possible steady states in this system, the dynamic trajectories converge on only two of them. The middle steady state is avoided: it is unstable, being unable to attract any trajectories. (B) Imagine that the system is at the upper-left steady state and experiences a perturbation (black arrows), such as a random fluctuation in the production rates of X and/or Y . If the perturbation is small (arrow 1), the system will return to the same steady state. On the other hand, a perturbation that drives the system beyond the unstable (middle) steady state (arrow 2) causes it to switch to the lower-right steady state. The set of perturbations that a system can withstand without switching from one steady state to the other is known as the region of attraction of that steady state.

steady state. For example, when there is a low cooperativity of protein X binding to the promoter of gene Y (that is, a small Hill coefficient, h_X , in Equation 8-11), the plot of $[Y]$ is less curved (Figure 8-83E), and it is less likely that there will be multiple intersections of the two curves.

We emphasized earlier that positive feedback typically generates a bistable system with two stable steady states. Why does the system modeled in Figure 8-83D have three? This conundrum can be explained by solving the reaction rate equations (Equation set 8-9; Figure 8-83B) for various different starting conditions of $[X]$ and $[Y]$, determining all values of $[X]$ and $[Y]$ as a function of time. Starting with each set of initial concentrations of $[X]$ and $[Y]$, these calculations produce a so-called *trajectory* of points, each indicated by a curved green line on Figure 8-84A. A fascinating pattern emerges: each trajectory moves across the plot and settles in one of two steady states, but never in the third (middle steady state). We conclude that the middle steady state is *unstable* because it cannot “attract” any trajectories. The system therefore has only two *stable* steady states. Thus, the number of stable steady states in a system need not be equal to the total number of its theoretically possible steady states. In fact, stable steady states are usually separated by unstable ones, as in our example.

Once this system adopts a fate by settling in one of the two steady states, does it have the ability to switch to the other state? The numerical solution of Equation set 8-9 can again provide an answer. In Figure 8-84B, we show the solution of this equation set for two perturbations from the upper-left steady state. For a small perturbation, the system returns to its original steady state. But the larger perturbation causes the system to switch to the alternate steady state. Thus, this system can be switched from one stable steady state to the other by subjecting it to an input (or a perturbation) that is large enough to make the other steady state more attractive. More generally, every stable steady state has a corresponding *region of attraction*, which can be intuitively thought of as the range of perturbations (of $[X]$ or $[Y]$ in this example) for which the dynamic trajectories converge back to that particular steady state, rather than switch to the other one.

The concept of a region of attraction has interesting implications for the heritability of transcriptional states and the transition rates between them. If the region of attraction around one steady state is large, for example, then most cells in the population will assume this particular state. Furthermore, this state is likely to be inherited by daughter cells, because minor perturbations, like those ensuing from an asymmetric distribution of molecules during cell division, will rarely be sufficient to induce switching to the other steady state. We should expect that the use of positive feedback, coupled to cooperativity, will quite often be associated with systems requiring stable cell memory.

Robustness Is an Important Characteristic of Biological Networks

Biological regulatory systems are exposed to frequent and sometimes extreme variations in external conditions or the concentrations or activities of key components. The ability of these systems to function normally in the face of such perturbations is called **robustness**. If we understand a complex system to the extent that we can reproduce its behavior with a computational model, then the

robustness of the system can be assessed by determining how well its normal function persists after changes in various parameters, such as rate constants and component concentrations. We have already seen, for example, how the presence of negative feedback reduces the sensitivity of the steady state to changes in the values of the system's parameters (see Figure 8-79). Considerations of robustness also apply to dynamic behaviors. Thus, for example, when discussing negative feedback, we described how the behavior of a system tends to become more oscillatory as the number of components that constitute the feedback loop increases. If we use different values of the parameters in models derived for systems like those in Figure 8-80, we find that the system with the longer loop tends to exhibit stable oscillations within a much broader range of parameters, indicating that this system provides a more robust oscillator. We can perform similar calculations to determine the ability of different systems to achieve robust bistability arising from positive feedback. Thus, one benefit of computational models is that they allow us to probe the robustness of biological networks in a systematic and rigorous way.

Two Transcription Regulators That Bind to the Same Gene Promoter Can Exert Combinatorial Control

Thus far, we have discussed how one transcription regulator can modulate the expression level of a gene. Most genes, however, are controlled by more than one type of transcription regulator, providing *combinatorial control* that allows two or more inputs to influence the expression of one gene. We can use computational methods to unveil some of the important regulatory features of combinatorial control systems.

Consider a gene whose promoter contains binding sites for two regulatory proteins, A and R , which bind to their individual sites independently. There are four possible binding configurations (Figure 8-85A). Suppose that A is a transcription activator, R is a transcription repressor, and the gene is only active when A is bound and R is not bound. We learned earlier that the probability that A is bound and the probability that R is not bound can be determined by the equations in Figure 8-86A. The product of these two probabilities gives us the probability of gene activation.

This example illustrates an AND NOT logic function (A and not R) (see Figure 8-85A). Maximal activation of this gene is accomplished when $[A]$ is high and $[R]$ is zero. However, intermediate levels of gene activation are also possible depending on the levels of A and R and also on the binding affinities of A and R for their respective sites (that is, K_A and K_R). When $K_A > K_R$, a small concentration of A is capable of overcoming repression by R . Conversely, if $K_A < K_R$, then much more A is needed to activate the gene (Figure 8-86B and C).

Many other logic functions can govern combinatorial gene regulation. For example, an AND logic gate results when two activators, A_1 and A_2 , are both required for a gene to be transcribed (Figure 8-85B and Figure 8-86D). In *E. coli* cells, the *AraJ* gene controls some aspects of arabinose sugar metabolism: its

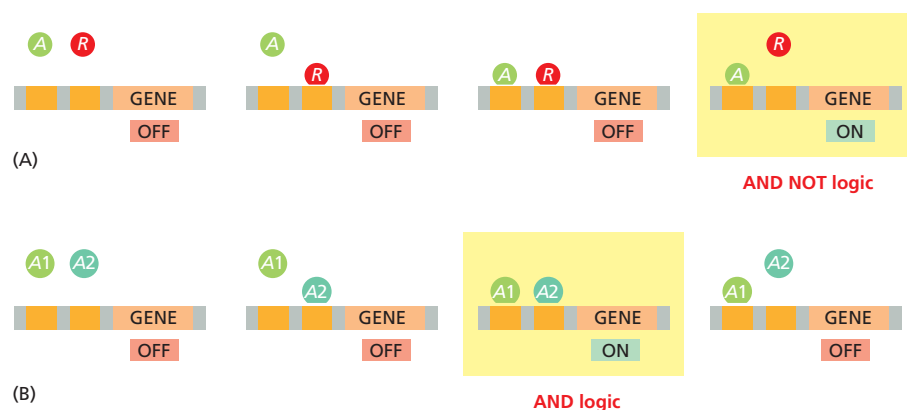


Figure 8-85 Combinatorial control of gene expression. There are many ways in which gene expression can be controlled by two transcription regulators. To define precisely the relationship between the two inputs and the gene expression output, a regulatory circuit is often described as a specific type of *logic gate*, a term borrowed from electronic circuit design. A simple example is the OR logic gate (not shown here), in which a gene is controlled by two transcription activators, and one or the other can activate gene expression. (A) In a system with an activator A and repressor R , if transcription is turned on only when A is bound and R is not, then the result is an AND NOT logic gate. We saw an example of this logic in Chapter 7 (Figure 7-18). (B) An AND gate results when two transcription activators, A_1 and A_2 , are both required to turn on a gene.

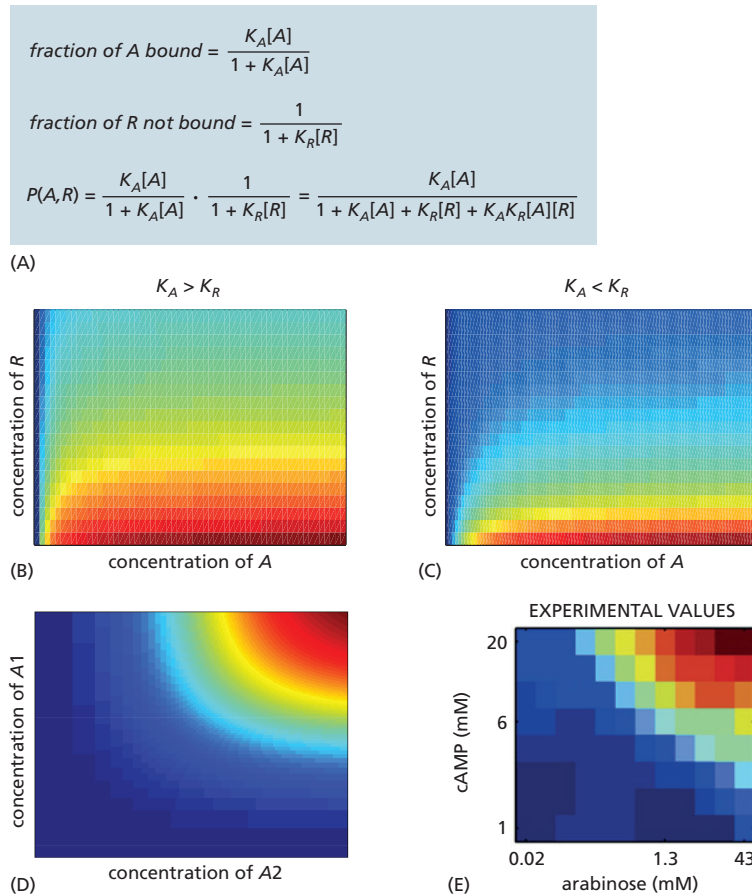


Figure 8–86 How the quantitative output of a gene depends on both its combinatorial logic and the affinities of transcription regulators. (A) In a combinatorial gene regulatory system like that illustrated in Figure 8–85A, the fraction of promoters bound by activator *A* and the fraction not bound by repressor *R* are each determined as shown here. The product of these probabilities provides the probability, $P(A, R)$, that a gene promoter is active. (B–E) In these four panels, *red* indicates high gene expression and *blue* indicates low gene expression. (B, C) Depictions of gene expression from the system described in panel A. The two panels demonstrate how the system behaves when the relative affinities of the two transcription regulators change as indicated above each panel. (D) Gene expression in a case where the gene turns on only at high levels of both activating inputs (*A1* and *A2*), as shown in Figure 8–85B. (E) Experimental data showing measured expression of a gene in *E. coli* that is combinatorially regulated by two inputs: arabinose and cAMP. Note the close resemblance to panel D. (E, adapted from S. Kaplan et al., *Mol. Cell* 29:786–792, 2008. With permission from Elsevier.)

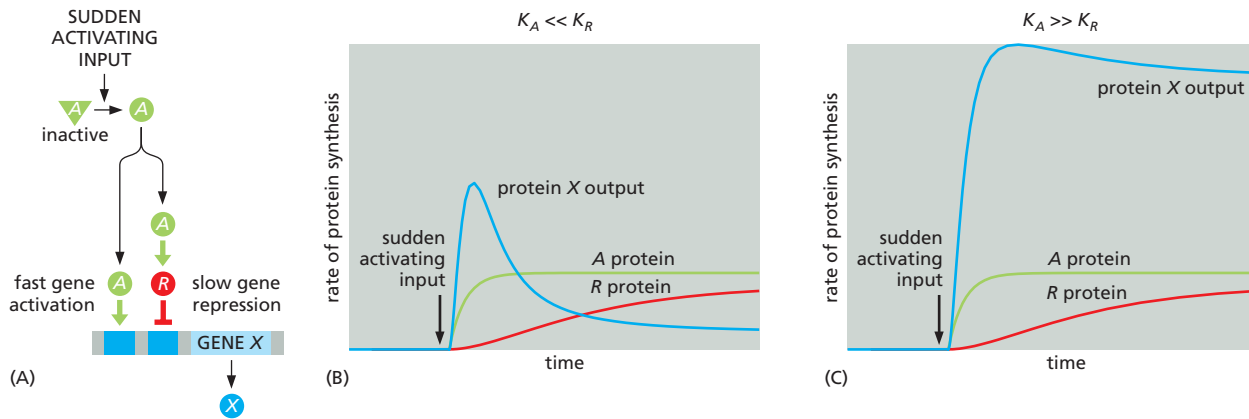
expression requires two transcription regulators, one activated by arabinose and the other activated by the small molecule cAMP (Figure 8–86E).

An Incoherent Feed-forward Interaction Generates Pulses

Imagine that a sudden input signal immediately activates a transcription activator *A* and that the same input signal induces the much slower synthesis of a transcription repressor protein *R* that acts on the same gene *X*. If *A* and *R* control gene expression by an AND NOT logic function like that described above, our intuition tells us that this system should be able to generate a pulse of transcription: when *A* is activated (and *R* is absent), the transcription of gene *X* will begin and cause an increase in the concentration of protein *X*, but then transcription will shut off when the concentration of *R* increases to a sufficiently high value.

Arrangements of this type are common in the cell. In *E. coli*, for example, galactose metabolic genes are positively regulated by the catabolite activator protein (CAP), which is activated at high levels of cAMP. The same genes are repressed by the GalS repressor protein, which is encoded by a gene whose transcription is likewise activated by CAP. Thus, an increase in input (cAMP) activates *A* (CAP), and transcription of the galactose genes begins. But activation of *A* also causes a subsequent buildup of *R* (GalS), which causes the same genes to be repressed after a delay. This results in an *incoherent feed-forward motif* (Figure 8–87A).

The response of the incoherent feed-forward motif will vary, depending on the parameters of the system. Suppose, for example, that the transcription activator protein *A* binds more weakly to the gene regulatory region than does the transcription repressor protein *R* ($K_A \ll K_R$). In this case, there will be a transient burst of protein synthesized by the affected gene (gene *X*) in response to a sudden activating input (Figure 8–87B). In contrast, the output will be more sustained if K_A is much larger than K_R , because the repression will be too weak to overcome



the gene activation (Figure 8-87C). Other properties of this network, such as the dependence of the amplitude of the pulse on the various rate constants in the system, can be explored with the same computational tools. Thus, our intuitive guess about how this system would behave was only partially correct; even the simplest of networks depends on precise interaction strengths, demonstrating yet again why mathematics is needed to complement cartoon drawings.

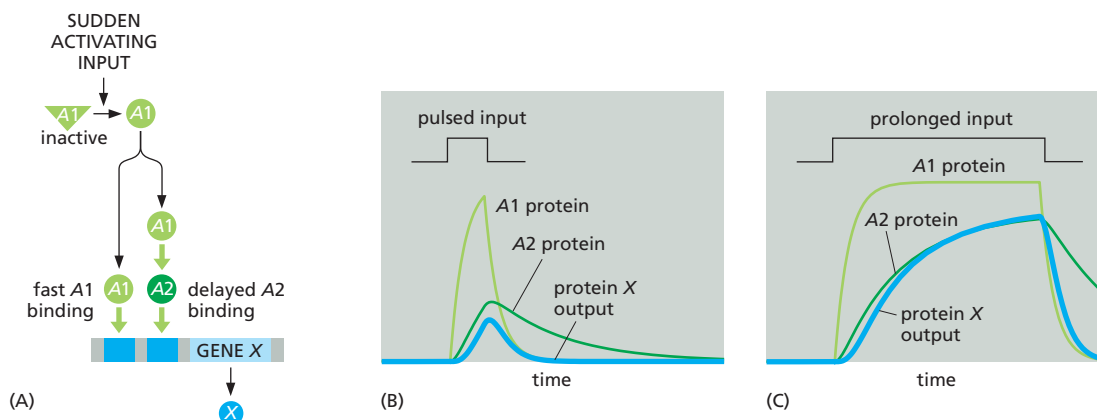
A Coherent Feed-forward Interaction Detects Persistent Inputs

In the bacterium *E. coli*, the sugar arabinose is only consumed when the preferred sugar, glucose, is scarce. The strategy that cells use to assess the presence of arabinose and absence of glucose involves a feed-forward arrangement that is different from the one just described. In this case, depletion of glucose causes an increase of cAMP, which is sensed by the CAP transcription activator protein, as described previously. In this case, however, CAP also induces the synthesis of a second transcription activator, AraC. Both activator proteins are necessary to activate arabinose metabolic genes (the AND logic function in Figure 8-85B).

This arrangement, known as a *coherent feed-forward motif*, has the interesting characteristics illustrated in Figure 8-88. Imagine that two activators, A1 and A2, are both required to initiate transcription of a gene. The input to the network activates A1 directly, but only activates A2 through this A1 activation. Thus, for a protein to be synthesized from this gene, long-term inputs are required that allow both A1 and A2 to be produced in active form. Brief input pulses are either ignored or produce small outputs. The requirement for a long input is important if assurances about a signal are needed before a costly cellular program is triggered. For example, glucose is the sugar on which *E. coli* cells grow best. Before cells trigger arabinose metabolism in the example above, it might be beneficial to be sure that glucose has been depleted (a sustained CAP pulse), rather than inducing the arabinose program during a transient glucose fluctuation.

Figure 8-87 How an incoherent feed-forward motif can generate a brief pulse of gene activation in response to a sustained input. (A) Diagram of an incoherent feed-forward motif in which the transcription activator A and the repressor R control the expression of gene X using the AND NOT logic of Figure 8-85A. (B) When $K_A \ll K_R$, this motif generates a pulse of protein X expression, such that the output goes back down even if the input remains high. (C) When $K_A \gg K_R$, the same motif responds to a sustained input by generating a sustained output.

Figure 8-88 How a coherent feed-forward motif responds to various inputs. (A) Diagram of a coherent feed-forward motif in which the transcription activators A1 and A2 together activate expression of gene X using the AND logic of Figure 8-85B. (B) The response to a brief input can be either weak (as shown) or nonexistent. This allows the motif to ignore random fluctuations in the concentration of signaling molecules. (C) A prolonged input produces a strong response that can turn off rapidly.



The Same Network Can Behave Differently in Different Cells Because of Stochastic Effects

Up to this point, we have assumed that all cells in a population produce identical behaviors if they contain the same network. It is important, however, to account for the fact that cells often show considerable individuality in their responses. Consider a situation in which a single mother cell divides into two daughter cells of equal volume. If the mother cell has only one molecule of a given protein, then only one daughter will inherit it. The daughters, though genetically identical, are already different. This variability is most pronounced for molecules that are present in small numbers. Nevertheless, even when there are many copies of a particular protein (or RNA), it is very unlikely that both daughter cells will end up with exactly the same number of molecules.

This is just one illustration of a universal feature of cells: their behaviors are often **stochastic**, meaning that they display variability in their protein content and therefore exhibit variations in phenotypes. In addition to the asymmetric partitioning of molecules after cell division, variability can originate from many chemical reactions. Imagine, for example, that our mother cell contains a simple gene regulatory circuit with a positive feedback loop like that shown in Figure 8-82B. Even if both daughter cells receive a copy of this circuit, including one copy of the initial transcription activator protein, there will be variability in the time required for promoter binding—and it will be statistically nearly impossible for the genes in the two daughter cells to become activated at precisely the same time. If the system is bistable and poised near a switching point, then variability in the response might flip the switch in only one daughter cell. Two daughter cells that were born identical can thereby acquire, by chance, a dramatic difference in phenotype.

More generally, isogenic populations of cells grown in the same environment display diversity in size, shape, cell-cycle position, and gene expression. These differences arise because biochemical reactions require probabilistic collisions between randomly moving molecules, with each event resulting in changes in the number of molecular species by integer amounts. The amplified effect of fluctuations in a molecular reactant, or the compounded effects of fluctuations across many molecular reactants, often accumulates as an observable phenotype. This can endow a cell with individuality and generate nongenetic cell-to-cell variability in a population.

Nongenetic variability can be studied in the laboratory by single-cell measurements of fluorescent proteins expressed from genes under the control of a specific promoter. Live cells can be mounted on a slide and viewed through a fluorescence microscope, revealing the striking variability in protein expression levels (Figure 8-89). Another approach is to use flow cytometry, which works by streaming a dilute suspension of cells past an illuminator and measuring the fluorescence of individual cells as they flow past the detector. Fluorescence values can be used to build histograms that reveal the variability in a process across a population of cells, with a broad histogram indicating higher variability.

Several Computational Approaches Can Be Used to Model the Reactions in Cells

We have focused primarily on the use of ordinary differential equations to model the dynamics of simple regulatory circuits. These models are called *deterministic*, because they do not incorporate stochastic variability and will always produce the same result from a specific set of parameters. As we have seen, such models can provide useful insights, particularly in the detailed mechanistic analysis of small regulatory circuits. However, other types of computational approaches are also needed to comprehend the great complexity of cell behavior. *Stochastic models*, for example, attempt to account for the very important problem of random variability in molecular networks. These models do not provide deterministic predictions about the behavior of molecules; instead, they incorporate random

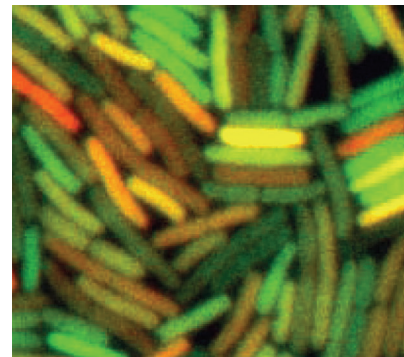


Figure 8-89 Different levels of gene expression in individual cells within a population of *E. coli* bacteria. For this experiment, two different reporter proteins (one fluorescing *green*, the other *red*), controlled by a copy of the same promoter, have been introduced into all of the bacteria. Some cells express only one gene copy, and so appear either *red* or *green*, while others express both gene copies, and so appear *yellow*. This experiment reveals variable levels of fluorescence, indicating variable levels of gene expression within an apparently uniform population of cells. (From M.B. Elowitz et al., *Science* 297:1183–1186, 2002. With permission from AAAS.)

variation into molecule numbers and interactions, and the purpose of these models is to obtain a better understanding of the probability that a system will exist in a certain state over time.

Numerous other modeling strategies have been or are being developed. *Boolean networks* are used for the qualitative analysis of complex gene regulatory networks containing large numbers of interacting components. In these models, each molecule is a node that can exist in either the active or inactive state, thereby affecting the state of the nodes it is linked to. Models of this sort provide insights into the flow of information through a network, and they were useful in helping us understand the complex gene regulatory network that controls the early development of the sea urchin (see Figure 7-45). Boolean networks therefore reduce complex networks to a highly simplified (and potentially inaccurate) form. At the other extreme are *agent-based simulations*, in which thousands of molecules (or “agents”) in a system are modeled individually, and their probable behaviors and interactions with each other over time are calculated on the basis of predicted physical and chemical behaviors, often while taking stochastic variation into account. Agent-based approaches are computationally demanding but have the potential to generate highly life-like simulations of real biological systems.

Statistical Methods Are Critical for the Analysis of Biological Data

Dynamics, differential equations, and theoretical modeling are not the be-all and end-all of mathematics. Other branches of the subject are no less important for biologists. Statistics—the mathematics of probabilistic processes and noisy data sets—is an inescapable part of every biologist’s life.

This is true in two main ways. First, imperfect measurement devices and other errors generate experimental noise in our data. Second, all cell-biological processes depend on the stochastic behavior of individual molecules, as we just discussed, and this results in biological noise in our results. How, in the face of all this noise, do we come to conclusions about the truth of hypotheses? The answer is statistical analysis, which shows how to move from one level of description to another: from a set of erratic individual data points to a simpler description of the key features of the data.

Statistics teaches us that the more times we repeat our measurements, the better and more refined the conclusions we can draw from them. Given many repetitions, it becomes possible to describe our data in terms of variables that summarize the features that matter: the mean value of the measured variable, taken over the set of data points; the magnitude of the noise (the standard deviation of the set of data points); the likely error in our estimate of the mean value (the standard error of the mean); and, for specialists, the details of the probability distribution describing the likelihood that an individual measurement will yield a given value. For all these things, statistics provides recipes and quantitative formulas that biologists must understand if they are to make rigorous conclusions on the basis of variable results.

Summary

Quantitative mathematical analysis can provide a powerful extra dimension in our understanding of cell regulation and function. Cell regulatory systems often depend on macromolecular interactions, and mathematical analysis of the dynamics of these interactions can unveil important insights into the importance of binding affinities and protein stability in the generation of transcriptional or other signals. Regulatory systems often employ network motifs that generate useful behaviors: a rapid negative feedback loop dampens the response to input signals; a delayed negative feedback loop creates a biochemical oscillator; positive feedback yields a system that alternates between two stable states; and feed-forward motifs provide systems that generate transient signal pulses or respond only to sustained inputs. The dynamic behavior of these network motifs can be dissected in detail with deterministic and stochastic mathematical modeling.

PROBLEMS

Which statements are true? Explain why or why not.

8-1 Because a monoclonal antibody recognizes a specific antigenic site (epitope), it binds only to the specific protein against which it was made.

8-2 Given the inexorable march of technology, it seems inevitable that the sensitivity of detection of molecules will ultimately be pushed beyond the yoctomole level (10^{-24} mole).

8-3 If each cycle of PCR doubles the amount of DNA synthesized in the previous cycle, then 10 cycles will give a 10^3 -fold amplification, 20 cycles will give a 10^6 -fold amplification, and 30 cycles will give a 10^9 -fold amplification.

8-4 To judge the biological importance of an interaction between protein *A* and protein *B*, we need to know quantitative details about their concentrations, affinities, and kinetic behaviors.

8-5 The rate of change in the concentration of any molecular species *X* is given by the balance between its rate of appearance and its rate of disappearance.

8-6 After a sudden increase in its rate of synthesis, a protein with a slow rate of degradation will reach a new steady-state level more quickly than a protein with a rapid rate of degradation.

Discuss the following problems.

8-7 A common step in the isolation of cells from a sample of animal tissue is to treat the tissue with trypsin, collagenase, and EDTA. Why is such a treatment necessary, and what does each component accomplish? And why does this treatment not kill the cells?

8-8 Tropomyosin, at 93 kilodaltons, sediments at 2.6S, whereas the 65-kilodalton protein, hemoglobin, sediments at 4.3S. (The sedimentation coefficient *S* is a linear measure of the rate of sedimentation.) These two proteins are drawn to scale in **Figure Q8-1**. Why does the bigger

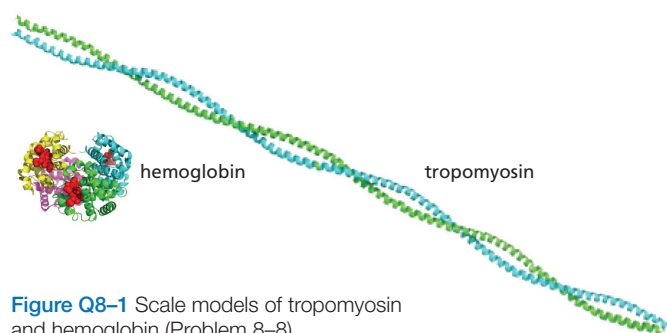


Figure Q8-1 Scale models of tropomyosin and hemoglobin (Problem 8-8).

protein sediment more slowly than the smaller one? Can you think of an analogy from everyday experience that might help you with this problem?

8-9 Hybridoma technology allows one to generate monoclonal antibodies to virtually any protein. Why is it, then, that genetically tagging proteins with epitopes is such a commonly used technique, especially as an epitope tag has the potential to interfere with the function of the protein?

8-10 How many copies of a protein need to be present in a cell in order for it to be visible as a band on an SDS gel? Assume that you can load 100 μg of cell extract onto a gel and that you can detect 10 ng in a single band by silver staining the gel. The concentration of protein in cells is about 200 mg/mL, and a typical mammalian cell has a volume of about 1000 μm^3 and a typical bacterium a volume of about 1 μm^3 . Given these parameters, calculate the number of copies of a 120-kilodalton protein that would need to be present in a mammalian cell and in a bacterium in order to give a detectable band on a gel.

8-11 You have isolated the proteins from two adjacent spots after two-dimensional polyacrylamide-gel electrophoresis and digested them with trypsin. When the masses of the peptides were measured by MALDI-TOF mass spectrometry, the peptides from the two proteins were found to be identical except for one (**Figure Q8-2**). For this peptide, the mass-to-charge (m/z) values differed by 80, a value that does not correspond to a difference in amino acid sequence. (For example, glutamic acid instead of valine at one position would give an m/z difference of around 30.) Can you suggest a possible difference between the two peptides that might account for the observed m/z difference?

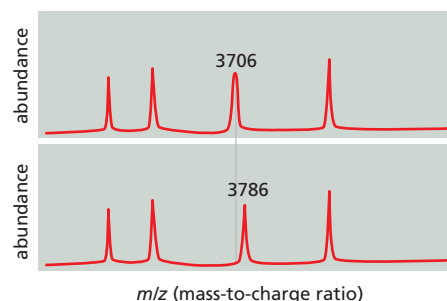


Figure Q8-2 Masses of peptides measured by MALDI-TOF mass spectrometry (Problem 8-11). Only the numbered peaks differ between the two protein samples.

8-12 You want to amplify the DNA between the two stretches of sequence shown in **Figure Q8-3**. Of the listed primers, choose the pair that will allow you to amplify the DNA by PCR.

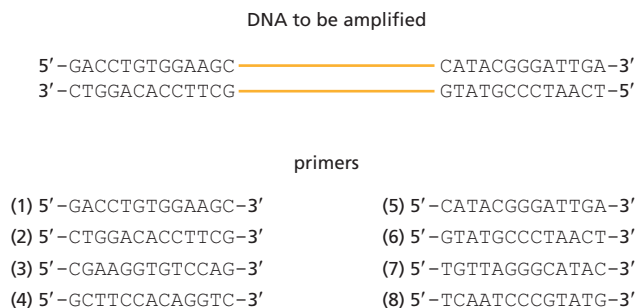


Figure Q8-3 DNA to be amplified and potential PCR primers (Problem 8-12).

8-13 In the very first round of PCR using genomic DNA, the DNA primers prime synthesis that terminates only when the cycle ends (or when a random end of DNA is encountered). Yet, by the end of 20–30 cycles—a typical amplification—the only visible product is defined precisely by the ends of the DNA primers. In what cycle is a double-strand fragment of the correct size first generated?

8-14 Explain the difference between a gain-of-function mutation and a dominant-negative mutation. Why are both these types of mutation usually dominant?

8-15 Discuss the following statement: “We would have no idea today of the importance of insulin as a regulatory hormone if its absence were not associated with the human disease diabetes. It is the dramatic consequences of its absence that focused early efforts on the identification of insulin and the study of its normal role in physiology.”

8-16 You have received the results from an RNA-seq analysis of mRNAs from liver. You had anticipated counting the number of reads of each mRNA to determine the relative abundance of different mRNAs. But you are puzzled because many of the mRNAs have given you results like those shown in **Figure Q8-4**. How is it that different parts of an mRNA can be represented at different levels?

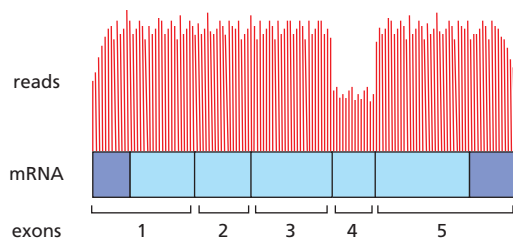


Figure Q8-4 RNA-seq reads for a liver mRNA (Problem 8-16). The exon structure of the mRNA is indicated, with protein-coding segments indicated in *light blue* and untranslated regions in *dark blue*. The numbers of sequencing reads are indicated by the heights of the vertical lines above the mRNA.

8-17 Examine the network motifs in **Figure Q8-5**. Decide which ones are negative feedback loops and which are positive. Explain your reasoning.

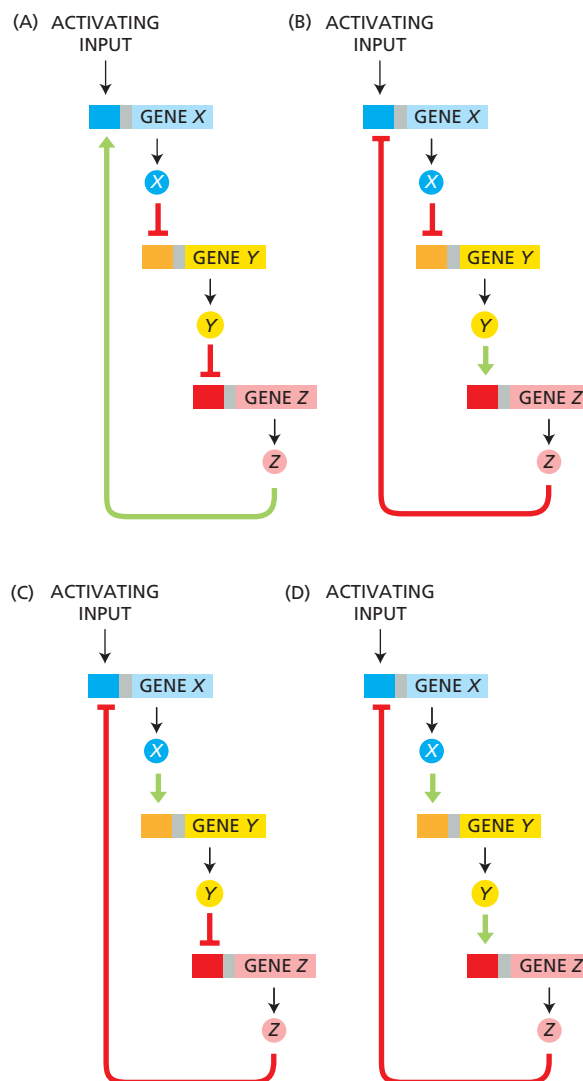


Figure Q8-5 Network motifs composed of transcription activators and repressors (Problem 8-17).

8-18 Imagine that a random perturbation positions a bistable system precisely at the boundary between two stable states (at the *orange dot* in **Figure Q8-6**). How would the system respond?

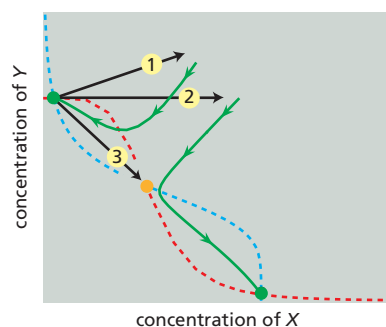


Figure Q8-6

Perturbations of a bistable system (Problem 8-18). As shown by the *green lines*, after perturbation 1 the system returns to its original stable state (*green dot* at left), and after perturbation 2, the system moves to the other stable state (*green dot* at right). Perturbation 3 moves the system to the precise boundary between the two stable states (*orange dot*).

8–19 Detailed analysis of the regulatory region of the *Lac* operon has revealed surprising complexity. Instead of a single binding site for the Lac repressor, as might be expected, there are three sites termed operators (O_1 , O_2 , and O_3) arrayed along the DNA as shown in [Figure Q8–7](#).

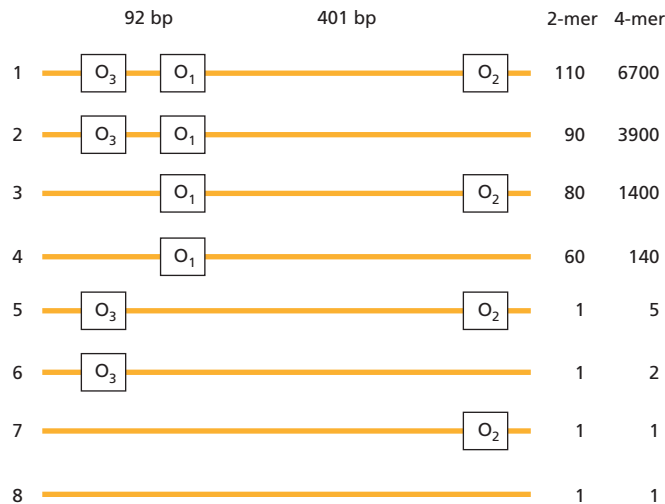


Figure Q8–7 Repression of β -galactosidase by promoter regions that contain different combinations of Lac repressor binding sites (Problem 8–19). The base-pair (bp) separation of the three operator sites is shown. Numbers at *right* refer to the level of repression, with higher numbers indicating more effective repression by dimeric (2-mer) or tetrameric (4-mer) repressors. (From S. Oehler et al., *EMBO J.* 9:973–979, 1990. With permission from John Wiley & Sons.)

To probe the functions of these three sites, you make a series of constructs in which various combinations of operator sites are present. You examine their ability to repress expression of β -galactosidase, using either tetrameric (wild type) or dimeric (mutant) forms of the Lac repressor. The dimeric form of the repressor can bind to a single operator (with the same affinity as the tetramer) with each monomer binding to half the site. The tetramer, the form normally expressed in cells, can bind to two sites simultaneously. When you measure repression of β -galactosidase expression, you find the results shown in [Figure Q8–7](#), with higher numbers indicating more effective repression.

A. Which single operator site is the most important for repression? How can you tell?

B. Do combinations of operator sites (Figure Q8–7, constructs 1, 2, 3, and 5) substantially increase repression by the dimeric repressor? Do combinations of operator sites substantially increase repression by the tetrameric repressor? If the two repressors behave differently, offer an explanation for the difference.

C. The wild-type repressor binds O_3 very weakly when it is by itself on a segment of DNA. However, if O_1 is included on the same segment of DNA, the repressor binds O_3 quite well. How can that be?

REFERENCES

Isolating Cells and Growing Them in Culture

- Greenfield E (2014) *Antibodies: A Laboratory Manual*, 2nd ed. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.
- Sherr CJ & DePinho RA (2000) Cellular senescence: mitotic clock or culture shock? *Cell* 102, 407–410.
- Spector DL, Goldman RD & Leinwand LA (eds.) (1998) *Cells: A Laboratory Manual*. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.
- Todaro GJ & Green H (1963) Quantitative studies of the growth of mouse embryo cells in culture and their development into established lines. *J. Cell Biol.* 17, 299–313.

Purifying Proteins

- Burgess R & Deutcher MP (eds.) (2009) *Guide to Protein Purification*, 2nd ed. *Methods in Enzymology*, Vol. 463. Cambridge, MA: Academic Press.
- De Duve C & Beaufay H (1981) A short history of tissue fractionation. *J. Cell Biol.* 91, 293s–299s.
- Scopes RK (1994) *Protein Purification: Principles and Practice*, 3rd ed. New York: Springer-Verlag.
- Simpson RJ, Adams PD & Golemis EA (2008) *Basic Methods in Protein Purification and Analysis: A Laboratory Manual*. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.
- Wood DW (2014) New trends and affinity tag designs for recombinant protein purification. *Curr. Opin. Struct. Biol.* 26, 54–61.

Analyzing Proteins

- Aebersold R & Mann M (2016) Mass-spectrometric exploration of proteome structure and function. *Nature* 537, 347–355.

- Baxevanis AD, Bader GD & Wishart DS (eds.) (2020) *Bioinformatics: A Practical Guide to the Analysis of Genes and Proteins*, 4th ed. Hoboken, NJ: Wiley.
- Goodrich JA & Kugel JF (2007) *Binding and Kinetics for Molecular Biologists*. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.
- Mehmood S, Allison TM & Robinson CV (2015) Mass spectrometry of protein complexes: from origins to applications. *Annu. Rev. Phys. Chem.* 66, 453–474.
- O’Connell MR, Gamsjaeger R & Mackay JP (2009) The structural analysis of protein-protein interactions by NMR spectroscopy. *Proteomics* 9, 5224–5232.
- Pollard TD (2010) A guide to simple and informative binding assays. *Mol. Biol. Cell* 21, 4061–4067.
- Wuthrich K (2003) NMR studies of structure and function of biological macromolecules (Nobel Lecture). *J. Biomol. NMR* 27, 13–39.

Analyzing and Manipulating DNA

- Gibson DG, Young L, Chuang RY . . . Smith HO (2009) Enzymatic assembly of DNA molecules up to several hundred kilobases. *Nat. Methods* 6, 343–345.
- Green MR & Sambrook J (2012) *Molecular Cloning: A Laboratory Manual*, 4th ed. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.
- Kosuri S & Church GM (2014) Large-scale de novo DNA synthesis: technologies and applications. *Nat. Methods* 11, 499–507.
- Metzker ML (2010) Sequencing technologies—the next generation. *Nat. Rev. Genet.* 11, 31–46.

- Mullis K, Faloona F, Scharf S . . . Erlich H (1986) Specific enzymatic amplification of DNA in vitro: the polymerase chain reaction. *Cold Spring Harb. Symp. Quant. Biol.* 51 Pt 1, 263–273.
- Mullis KB (1990) The unusual origin of the polymerase chain reaction. *Sci. Am.* 262, 56–61, 64–55.
- Nathans D & Smith HO (1975) Restriction endonucleases in the analysis and restructuring of DNA molecules. *Annu. Rev. Biochem.* 44, 273–293.
- Sanger F, Nicklen S & Coulson AR (1977) DNA sequencing with chain-terminating inhibitors. *Proc. Natl. Acad. Sci. USA* 74, 5463–5467.
- Shendure J & Lieberman Aiden E (2012) The expanding scope of DNA sequencing. *Nat. Biotechnol.* 30, 1084–1094.
- Watson JD, Myers RM & Caudy AA (2007) Recombinant DNA: Genes and Genomes—A Short Course, 3rd ed. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.

Studying Gene Function and Expression

- Agrotis A & Ketteler R (2015) A new age in functional genomics using CRISPR/Cas9 in arrayed library screening. *Front. Genet.* 6, 300.
- D'Haeseleer P (2005) How does gene expression clustering work? *Nat. Biotechnol.* 23, 1499–1501.
- Doudna JA (2020) The promise and challenge of therapeutic genome editing. *Nature* 578, 229–236.
- Fellmann C & Lowe SW (2014) Stable RNA interference rules for silencing. *Nat. Cell Biol.* 16, 10–18.
- Ingolia NT, Hussmann JA & Weissman JS (2019) Ribosome profiling: global views of translation. *Cold Spring Harb. Perspect. Biol.* 11.
- Luecken MD & Theis FJ (2019) Current best practices in single-cell RNA-seq analysis: a tutorial. *Mol. Syst. Biol.* 15, e8746.

- Mali P, Esvelt KM & Church GM (2013) Cas9 as a versatile tool for engineering biology. *Nat. Methods* 10, 957–963.
- Rio DC, Ares M, Hannon GJ & Wilson TW (2011) RNA: A Laboratory Manual. Cold Spring Harbor, NY: Cold Spring Harbor Laboratory Press.
- Stark R, Grzelak M & Hadfield J (2019) RNA sequencing: the teenage years. *Nat. Rev. Genet.* 20, 631–656.
- Strachan T & Read AP (2018) Human Molecular Genetics, 5th ed. Boca Raton, FL: CRC Press.
- Wilson RC & Doudna JA (2013) Molecular mechanisms of RNA interference. *Annu. Rev. Biophys.* 42, 217–239.

Mathematical Analysis of Cell Function

- Alon U (2007) Network motifs: theory and experimental approaches. *Nat. Rev. Genet.* 8, 450–461.
- Alon U (2019) An Introduction to Systems Biology: Design Principles of Biological Circuits, 2nd ed. Boca Raton, FL: Chapman & Hall/CRC.
- Ferrell JE, Jr. & Ha SH (2014) Ultrasensitivity part III: cascades, bistable switches, and oscillators. *Trends Biochem. Sci.* 39, 612–618.
- Gunawardena J (2014) Models in biology: “accurate descriptions of our pathetic thinking.” *BMC Biol.* 12, 29.
- Ingalls BP (2013) Mathematical Modeling in Systems Biology: An Introduction. Cambridge, MA: MIT Press.
- Lewis J (2008) From signals to patterns: space, time, and mathematics in developmental biology. *Science* 322, 399–403.
- Novak B & Tyson JJ (2008) Design principles of biochemical oscillators. *Nat. Rev. Mol. Cell Biol.* 9, 981–991.
- Xia PF, Ling H, Foo JL & Chang MW (2019) Synthetic genetic circuits for programmable biological functionalities. *Biotechnol. Adv.* 37, 107393.